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## **Innovative ways to encourage household savings for financial wellbeing in South Africa: A township and national level empirical analysis.**

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**INNOVATIVE WAYS TO ENCOURAGE HOUSEHOLD SAVINGS FOR  
FINANCIAL WELLBEING IN SOUTH AFRICA: A TOWNSHIP AND  
NATIONAL LEVEL EMPIRICAL ANALYSIS**

by

**NHLONIPHO PROSPERITY BALOYI**

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A dissertation submitted in fulfilment of the requirements for the degree

**Master of Business Studies in Finance**

in the

Department of Finance and Investment

**FACULTY OF ECONOMICS AND FINANCE**

at the

**TSHWANE UNIVERSITY OF TECHNOLOGY**

**Supervisor:** Prof. Mulatu F. Zerihun (PhD, DBA)

**September 2023**

## DECLARATION

I, Nhlonipho Prosperity Baloyi, declare that this work is wholly my own work, and has not been submitted to any other university or other institution of higher education for the purpose of obtaining a degree. Where help has been sought, such help has been properly acknowledged.



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**Nhlonipho Prosperity Baloyi**

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**Date**

2024/01/17

**Submission approved by:**

Prof. Mulatu Fekadu Zerihun (PhD, DBA)



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**Signature and date**

2024/01/17

## **DEDICATION**

I dedicate this work to my God, who is the God of Abraham, Isaac, and Jacob; the God of the prophets and apostles. The father of my Lord Jesus Christ who always causes me to triumph. This work is also dedicated to my family and friends.

## **ACKNOWLEDGEMENTS**

Firstly, I would like to thank the Almighty God, my father, and the father of my Lord Jesus Christ, for granting me the strength, ability, and wisdom to complete this dissertation. Dear Lord Jesus, I am grateful for your love and grace and thank you for your kindness. You have beautified my life with joy, goodness, and grace.

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I would also like to acknowledge the Tshwane University of Technology for offering me the Research and Innovation (RnI) scholarship to fund my master's tuition.

## **ABSTRACT**

South Africa is a country with a poor saving culture, which is detrimental to households in townships and at the national level. The gross household savings was 2.2% as of the fourth quarter of 2022 (South African Reserve Bank, 2023:22). Lack of household savings results in many households being adversely affected by financial uncertainty. Predominantly, challenges are encountered in addressing financial emergencies and/or long-term financial needs. The overall objective of this study was to craft innovative ways that can be used to encourage savings for financial wellbeing at the household level in townships and at the national level. The study collected both primary and secondary data. The study collected data through questionnaires and focus group discussions. Multiple logistic regression and linear regression models were estimated to assess the relationship between the household savings variable (for logistic regression) and monthly household savings in rands (for linear regression).

The secondary time series data were analysed using autoregressive distributed lag, which assisted in interpreting the long-run relationships between the household savings ratio and independent variables, with the data spanning 1980 to 2021. The study found inadequate household savings at both the macro and micro levels, and where household savings did exist, the household savings were decreasing at both the micro and macro levels. The study suggests innovative ways to encourage household savings for financial wellbeing, at both the macro and micro levels, which are a state-owned commercial bank; introducing a financial literacy curriculum into early childhood development; the government, through the National Treasury, to allocate incentives for household savings, such as tax incentives; commercial banks to cater for households in townships by offering them services specific to townships

and their traits; and advertisements on various platforms on different household savings packages as innovative ways to encourage household savings for financial wellbeing at both the micro and macro levels. The outcomes of this study, in the form of the innovative ways to encourage household savings for financial wellbeing at both the household (micro) and national (macro) levels, are very significant and may be adopted by policymakers as a tool that they can utilise to comprehend the root causes of low, and in many instances a lack of, household savings. Policymakers can also adopt crafted innovative ways to encourage household savings for financial wellbeing at both the household (micro) and national (macro) levels as a vehicle to achieve the National Development Plan's (NDP) objective of increasing household and other types of savings. The findings can therefore be adopted as a roadmap to fulfil the NDP's mandate on savings by 2030, and they are also aligned to meeting the United Nations' Sustainable Development Goals 8 and 10.

**Keywords:** Financial Wellbeing, Households, Savings, Townships

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## LIST OF ABBREVIATIONS

|         |                                                        |
|---------|--------------------------------------------------------|
| ADF     | Augmented Dickey-Fuller                                |
| ANC     | African National Congress                              |
| ANOVA   | Analysis of variance                                   |
| AOR     | Adjusted odds ratio                                    |
| ARDL    | Autoregressive distributed lag                         |
| ATM     | Automated teller machine                               |
| COSATU  | Congress of South African Trade Unions                 |
| CUSUM   | Cumulative sum of recursive residuals                  |
| CUSUMSQ | Cumulative sum of squared recursive residuals          |
| ECD     | Early Childhood Development                            |
| ECM     | Error correction model                                 |
| ECT     | Error correction coefficient                           |
| FDIC    | Federal Deposit Insurance Corporation                  |
| FinTech | Financial technology                                   |
| GDP     | Gross domestic product                                 |
| IMF     | International Monetary Fund                            |
| LM      | Lagrange multiplier                                    |
| NDP     | National Development Plan                              |
| NEHAWU  | National Education, Health and Allied Workers' Union   |
| NPC     | National Planning Commission                           |
| OECD    | Organization for Economic Co-operation and Development |
| PLS     | Prize-linked savings                                   |
| PoPI    | Protection of Personal Information [Act]               |
| RESET   | Regression equation specification error test           |
| SARB    | South African Reserve Bank                             |

|          |                                     |
|----------|-------------------------------------|
| SDG      | Sustainable Development Goal        |
| SMME     | Small, medium, and micro enterprise |
| Stats SA | Statistics South Africa             |
| TUT      | Tshwane University of Technology    |
| UN       | United Nations                      |
| VBS      | Venda Building Society              |

## **CHAPTER 1**

### **INTRODUCTION AND CONCEPTUAL FRAMEWORK**

#### **1.1 GENERAL INTRODUCTION**

Household savings are a paramount domestic source of funds to finance capital investment, which is a main driver of long-term economic growth. Household savings rates differ considerably between countries because of institutional, demographic, and socio-economic differences (Organisation for Economic Co-operation and Development [OECD], 2023:1). For example, government provisions for old age pensions and the demographic age structure of the population will affect the rate at which populations save (older persons tend to run down their financial assets during their retirement, to the detriment of their savings). Equally, the availability and cost of credit, as well as attitudes towards debt, may impact the decisions made by individuals regarding whether to spend or save. Household savings are estimated by subtracting household consumption expenditure from household disposable income plus an adjustment for the change in pension entitlements (OECD, 2017:1). Disposable income refers to the net income after all compulsory deductions (Ralarala & Masipa, 2021:28). Financial wellbeing is a state of being financially sound where households can fully meet current and ongoing financial obligations, can feel secure in their financial future, and are able to make choices that allow them to have leisure (Cole, Iverson & Tufano, 2014). The link between household savings and financial wellbeing is that financially healthy households are able to eradicate household financial concerns and also enable households, through the reduction of the inequality gap, to alleviate poverty and reduce unemployment, and it aids households to invest.

Unemployment, poverty, and inequality are the main challenges that plague the South African economy (World Bank, 2018:42). Policymakers have introduced policies such as the Growth, Employment and Redistribution policy in 1996, the National Development Plan (NDP) Vision 2030 in 2012 (National Planning Commission [NPC], 2012), and South Africa's Economic Reconstruction and Recovery Plan in 2020 to address these challenges; however, without significant success (International Monetary Fund [IMF], 2023:6). In addition, according to the World Bank (2022:60), South Africa has a high level of inequality, with a Gini coefficient for income distribution of 0.7. Wealth is distributed even more unequally, with only 1% of the population owning half of the wealth in the country, while the top 10% of the population owns at least 80% of the nation's wealth. The problems of high levels of unemployment, poverty, and inequality are partly attributed to the colonial disenfranchisement policies that excluded a high percentage of the general public from the mainstream economy and barred them from formal savings instruments (Zwane, Greyling & Maleka, 2016:209). Moreover, apartheid policies were detrimental to many people's livelihoods, primarily black people, which deprived them of their higher-yielding assets, especially livestock and land (Kloppers & Pienaar, 2014:692). Encouraging households to save and reviving a household saving culture can play a significant role in reversing these problems. However, a recent South African Reserve Bank (SARB, 2023:27) report stipulated that the country's national savings rate decreased by 1.9%, from 16.5% in 2021 to 14.6% in 2022. This was accompanied by the perpetual decline of the savings rate of households, which decreased from 2.3% in Quarter Three of 2022 to 2.2% in Quarter Four of 2022. As of 2017, Ntsalaze (2017:2) indicated that South African households spent 45% of their household income on debt repayments, which is beyond levels that are considered sustainable. TransUnion (2020:2-5) stated

that in the year 2020, only 12% of consumers indicated that their household finances were planned and 69% said their finances were worse than planned, while consumers' inability to pay their bills and loans was at 84%. As long as a decade ago, Cronjé and Roux (2010:22) indicated that the savings levels in South Africa were at their lowest. As a result, the country depends on foreign investment to invest in enablers for future growth (PwC, 2023:5). It is very detrimental and unhealthy for a country to depend on foreign investors, as they have a record of withdrawing at the most subtle forecast of bad news (Cronjé & Roux, 2010:22-23).

This study conducted an analysis of savings at both the household and national levels in South Africa. For both national-level analysis and the household-level analysis, the methodologies used were mixed methods which consist of both qualitative and quantitative research methods. Multivariate regression analysis, questionnaires, and focus group discussions for household-level analysis, whereas the national-level analysis followed autoregressive distributed lag (ARDL) model.

For the purpose of exploring innovative ways to encourage household savings in the study area, this study utilised prominent theoretical frameworks, namely the life cycle hypothesis, the precautionary savings theory, Ricardian equivalence, and the permanent income hypothesis.

The motivation for this study was to suggest innovative ways to encourage household savings for financial wellbeing to enable the reduction of the inequality gap, alleviating poverty, reducing unemployment, and aiding households' ability to save. To this end, the study aimed to craft innovative ways that can be used to encourage household savings for financial wellbeing at the household level in townships, using the township of Atteridgeville as the case study, and at the national level using secondary data.

The study also sought to analyse the extent to which savings can be improved, especially in townships, with the main drive of encouraging financial wellbeing in all households in the nation, as well as financial wellbeing at the macro level. In an attempt to achieve these objectives, the study collected primary questionnaire and focus group data and secondary time series data. Descriptive and analytical statistics were used to draw inferences from the primary data, data from the questionnaire were also analysed by the econometric regressions. and econometric regressions were used for the secondary data.

The study found both a lack of and inadequate household savings at both the macro (national) and micro (household) levels, which lead to a decline in household savings. The study suggests innovative ways to encourage household savings for financial wellbeing at both the macro and micro levels, namely a state-owned commercial bank; introducing a financial literacy curriculum into early childhood development (ECD); the government, through the National Treasury, to allocate incentives for household savings such as tax incentives; commercial banks to cater for households in townships by offering them services that cater specifically for townships and their traits; and advertisements on various platforms on different household savings packages. The findings can be adopted as a roadmap to fulfil the NDP's mandate on savings by 2030 and to meet the United Nations' (UN) Sustainable Development Goals (SDGs) 8 and 10.

## **1.2 PROBLEM STATEMENT**

It is well documented that South Africa has low savings rates. This has led the country to depend on foreign investment to fund elements that result in both economic growth and development (IMF, 2023:1; Statistics South Africa [Stats SA], 2020:31). As of

March 2023, the SARB (2023:22) stipulated that the national savings rate decreased by 2.9% from 15.9% in Quarter three of 2021 to 13.0% in Quarter Four of 2022. Low household savings and a declining trend in savings among South African households leave the country vulnerable to economic shocks as people lack buffer resources to protect and sustain themselves during financial uncertainties. The COVID-19 pandemic exposed this vulnerability as many lower-income households lost their income, which left them in dire financial need without savings to fall back on (OECD, 2020:33). Furthermore, the overdependence on foreign direct investment for several years means that fewer national resources are available for counteracting the fight against the three major economic challenges of unemployment, poverty, and inequality (National Treasury, 2019:52). Similarly, South Africans' economic participation is significantly inhibited by their lack of savings, which would allow households to participate in financial markets (World Bank, 2014:9). Most studies so far (Ögren, 2018; Shruthi & Murugan, 2018; Mbukanma *et al.*, 2020; Reyers, 2019) have attempted to solve the problem of low household savings and the poor saving culture. However, most studies have failed to suggest innovative ways to address low household savings. In view of the inability to devise solutions to address low household savings, there is an urgent need to craft innovative ways to foster a saving culture among South African households. These challenges are more pronounced among township households, who find themselves comparably more disadvantaged in times of financial crises than other social groups (OECD, 2020:18), which is also reflected by a perpetual decline in the national savings rate (SARB, 2023:31). This study noted both a social gap and a methodology gap in addressing the challenges of low household savings in the country.

## **1.3 RESEARCH OBJECTIVES**

The study's research objectives are presented below.

### **1.3.1 Primary objective**

The primary objective of the study was to craft innovative ways that can be used to encourage savings at the household level in townships, using Atteridgeville township as the case study, and at the national level, using secondary data.

### **1.3.2 Specific objectives**

To achieve the primary objective, the study pursued the following specific objectives:

- To determine the perceived effect demographic characteristics on household.
- To analyse the extent to which savings can be promoted among households.
- To understand the determinants of household savings and using the results to construct innovative policy recommendations to encourage household savings in townships and at the national level.
- To perceive the effect of commercial banks/financial institutions on household saving
- To identify innovative factors that promote household savings at the household (micro) level and the national (macro) level.

## **1.4 RESEARCH QUESTIONS**

The study's research questions are presented below.

#### **1.4.1 Primary research question**

The study sought to answer the following primary research question: What are innovative ways that can be used to encourage savings at the household level in townships and at the national level?

#### **1.4.2 Secondary research questions**

To assist in answering the primary research question, the following secondary research questions were posed:

- What determine the perceived effect demographic characteristics on household?
- To what extent can savings be promoted among households?
- To what extent does the effect of commercial banks/financial institution have household saving
- What are the determinants of household savings?

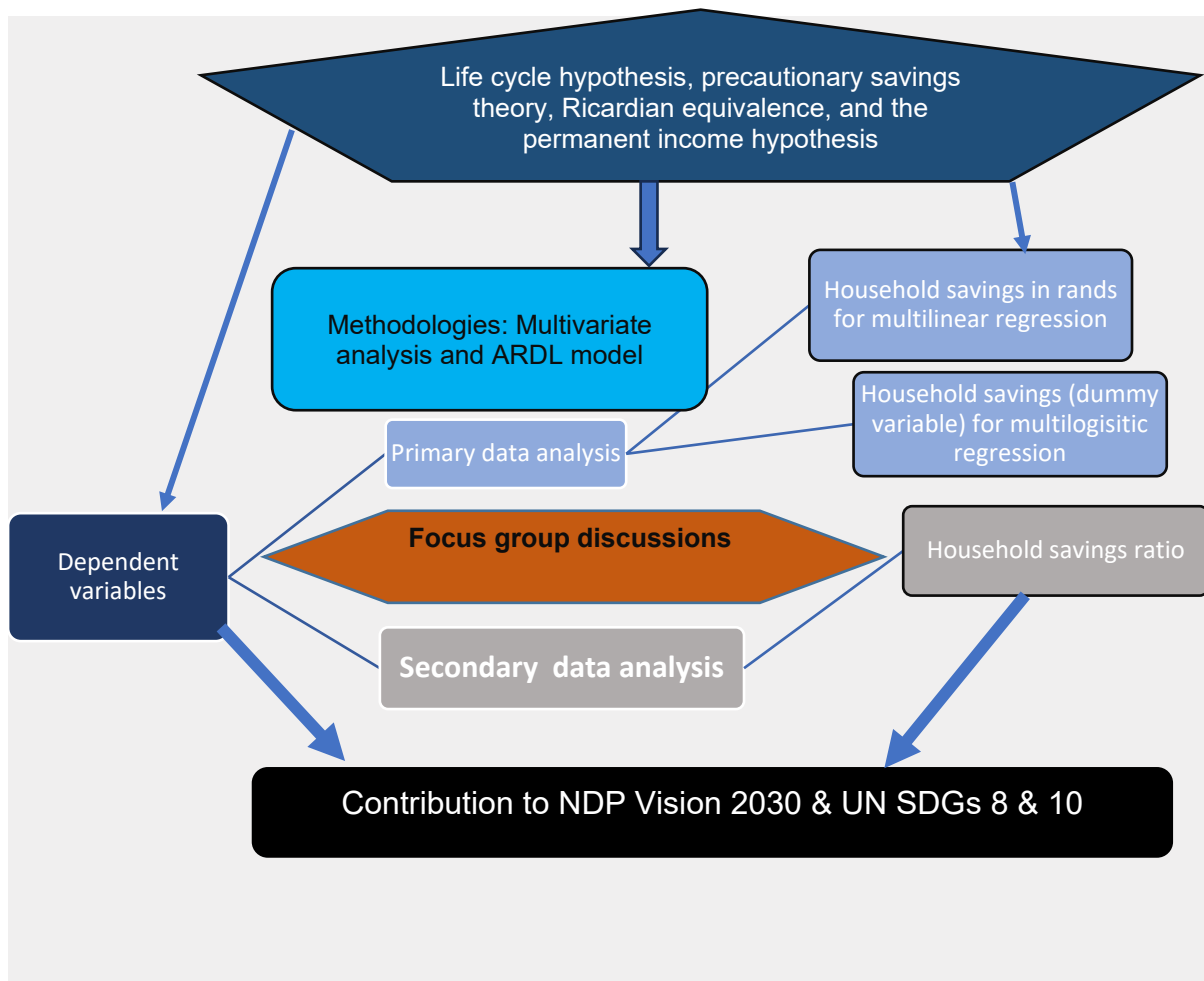
### **1.5 SIGNIFICANCE OF THE STUDY**

The significance of the study is that it aimed to provide in-depth insight on the determinants of households' savings among South African households. The study also sought to investigate and recommend innovative ways to encourage savings for household and national financial wellbeing. Furthermore, the study aimed to aid in ascertaining the relevance of policies and possible policy instruments/strategies that have and may be carried through in order to improve low household savings. The study is also important because it sought to contribute to the achievement of the NDP Vision 2030 and the UN's SDGs 8 and 10 in terms of promoting sustainable and

inclusive economic growth and productivity, employment and decent work for all, and reducing inequalities.

## **1.6 CONCEPTUAL FRAMEWORK OF THE STUDY**

The conceptual framework of this study is depicted in Figure 1.1. It shows how the variables of interest are related with prominent theories and methods of data analysis. Household savings as the dependent variable consist of various independent variables. The independent variables also assisted in developing solutions to close the gap in the literature, which supported drawing proper analyses to aid the achievement of the main aim of crafting innovating ways to encourage household savings for financial wellbeing. These independent variables would not result in the number of parameters being greater than the quantity of observations, as that could lead to the outcome of overfitting and a lack of a degree of freedom. The concepts presented in Figure 1.1 include prominent theories on savings, methodologies employed in both primary and secondary data collection, and the dependent variables utilised. Furthermore, the concepts highlight the contribution that the study makes to achieve NDP Vision 2030 and the SDGs.



**Figure 1.1: Conceptual framework of the study**

Source: Synthesised by the researcher from the literature review

## 1.7 ETHICAL CONSIDERATIONS

This study was conducted following the Tshwane University of Technology's (TUT) research ethics policy. According to this policy, the research proposal and questionnaire were submitted for approval (see Annexure 3). The TUT research ethics policy observes the South African Protection of Personal Information (PoPI) Act; the research thus observed the PoPI Act. The study followed the requirements for the lawful processing of all personal information and the release of such information to other parties. Participation in the study was on a voluntary basis as it was strictly based on the participants' consent. If a participant felt uncomfortable, they could withdraw from the study at any time. The participants were sufficiently informed of the motive of

the study and what it sought to achieve, especially their significance in participating on the surveys. This assisted by enabling the participants to make a well-informed decision regarding participating in the survey. The participants' personal information was held strictly confidential and the survey did not require the participants' identities because the processing of a person's information must be presented in aggregate and average forms only for academic purposes, according to PoPI Act (No. 4 of 2013). The collection and analysis of data were limited to Atteridgeville township; however, the study's recommendations can be adapted to other townships across South Africa.

## **1.8 ORGANISATION OF THE DISSERTATION**

The dissertation commences with Chapter 1, which entails the introduction and conceptual framework. This chapter is very important because it provides in detail the reasons why this study was conducted and what it sought to achieve. The problem statement and the significance of the study are also clearly defined in Chapter 1. Chapter 2 consists of the theoretical and empirical literature review, which examines various theoretical frameworks and previous studies. Chapter 3 provides insight into and details on the methodology that this study followed. Chapter 4 presents the data analysis and findings, and Chapter 5 contains the conclusion, policy implications, and suggestions for further research.

## **CHAPTER 2:**

### **LITERATURE REVIEW**

#### **2.1 INTRODUCTION**

Chapter 1 explored household savings in South Africa, the problem of a lack of a saving culture, and a declining trend in savings among South African households, which leaves the country vulnerable to economic shocks as people lack buffer resources to protect and sustain themselves during financial uncertainties. The COVID-19 pandemic exposed this vulnerability as many lower-income households lost their livelihoods and found themselves in dire financial need without savings to fall back on (OECD, 2020:33). This chapter examines various studies that have been conducted to derive comprehension of previous research conducted on the topic and/or applicable debates on households and savings. The theoretical and empirical framework is thoroughly investigated; the former explores the existing theories and their relationships while the latter focuses on previous research conducted with the aim of addressing the lack of household savings in the country. There is a gap in the literature as household savings are low and continually diminishing; there is thus a need to craft innovative ways to encourage household savings for financial wellbeing, especially in townships. The objective is to craft innovative ways to encourage savings for financial wellbeing at the household level and at the national level. The literature review therefore focuses on both the micro and macro levels.

#### **2.2 OVERVIEW OF THE SOUTH AFRICAN SAVING CULTURE**

South Africa has a dual economy, which means that the country is divided into two autonomous economies. The dualist interpretation is presented as a simple measure

of comprehending South Africa and the difficulties that plague it. The interpretation indicates that the dual economy structure in South Africa amalgamates features of a relatively advanced capitalist economy with many of the attributes of third-world underdevelopment. In other words, in the dualist approach, South Africa's economy is defined by two parallel economies, namely the first economy and the second economy. Based on this view, the second economy is adversative to the first economy. Mbeki (2004:29, cited by Bojabotseha, 2011:2) contended that the first economy, on the one hand, is modern, produces the bulk of the country's wealth, and is integrated with the global economy. The second economy, also known as the marginalised economy, is characterised by underdevelopment, contributes little to the gross domestic product (GDP), contains a big percentage of the country's population, incorporates the poorest of the rural and urban population, is structurally disconnected from both the first and global economy, and is incapable of self-generated growth and development (Mbeki, 2004:29, cited by Bojabotseha, 2011:2). Neoliberalism is thriving in the South African economy, which consists of a variety of private freedoms, combined with centralised economic planning and government regulation (National Treasury, 2020:1). It is confronted by various challenges that have a negative impact on the country's financial wellbeing. The country's financial wellbeing is in a dire state as the government's debt-to-GDP ratio is expected to increase to 71.4% in 2023/2024 from 48.5% in the 2017/2018 financial year. The debt-to-GDP ratio is projected to increase to 73.6% of the GDP by 2025/26 (National Treasury, 2023:28). The country's economic status commonly reflects the citizen's economic status; this is therefore a correct explanation of the low household savings identified at both the township and national levels. It is thus safe to conclude that as township households' financial wellbeing is appalling as reflected by their lack of household savings, it follows from

the country's financial wellbeing; hence the need to craft innovative ways to enable and improve the country's financial wellbeing. The study accepts that when households are financially healthy, the country's financial wellbeing will subsequently improve and it will be in good financial health.

### **2.3 REVIEW OF TOWNSHIP ECONOMIES IN SOUTH AFRICA: CHALLENGES AND PROSPECTS**

This study adopted the National Treasury's (2020:1) definition of a township, which defines a township as a dense urban settlement usually built at a distance from centres of commercial and industrial activity. In the apartheid era, this was by design with townships established as dormitory towns for black workers in mines and factories, with no internal economic logic and limited social services (National Treasury, 2020:2). In the post-apartheid period, this pattern has tended to be reproduced because large-scale projects on cheap but poorly located land offer lower unit costs, despite being far from economic opportunities in the core economy. The social, infrastructural, and economic costs of this ineffective urban planning were underestimated and became set in stone (National Treasury, 2020:1). South Africa has approximately 532 townships that house Africans, coloureds, and Indians. These townships have been recorded as having a larger demographic area than Johannesburg and Durban combined (Schwabe, 2020:1). In 2019, 21.7 million people lived in these townships. The Gauteng province, with 8.9 million people, has the largest township population, followed by the Western Cape, with 3.2 million people (Schwabe, 2020:1). Generally, townships are characterised by various common economic activities such as spaza shops, hair salons, shisanyamas, and many other business activities. The single biggest category of activity is retail, which comprises mainly street traders, spaza

shops, and shebeens; many of which operate from homes on a very small scale (National Treasury, 2020:1). Hardware and building supply stores are also dominant. In the services sector, the taxi industry thrives and is the backbone of related service activities – from tyre repair to carwashes. The rest of the service sector, however, tends to focus on personal and social services, including food preparation, hair salons, and ECD centres. Manufacturing activities are notably limited, along with the jobs and business services often associated with their presence. Those that exist are rarely linked to value chains or markets outside the township. Most jobs are informal and precarious.

Irrespective of the favourable prospects in townships, most economic activities in townships are trapped within a narrow band of informal activities. The returns are often low, and so are the employment multipliers (National Treasury, 2020:1). Townships also depict high levels of urban poverty and unemployment, particularly for the youth (National Treasury, 2020:2). The lack of jobs in townships means many people depend on economic opportunities far away. For workers, this adds hugely to the costs of commuting, which erodes the value of their wages and also encroaches on the time they have available outside of work. These factors raise the cost for township populations and reduce productivity, with economy-wide impacts. For the unemployed, the costs of work-search activities are often simply unaffordable. The other concerning challenge of the township economy is financial exclusion. The World Bank (2014:115) indicated that when it comes to businesses, small, medium, and micro enterprises (SMMEs) in townships are 12 percentage points less likely to use a bank account than SMMEs in non-township urban areas. Even if they do have a bank account, they are more likely to use a personal one than one in the business' name. While credit penetration is generally very low (less than 5% of SMMEs in either

townships or non-township urban areas report any borrowing), most credit in townships originates from family and friends, whereas in non-township urban areas it does so from banks. The Sustainable Livelihoods Foundation (2016:1) indicated that despite the need to support the township economy, informality can also function as a poverty trap and a barrier to business survival and growth. Informal enterprises are vulnerable to both crime and police harassment. Informality limits access to credit and insurance, while insecure land tenure limits access to capital required for growth. Without licences, shebeens cannot become taverns and are forced to operate in the shadows, and without formal registration, informal crèches cannot access the government subsidies available to ECD centres; hence the need to craft innovative ways to encourage household savings for financial wellbeing at both the micro and macro levels, because of the importance of households in townships. This will enable households to fulfil current and ongoing financial obligations, secure their financial future, and acquire financial freedom.

## **2.4 HOUSEHOLD SAVINGS AND ITS ROLE IN THE ECONOMY**

Savings may benefit households by enabling them to finance their children's tertiary education, build houses, and sustain them during unexpected financial downturns. The importance and urgency of household savings cannot be overemphasised as household savings are not only beneficial to households, but also contribute to the growth of the economy and the development of the country. Prinsloo (2000:2) and Lerner (1938:297) stated that household savings improve the country's investments. Household savings are also important as they enable financial wellbeing, which reduces unplanned expenses and borrowing by households (Bufe *et al.*, 2021:8) and enable households to finance unforeseen disruptions of household expenditure and

unanticipated consumption needs (OECD, 2020:9). The importance of household savings has also been amplified by the increase in socio-economic crises around the world (World Bank, 2018:10).

According to Shruthi and Murugan (2018:399), household savings mainly depend on aspects such as household income, status, and position at the societal level. However, innovative ways to encourage household savings seek to enable households to participate by having savings at any level of their socio-economic status. Households are able to save by utilising various savings instruments, from bank savings accounts, to community-based savings schemes such as stokvels and burial societies, with the future goal of enabling households, especially those in townships, to make investments. Santamicone (2013:30) mentioned other types of household savings instruments, such as emergency savings, educational savings, retirement savings, and leisure savings. Households can therefore have savings for both insurance and assurance purposes. However, these different savings instruments have proven to be ineffective, which could be because households do not utilise them effectively; hence the urgency to introduce and enable awareness of innovative ways to encourage savings for financial wellbeing at the household level, especially in townships.

The available literature on households and savings has not been able to craft innovative ways to encourage households to save for their financial wellbeing. The South African household savings rate has constantly been diminishing (Netshitenzhe & Lings, 2020:11-12), which provoked the need to craft innovative ways to encourage household savings for financial wellbeing in South Africa. Many households in the townships do not have savings, irrespective of their socio-economic status (OECD, 2020:1), although a lack of household savings and financial fragility are more severe

among low-income households with low educational attainment, families with many dependants, and the unemployed (Lusardi, Schneider & Tufano, 2011:1). Consequently, these households may not be able to finance emergency spending, nor have the ability to be financially independent, especially upon their retirement. When the working class retires, they are most likely to depend on the country's social security protection system for an income, which is mainly in the form of receiving social grants. This dependency on the social security protection system is as a result of lack of sufficient household savings (Sanlam, 2021:46). In 2021, the National Treasury issued a document to encourage South African households to save because the National Treasury (2021a:2) discovered that many households lack sufficient retirement savings. These households are also without sufficient savings to cater for their short-to medium-term financial needs. Households with retirement savings are mainly those with contractual and compulsory retirement savings. Because they are binding, they enable households to maintain their savings, and they gain access to those savings when departing from their jobs (National Treasury, 2021a:31). The 2021 South African savings for households recorded 2% of the GDP (National Treasury, 2021a:3), of which notably 2% were retirement funds. Consequently, retirement replacement values are very shallow. In South Africa, the Sanlam Benchmark Survey (2021:47-48) showed that, on average, the replacement ratio is approximately 25% to 30%; as a result, those who are retiring indicate that they are unable to live on the starting pension offered by a guaranteed annuity. Most individuals receive their pension or retirement savings after retirement; however, due to the inadequacy of the funds, it depletes quickly, which leads them to become recipients of social security grants and depend on the country's social security protection system for income.

Non-compulsory savings are always the lowest. One-third of the participants in the Old Mutual Savings and Investment survey of 2021 indicated that if they lost their income, the savings that they had acquired would not be insufficient to cover one month's expenses. The survey also found that 56% of the participants had "high and overwhelming" financial distress levels in 2021 (Old Mutual, 2021:13). It is therefore safe to conclude that the South African working class is mainly disadvantaged when working and they are also vulnerable during their retirement phase. Many households lack savings because they lack sufficient knowledge of investment and compulsory and non-compulsory savings. Both individuals and households do not have savings for future financial needs. In many households, especially in townships, the children are left without an inheritance and properties are not insured. The SARB (2020:27) indicated that for the economy to maintain the target for growth of 4% or greater annually, the government has set a target of national savings of 23% of the GDP. The national gross savings was 15.4% of the GDP in 2020, with household savings of 1.2% (SARB, 2020:17). Financial literacy and knowledge are vital for all South Africans to assist in alleviating financial distress during retirement or unexpected financial crises. Financial literacy and knowledge may also assist in increasing households' income in times of need. When people start working at an early age and are better informed financially, their life cycles depict better financial results. The COVID-19 era exposed the vulnerability of household savings, which requires urgent and intentional attention to bring about innovative ways to encourage household savings. South Africa is a developing country, and among other traits are lower household savings (National Treasury, 2021a:2). These lower savings also reflect in South Africa's domestic savings, which have declined over the past three decades (OECD, 2020:10).

## **2.5 REVIEW OF HOUSEHOLD SAVINGS IN SOUTH AFRICA**

When conducting the literature review on household savings in South Africa, the study examined households' saving behaviour and the determinants of household savings in the country. The findings of various research studies that were explored pointed out the lack of sufficient research conducted in the area of households and savings in the country; hence the need to urgently fulfil the objective of this study of crafting innovative ways to encourage household savings for financial wellbeing.

### **2.5.1 Household saving behaviour in South Africa**

This section examines various literature views to determine if South Africa being commonly known as a credit economy (Mbukanma *et al.*, 2020:14660) has an impact on the lack of household savings confronting the country. Many countries with favourable household savings have investments from foreign countries; they therefore attract foreign investment. Comes *et al.* (2018:238), Mızrak and Dastan (2018:235), and Steinert *et al.* (2018:240) indicated that in a credit economy, the availability of financial products through credit promotes the lack of household savings. This is also seen in households' inability to manage their finances, as they depend on credit liabilities available to them. This has resulted in a huge hindrance to households' capacity and ability to make prudent financial decisions, especially among financially illiterate households. The credit economy has positive benefits that enable households to access credit and financial products and services, which allow households the opportunity to have an instant financial instrument or loan, usually through bank finance. The credit economy has produced an economic environment that puts financially illiterate households in perpetual debt (Klapper, Lusardi & Van Oudheusden, 2015:18; Lusardi & Mitchell, 2014:157). Households possibly do not

perceive the importance and the need of having savings in a credit economy as there is a perpetual decline in South Africa's household savings. In the African context, Mensahklo, Kornu and Dom (2017:9) examined the saving behaviour of households in Ghana and drew similar conclusions as Ögren (2018:24) that few households with savings were encouraged by high interest rates as a reward for savings. Santamicone (2013:1) and Frisancho (2016:29) found statistical evidence that educational level and employment status affect household saving behaviour.

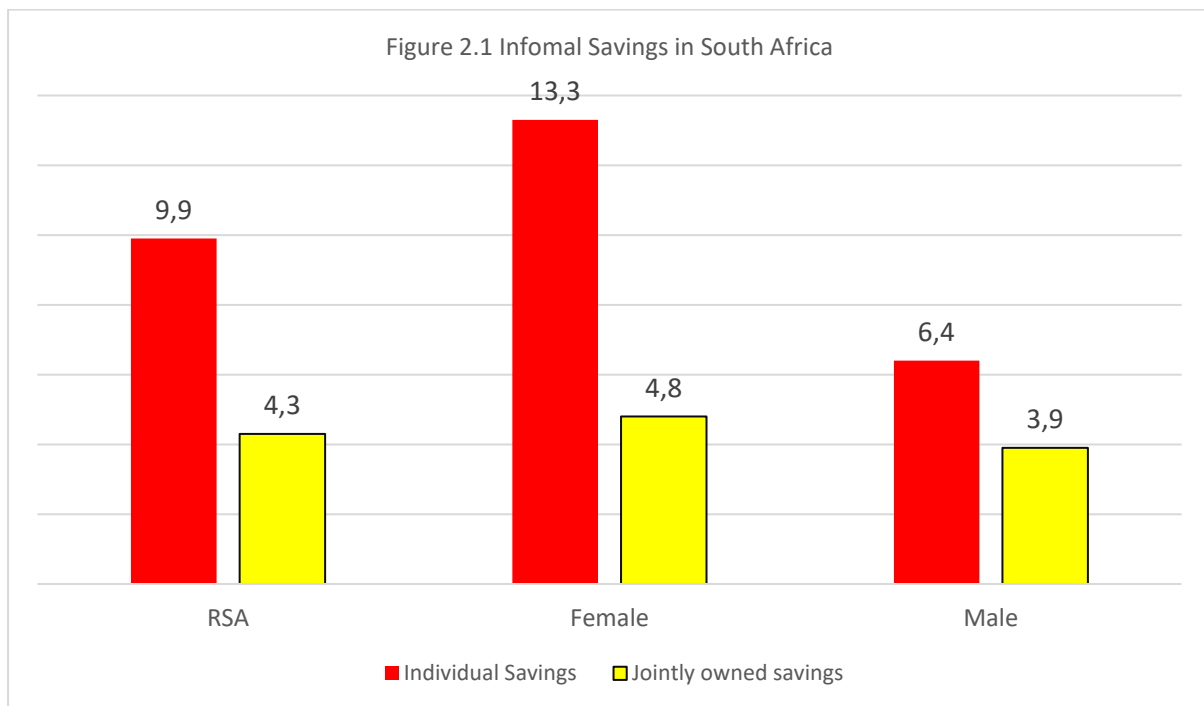
Table 2.1 indicates the perpetual decline in South African saving behaviour. The SARB (2023:22) stipulated that the country's national savings rate decreased by 1.9%, from 16.5% in 2021 to 14.6% in 2022. This was accompanied by the perpetual decline of the savings rate of households, which diminished from 2.3% in Quarter Three of 2022 to 2.2% in Quarter Four of 2022.

**Table 2.1: South African formal saving behaviour**

| Savings                         | 2021        |             |             |             |             | 2022        |             |             |             |             |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                                 | Q1          | Q2          | Q3          | Q4          | Year        | Q1          | Q2          | Q3          | Q4          | Year        |
| Corporate                       | 18.2        | 13.9        | 15.9        | 15.2        | 15.8        | 17.0        | 10.3        | 16.3        | 13.0        | 14.1        |
| General government              | -4.3        | 0.4         | -1.7        | -2.6        | -2.1        | -3.1        | 0.7         | -2.7        | -2.3        | -1.9        |
| <b>Household savings</b>        | 2.8         | 2.6         | 2.9         | 2.7         | 2.8         | 2.5         | 2.3         | 2.9         | 2.2         | 2.4         |
| <b>Total (national savings)</b> | <b>16.7</b> | <b>16.8</b> | <b>17.1</b> | <b>15.3</b> | <b>16.5</b> | <b>16.4</b> | <b>13.3</b> | <b>15.9</b> | <b>13.0</b> | <b>14.6</b> |

Source: SARB (2023)

Figure 2.1 indicates that the overall informal savings in South Africa are 9.9%, while jointly owned savings are only 4.3%. Household savings will enable the reduction of the inequality gap, alleviate poverty, reduce unemployment, and aid households' ability to invest.



**Figure 2.1: South African informal saving behaviour**

Source: Stats SA (2020)

## 2.5.2 Determinants of household savings in South Africa

Although literature on the determinants of household savings in South Africa is currently rare, Ralarala and Masipa (2021:28) found that the most important determinant of households saving is income, and Zwane *et al.* (2016:215) conducted a study on the determinants of South African household savings, which reiterated income as the main determinant of household savings. These findings indicate that South African household savings are vigorously income driven; however, the findings also revealed other factors to consider that contribute to household savings, such as age structure, education achievement, and employment status. Yet the causal nexus

between savings and household size was found to be negative, which is a sign that larger families compromise household savings prospects. Simleit, Keeton and Botha (2011:10) examined the determinants of household savings in South Africa and their findings suggested that interest rates, a wealth effect, and upturns in the business cycle all contribute to explain the decline in household savings. The presence of a partial offset between household savings and government savings also has important implications for the effectiveness of using the fiscal position of the South African government to boost savings. Based on the examined literature, it can be concluded that income is a significant determinant of household savings.

## **2.6 FINANCIAL LITERACY AND HOUSEHOLD SAVINGS**

Existing literature was explored with the focus on the relationship between household savings and financial literacy (Frisancho, 2016:12). The findings emphasised the significance of encouraging household savings, especially among households with a lower income range as it contributes to urgent financial relief. Current evidence of financially illiterate households is demoralising. Reyers (2019:336) emphasised that financial literacy can result in household savings, as it enables households to comprehend and to be in charge of their finances. Although it is vital to understand the relationship between household savings and being financially informed, it is equally important for both individuals and households to be eager and intentional to become financially literate to be able to comprehend concepts of financial management. These efforts will enable individuals and households to improve their future financial wellbeing. According to Santamicone (2013:8), researchers in both developed and developing countries are working towards finding significant saving strategies and methods that will assist individuals and households to acquire savings with the

purpose of attaining financial liberty. The literature indicates that there is now a transformation in South African households because many young people have entered the labour market, which leads to a diminishing high dependency ratio (Cronjé & Roux, 2010:23). The high dependency ratio negatively affects household savings, which means that South African households' ability to save has slightly increased due to the diminishing dependency ratio (World Bank, 2017:45). However, South African households are still characterised by a debt culture rather than a saving culture as indicated by DebtSafe (2018). The South African economy is consumption led (National Treasury, 2021b:5) and South Africa is defined as having a marginal propensity to consume rather than a marginal propensity to save; hence the importance of innovative ways to encourage household savings, especially in townships. In an effort to encourage household savings, Cole *et al.* (2014) suggested the inception of prize-linked savings (PLS) products in South Africa, as a way to initiate savings by low-income households. PLS products allow those who save to yield more returns than the normal returns or rewards on savings, such as interest, capital yields, or income received from dividends. PLS products also allow receiving huge prizes. The possibility of winning is increased by account balances. In countries where PLS products have been introduced, savings are said to have improved, especially among individuals and households that commonly had no savings. Financial literacy is very significant and can be an important tool to improving household savings. Sibanda and Sibanda (2016:10) indicated that financial literacy skills are limited, especially in South Africa. Their study discovered that in South Africa and other nations with emerging economies, financial literacy is not common for the majority of the population. However, South Africa is distinguished by having well-developed, formal financial institutions and, according to the Global Findex Report (World Bank, 2015), the South

African population has the highest ranks of indebtedness because during 2013 and 2014, 86% of the South African population borrowed money from various financial institutions. Consequently, South Africa is recorded as having many citizens who have debt relative to the citizens in other nations. According to the National Credit Regulator's 2014 annual report, South Africa has a population that is very poor in terms of loan repayments. It indicated that 44.2% of their active consumers, which is equivalent to 9.6 million, have very poor credit records.

## **2.7 SOCIO-ECONOMIC STATUS AND HOUSEHOLD SAVINGS**

This section discusses household income and savings; the link between household savings and wealth; household assets, liabilities, and savings; education and savings; and employment status and savings.

### **2.7.1 Household income and savings**

Literature on household income and its relationship and relevance to household savings was extensively reviewed, which established that different studies (Mbuthia, 2011; Ralarala & Masipa, 2021; Chitema & Chitongo, 2020) have discovered that the most common factor that controls household savings is the income that households are left with after compulsory deductions. Ralarala and Masipa (2021:28) explained that it is not only the disposable income of a household that influences its savings but also the household's decision to postpone consumption. Mbuthia (2011) and Ralarala and Masipa (2021) stated that household savings stem from income that could have been consumed but was deferred for the future. Deferring income for future consumption is very vital as it contributes very much not only to households having savings but also to the development and growth of the country. Chitema and Chitongo

(2020), Lydon and McIndoe-Calder (2021), and Aidoo-Mensah (2018) indicated that low savings are detrimental to the nation because a nation's investments are very dependent on the nation's savings. They indicated that low savings result in low investment. Household savings and household income have a positive relationship with the economy; however, Azzolini *et al.* (2020:16) held a different view and explained that although it seems like households with high income save more and households with lower income save less, households with low income must not be excluded from savings. Azzolini *et al.* (2020:16) emphasised the importance of creating an equitable saving platform for low-income households with benefits and/or incentives. This is relevant to this study because saving designs for different households with different socio-economic statuses can be employed as an innovative way to encourage household savings.

### **2.7.2 The link between household savings and wealth**

Literature on household savings and wealth was examined with the purpose to determine the relationship between them. Joubert and Van der Merwe (2021) conducted research on this topic in South Africa. Their study analysed household savings, using both an income statement (savings) and a balance sheet (savings or wealth) approach, while also exploring the link between the two. Their findings revealed a long-term decline in net household savings to GDP, which turned negative in 2006, so that no saving occurred. In contrast, household wealth recorded an average nominal increase of 10.3% in the period 1995 to 2018. Focusing on the link between the two concepts, it is confirmed that net revaluation of assets plays a prominent role in supporting the rise in wealth, while the role of saving diminishes over time. Lupton *et al.* (2004) also investigated the link between household savings and

wealth, which suggested that the decline in the personal savings rate since 1984 is largely due to the significant capital gains in corporate equities experienced over this period. Over five-year periods, the effect of capital gains in corporate equities on saving is substantially larger than the effect of capital gains in housing or other assets. Failure to differentiate wealth effects across asset types results in a significant understatement or overstatement of the size of their impact, depending on the asset. Both the studies conducted by Joubert and Van der Merwe (2021) and Lupton *et al.* (2004) recommended the need for additional methodological analyses and more research on the various possible factors that drive household income and expenditure trends, as well as their sustainability. The contribution of household savings to total savings should be evaluated in detail.

### **2.7.3 Household assets, liabilities, and savings**

Various studies have extensively reviewed the impact that household assets and liabilities have on household savings. The OECD's (2022) report on the national accounts of OECD countries indicated that the household savings account is positively impacted by the financial assets that households acquire, such as savings deposits, investments in equity, shares, and bonds. These form an important part of the overall wealth of households, and are an important source of revenue, either through the sale of these assets, or as a source of property income, such as interest and dividends. However, the OECD's (2022) report on household liabilities reflected limited data as households' liabilities mainly focus on outstanding household liabilities, including non-profit institutions serving households. The OECD's (2022) indicator only took into account the value of dwellings, and no other types of non-financial assets. However, it still reflected a negative relationship between household liabilities and household

savings. Based on the OECD's (2022) report, it can be concluded that households with assets save more than households with liabilities. Joubert and Van der Merwe (2021:8-10) indicated that households with wealth are in a better position to acquire household savings than households that are in debt. Prinsloo (2002:63) defined household liabilities as obligations or liabilities arising from borrowing money or taking goods or services on credit, which is an obligation to pay later, and concluded by highlighting the negative relationship between household liabilities and household savings.

#### **2.7.4 Education and savings**

Literature on education and households was examined with the main aim of establishing if the level of academic education is relevant to household savings to enable this study to comprehend the impact of education on the financial and saving behaviour of households. Aydermir (2021) conducted research on the impact of education on savings and financial behaviour and found that there was really no evidence of the effects of schooling on men in relation to behaviour pertaining to finances and saving. However, the study revealed that the majority of women were significantly impacted by education regarding their savings and financial behaviour. The higher women's level of education, the more they greatly increased their savings. According to Ding, Lugauer and Bollinger (2019:32), based on their research in China on whether the expansion of the level of higher education has an impact on household savings, higher education remarkably increased household savings. They concluded their study by utilising a simple two-period model of household savings for college to motivate their work. They also utilised China Household Income Project data to estimate the impact of a change in the probability of college on the affected

households' savings rates before and after reform. The study findings indicated that 10-point rises in the probability of going to college increased the savings rate for a household by more than seven percentage points.

#### **2.7.5 Employment status and savings**

Kibet *et al.* (2017) used a macro-economic approach as the methodology to identify the determinants of household savings. They found that employment status and employment industries are the main determinants of household savings. When households have individuals who are employed, they automatically have the capacity and the ability to save; however, unemployed households do not have the ability to accrue household savings. Innovative ways to encourage household savings are a tool that seeks to allow unemployed households to save because savings are autonomous income that these households can depend and survive on. Innovative ways to encourage household savings also aim to introduce ways that will enable all households, irrespective of their employment status, to save.

### **2.8 HOUSEHOLDS' SAVING CULTURE**

Firth and Madinginye (2015:29) indicated that although South Africa is known to have the poorest savings rate in the world, the poor saving culture is not the only challenge and it is not limited to the household level. Rather, it is symptomatic of a deeper issue that is confronted by the country. Firth and Madinginye's (2015:29) research on national savings aimed to determine what could be done to improve the poor saving culture in the country. Their findings emphasised the significance of establishing a saving culture, in form of individuals and households participating in savings schemes such as stokvels and tax-free savings accounts. According to an article published in

the *Mail & Guardian* (2020), society schemes, also known as stokvels, are a savings vehicle that is accepted and desired by various households of various socio-economic statuses. Young people who are entering the labour market are attracted by and flocking to stokvels as a savings mechanism for their various benefits (*Mail & Guardian*, 2020:2). Although the stokvel instrument is evolving, its purpose has remained the same, which is to ensure that the participants save fixed amounts frequently and obtain a lump sum at the end of the year or at an agreed-upon appointed time. Many young and ambitious people are now utilising stokvels as savings vehicles for various objectives such as property investments and other class assets that would have been difficult to achieve had a person not participated in a stokvel. However, some township households use stokvel savings for the purposes of purchasing and sharing bulk food and for short-term financial relief. Various financial institutions have indicated that one of the common tools to instil and improve a household saving culture in South Africa is through tax-free savings accounts. This savings vehicle or instrument allows individuals and households to save a maximum of R36 000 per year (or a lifetime savings of R500 000) without being taxed. This is also a way to encourage savings among households; however, Cole *et al.* (2014) emphasised that South Africa has a poor saving culture even when tax-free savings instruments are available. This illustrates that there have not been significant and tangible improvements in the poor saving culture; hence the need to craft innovative ways to encourage household savings, especially at the household level. Without a saving culture, the country faces huge difficulties. Historically, South Africa was able to produce a savings rate of between 25% and 30%, which equates to 25% to 30% investment spending, which would have resulted in booming economic growth (SARB, 2021:1). However, over the past two decades, the South African savings rate has

hugely declined and is currently at 15.2%. Consequently, South Africa is ranked as the country with the lowest savings rate in the world. The investment rate is also at 13.0%, which is insufficient to invest and fund economic growth of 2% (SARB, 2021:9). With these alarming figures, it this study aimed to determine solutions that will assist in improving the poor saving culture and move the country towards sustained positive household savings rates. Various elements challenge the economic growth and savings rates in South Africa, such as the triple challenges of high unemployment, extreme poverty, and inequalities. Cronjé and Roux (2010) conducted a study on the saving culture of black middle-class South Africans. The study revealed that South Africa does not have a saving culture, and it could rather be described as a debt culture. However, other developing countries, such as China and India, do have a saving culture, and those countries' households and middle class are the drivers of the upward trend of their savings rate. Cronjé and Roux (2010:25) reiterated that the increasing household savings in India and China are not as a result of policies introduced to encourage savings but as a result of other elements, such as a lack of adequate financial sector development and a feeble social safety net. This has encouraged many of their households to save to make sufficient provision for forecasted financial requirements. Momanyi and Wainaina's (2016) study established that Kenya has a progressive saving culture and their savings are perpetually rising. Their study also revealed that the determinants and inspiration of Kenya's saving culture are factors such as savings for emergencies, to start a business, to make a down payment for durable goods, to finance children's future needs, to finance forecast future financial needs, and to achieve financial dependence. The least significant reason for saving was savings with the purpose of a leaving an inheritance to children.

## **2.9 DETERMINANTS OF HOUSEHOLD SAVINGS: INTERNATIONAL PERSPECTIVE**

The purpose of examining literature on the determinants of household savings, especially in developing and developed countries, is because currently in South Africa there is not sufficient, tangible, and relevant literature on household savings, especially during the post-apartheid era (Ralarala & Masipa, 2021:28). The knowledge of the determinants of household savings in both developed and developing countries assist in obtaining a real comprehension of these factors, which assisted this study to craft innovative ways to encourage household savings. Many studies examined in this chapter indicated that the most common factor that controls household savings is the income that households are left with after compulsory deductions such as taxes (Mbuthia, 2011; Ralarala & Masipa, 2021; Chitema & Chitongo, 2020). This part of the literature review sought to determine if the above statement applies to an international perspective by examining both developed and developing countries. Developing countries are characterised by high levels of unemployment, poverty, and inequality (OECD, 2020:19). These are countries with dire standards of living, which are reflected by various traits such poverty, poor health conditions, lack of proper infrastructure, lack of proper technological innovations, and a low GDP; among others. GDP is the total value of goods and services produced within the boundaries of a country over a period of time (OECD, 2020:23). Developing countries have low income per capita and mainly depend on foreign aid to enable economic growth. Developing countries are also known as emerging economies. Developed countries are countries that are more advanced than developing countries. These are countries with proper infrastructure and that have advanced in technology. They are also characterised by a matured economy and high standards of living (OECD, 2020:75).

### 2.9.1 Studies in developing countries

Ralarala and Masipa (2021) found that disposable income and consumption are the most vital contributors to determining household savings in South Africa. Their research utilised the ARDL methodology to analyse determinants of household savings. Their study concluded by advocating innovative ways that can be utilised to encourage household savings, as household savings significantly contribute to the economic vitality of the country. Simleit *et al.* (2011) conducted a similar study on the determinants of household savings in South Africa and their findings differed from Ralarala and Masipa's (2021) findings. Simleit *et al.* (2011) indicated that the determinants of household savings in South Africa were factors such as interest rates, the wealth effect, and upturns in the business cycle. They also illustrated that these factors also contributed to the decline of household savings. Their research was conducted using the vector error correlation model. Drawing observations from other developing countries that have similar traits as South Africa, using the case of Kosovo, Ribaj and Mexhuani (2021), in terms of savings and economic growth in developing countries, concurred with Ralarala and Masipa (2021) by pointing out the significance of disposable income as a factor that contributes to household savings. Their study also emphasised the significance of household savings by indicating that savings boost economic growth, vitalise investments, and enable production, which also limit countries' dependency on foreign aid. Kibet *et al.* (2017) conducted research on the determinants of savings in Kenya. Their study indicated that many studies conducted on the determinants of savings in developing countries have not been conclusive on the factors that determine savings. Kibet *et al.* (2017:137) therefore utilised the micro-economic approach to identify determinants of household savings, which was done by focusing on specific groupings such as teachers, entrepreneurs, and farmers.

Their study indicated that the determinants of household savings are based on the type of occupation, household income, age and gender of the household head, level of education, dependency ratio, service charge, transport costs, and credit access. Al-Jormard and Almula-Dhanoon (2019:4), in their study on determinants of personal savings in developing countries, studied the different types of incomes people can get a hold of and found that current income and transitory incomes were the vital types of income that determined savings, unlike permanent incomes. Saqid *et al.* (2016:60) also conducted a study on the determinants of household savings and concluded that the income that households receive is the main factor that determines savings; however, they also indicated the significance of employment and age, which were the most positive factors that contribute to household savings (Saqid *et al.*, 2016:54). Nghiem Le *et al.* (2022:42) conducted a study on determinants of household savings in the developing country of Vietnam. Their findings were similar to the studies mentioned above, and they concluded that factors that determine household savings were household income, householder age, number of dependants in the household, and the area in which the household is located. Nghiem Le *et al.*'s (2022:49) results were drawn from Tobit regression analysis.

### **2.9.2 Studies in developed countries**

Developed countries are countries that are advanced in technology and infrastructure and have a mature economy (World Bank, 2014:16). The standard of living of those who reside in developed countries is very high relative to those in emerging economies. The literature on households and savings in developed countries is limited. The overall net savings in developed countries are high, which are relative to their investment spending (OECD, 2019:5). Developed countries focus on assisting

emerging economies with foreign aid. Foreign aid was introduced with the purpose of allowing rich countries to assist poorer countries with funds so that they can improve their socio-economic status (Briggs, 2021:2). Such financial assistance can be used as a stimulus package to enable the recipient country to boost its economic growth, invest in infrastructure development, create employment, alleviate poverty, and cater for people's most basic needs. The main purpose of foreign aid is applaudable because it aims to improve economies in countries that are characterised by high levels of poverty, a high unemployment rate and inequality, and high disease and mortality rates. Foreign aid attempts to close the gap by assisting poor economies to return to their former glory or to improve by surpassing their former status, as explained by Abdullahi and Amzat (2011:108). However, it is said that foreign aid often has conditions that are filled with caveats. Some issuers of foreign aid give grants to developing countries with hidden motives to fulfil their own economic interests. The richer countries sometimes attempt to influence the policy positions of the countries to which they provide financial aid. They impose conditions that enable those countries to be fully dependent on foreign aid. On the other hand, the leaders of countries that receive foreign aid tend to forget the purpose of the foreign aid funds, which is to improve their countries' wellbeing. Governments that receive foreign aid often mismanage the money through corruption and unfruitful and wasteful expenditure. Countries that receive foreign aid often continue to be characterised by massive poverty as the foreign aid investment does not make a difference to improving the standards of living of the majority. The IMF and the World Bank are the most commonly known institutions that focus on foreign aid. They are also responsible for channelling foreign aid funds to countries in need. In March 2020, when the global economies were faced with the COVID-19 outbreak, South Africa implemented a strict lockdown,

which negatively affected the economy. In order to keep the country's economy from collapsing, the South African government applied for foreign aid funds of R500 billion from different international financial institutions, including the IMF and the African Development Bank, with the intention that the funds were to be utilised to assist the South African economy to recover from the negative impact of the lockdown, to boost economic performance, and to lift the economy out of recession. China is said to provide significant economic infrastructure and support for social development, especially to developing African countries, whereby, in return, China becomes the privileged buyer of African raw materials for China's growing economy. This is China's condition for providing foreign aid, as indicated by Briggs (2021:9). Foreign aid can be in the form of bilateral or multilateral agreements, which means that one or more parties can provide foreign aid to any society.

## **2.10 DOES BANKING PROMOTE HOUSEHOLD SAVINGS?**

In order to determine if banking promotes household savings, different kinds of banking, such as community savings, mutual banks, private commercial banks, and financial technology (FinTech) innovations, were examined. In the following subsections, each type of banking is explained and their characteristics are outlined. The strengths and weaknesses of each banking type are explored in relation to households and savings.

### **2.10.1 Experiences with community savings and informal savings schemes in townships**

There is no formal definition of community savings; this study therefore refers to community savings as both informal and formal ways of banking that are initiated to

cater to the financial needs of a certain community, society, or environment. Informal community savings refer to informal ways of savings and/or credit groups that were mainly initiated by women of different age groups in communities. However, these initiatives are currently popular and are also adopted by men (Coetzee & Cross, 2002:12). These informal community savings and credit groups work on a cyclical or rotational basis. Members frequently contribute a fixed amount, which contributes to a lump sum that is paid out at the end of the year or on an agreed-upon appointed time. Informal community savings are also known as community savings and lending groups. These groups allow the group members to borrow funds from the fixed amount that they frequently contribute and to return the funds with interest at an agreed time. This method of lending funds to group members is one of the ways they utilise to grow their funds, which they equally share at the end of the year with interest. Obtaining the lump sum with interest at the end of the year is one of the tools to encourage members to participate in these groups. However, this method varies based on the group's specifications. In some groups, all members contribute a specified amount to make up the lump sum, which is paid out to each member in turn until all the members have received their lump sum, which could be weekly, fortnightly, or monthly, and then starts revolving again. The latter is popular in the townships of South Africa and are known as stokvels. According to Matuku and Kaseke (2014:509), stokvels can be defined as a self-help strategy that aims to tackle challenges such as income inequality and poverty in various communities.

According to the Federal Deposit Insurance Corporation's (FDIC, 2020:122) report on community banking, formal community banking is common in the United States of America. The purpose of community savings is to enable simplified access to finances by communities while catering for the specialised needs of such communities.

Some community savings have a special relationship with their customers; they know where they live, how they intend to repay the funds, and what processes to follow in the event that a community member might be unable to repay the loan. The FDIC (2020:9) illustrated the strength of community savings as a key method to support community/local small businesses and consequently growing the economy of where the community bank is situated. The main strength or advantage of community savings is that they have lower interest; the cost of borrowing is therefore very low relative to commercial banks, which is beneficial to communities. Community savings can also contribute to the development of the communities in which they are situated, which is something that commercial banks do not normally participate in. However, community savings also have disadvantages (Coetzee & Cross, 2002:11). These disadvantages contribute to some members of the communities not participating in initiatives that are meant to improve their lives. Many community savings are situated in developing communities such as rural areas and urban-rural townships. Informal community ways of banking are widespread in these communities, which are characterised by high levels of unemployment, poverty, and inequality. According to DeYoung and Duffy (2003:14), default risk is one of the common risks associated with community savings. The likelihood of the risk of non-repayment of loans is also associated with weak social capital and high mobility as people tend to move into these communities for improved socio-economic status. These factors disadvantage community savings and hinder them from prospering as a significant component of the financial landscape in these communities. Community savings are supposed to provide a path to accessing loans and enable means for savings to community members as they are more formalised than the savings and lending groups that are commonly known as stokvels but less formalised than commercial banks. Savings and lending groups also have

weaknesses that discourage community members from participating, such as some members of these groups tending to pull out when they have received their lump sum and refusing and/or being unable to repay other members voluntarily due to non-compliance or involuntary because of a loss of income. Such practices could lead to these groups collapsing or experiencing break-aways. Community groups are mainly based on trust, where the community group chooses the most prominent person in the community, or the one with the best socio-economic status, to be their leader. This is a mechanism to avoid loan defaulting or failure to repay a loan; however, some leaders defraud their community groups. Innovative ways to encourage household savings are urgently needed and must be relevant to cater for the issues that challenge households in communities, especially their efforts to save. In South Africa, especially during the Christmas season, many journalists go around the country to cover stories of communities and members of stokvels who are victims of savings and lending community groups or members of community savings who were not repaid their monies that were deposited into those community savings on time.

### **2.10.2 Experiences with mutual banks**

In South Africa, mutual banks were inaugurated in 1993 as a banking group with less rigid capital requirements but similar risk management requirements relative to existing commercial banks (Verhoef, 2012:568). However, very few institutions have opted to become mutual banks, and the capital and reporting requirements are seen as rigid. Mutual banks render the same services and issue the same products as commercial banks, such as different consumer loans, be it a personal loan, mortgage bond, vehicle financing, educational loans, or others as per customer needs. Mutual banks also offer savings and transactional accounts and issue credit cards.

However, they are usually community focused. Mutual banks are formal banks that are owned by their members (customers). The profits that mutual banks make are utilised to benefit the members/customers. These profits can be traced to the lower lending interest that the members enjoy, as well as lower banking fees and charges (National Treasury, 2021a:5). Each and every member of the mutual bank is deemed to have an equal share irrespective of the type of account they have and/or the value of the funds in their accounts. The profits of mutual banks should be reinvested in the bank for the benefit of its members/customers. The purpose of mutual banks is to improve communities and their members as the profits made by the banks are supposed to be reinvested to benefit the members. Mutual banks should also allow increased access to banking services, in both the short and long term, to enable the financial stability of community members. Due to the low savings by the communities and in the country as a whole in the midst of the mutual banks in the country, there is still a gap that requires innovative ways to encourage household savings in the country (Verhoef, 2012:570). A brief case study of one of South Africa's mutual banks called Venda Building Society (VBS) is presented next. The purpose of this case study is to highlight the difficulties that communities are confronted with in terms of mutual banks. VBS was founded in 1982 and became a formal mutual bank in 1992. During the year 2016, it was recorded to have approximately 30 000 depositors with R800 million in deposits. VBS intended to be listed on the Johannesburg Stock Exchange in 2016; however, this plan did not come to fruition as it was declared bankrupt and insolvent and placed under curatorship, with the customers/members defrauded of approximately R2 billion. The 2018 report released by the SARB found evidence of wide-scale looting, fraud, and corruption. These are common experiences with various mutual banks across the country.

### **2.10.3 Experiences with private commercial banks**

Nawabzada (2017:3) defined a private commercial bank as a profit-seeking financial institution that accepts deposits, offers checking account services, provides various loans, and offers basic financial products such as certificates of deposits and savings accounts to individuals and small businesses. A private commercial bank is where most people do their banking. Commercial banks make money by providing loans and earning interest from mortgages, vehicle loans, business loans, and personal loans (Nawabzada, 2017:4). Customer deposits provide banks with the capital to make these loans. As a developing country, South Africa has been said to have a sophisticated, proper, effective, and large financial sector relative to other developing countries' financial sectors (National Treasury, 2021a:7). According to Roberts *et al.* (2018), the country has a large asset-to-GDP ratio that is well above that of most emerging markets. The bulk of these markets are recorded in the banking sector. South Africa has made significant progress in expanding access to financial services as a significant number of individuals own bank accounts; however, active usage remains low. Despite the increase in new entrants to the banking sector, the market remains highly concentrated, with the largest banks holding over 85% of the industry's deposits in 2020. South Africa's credit market is highly developed and formally provides credit to almost half of the South African population; however, the informal credit market remains endemic. Unregistered lenders have long been operating in South Africa and are considered to be socially embedded in lower-income communities. Commercial banks do not promote savings among poor people because of the low interest on savings products that they can afford. They offer no initiatives to encourage individuals and households to save but instead promote indebtedness to people. Roberts *et al.* (2018) indicated that overindebtedness remains an issue, with

more than 50% of South Africa's credit-active consumers being regarded as overindebted. Since 2009, the value of consumer credit has increased steadily as more individuals acquire credit. The number of individuals with an active credit account saw a peak in 2020 as heightened financial distress due to the COVID-19 pandemic drove an increasing number of consumers to acquire credit. The payment sector in South Africa is well developed and compares favourably with more developed economies.

#### **2.10.4 Experiences with state-owned commercial banks**

South Africa currently does not have a state-owned bank. However, the government is finalising plans to transition Postbank into a full-service bank. The responsible minister has announced plans to introduce the South African Postbank Amendment Bill to the National Assembly. The Bill proposes amending the shareholding arrangements of Postbank by transferring shareholding from the South African Post Office to the government through the creation of a new bank. The state-owned Postbank already accepts deposits and it is expected that the new state-owned bank will offer full-service features and focus on previously disadvantaged South Africans. In 2021, during the presentation to parliament, the department responsible indicated that the new state-owned bank is expected to provide for the inclusion of financially excluded members of the public to participate meaningfully in economic activities. It is also expected to provide a facility for some of the government's financial services. The department indicated that it is lobbying for a state-owned bank but not a state bank. While the new Postbank will be state-owned, the process is separate from the African National Congress (ANC) government's recent push to introduce a new state bank. The 2020 State of the Nation Address mentioned that the government was moving

forward with the establishment of a state bank as part of its effort to extend access to financial services to all South Africans and to help support the emergence of black agriculturists and black industrialists. This study identified the introduction of a state-owned bank or state bank as one of the innovative ways to encourage household savings in the country.

## **2.11 THE ROLE OF FINANCIAL TECHNOLOGY (FINTECH) INNOVATIONS IN PROMOTING SAVINGS**

FinTech innovations are utilised as a tool to transform the financial services sector across the globe. They have the potential to improve efficiency and to reduce operating costs for providers; allow seamless, real-time customer transactions; and drive hyper-personalisation by improving the understanding of customers' needs and behaviour (Becker, 2017:39). This is set to present significant benefits not only in improving the efficiency of the financial sector, but also in enhancing the financial health of South Africans. These include those previously excluded from financial segments by meeting their needs to manage risk and grow their wealth. According to the World Bank (2017:9), the development of FinTechs aims to provide an opportunity not only to solve payment, credit, and remittance problems, but also provides accessible and appropriate financial products such as investments, savings, and insurance at scale. South Africa's financial services sector is internationally recognised as one of the most sophisticated in the last decade, which has been complemented by a small but fast-growing FinTech community. While South African FinTechs are world class, with three being listed in the FinTech 100 list in 2016, incumbents continue to dominate the financial sector, which makes it challenging for FinTechs to find scale. This is further exacerbated by a risk-averse funding environment, a shortage of

entrepreneurial skills, and South Africa's complex regulatory environment. More than 200 FinTechs operate in South Africa. Rand Merchant Bank (2019:5) indicated that the flourishing FinTech industry in South Africa is mainly found in the financial and technology sectors, and that the number of FinTechs is expected to grow through support from innovation hubs and the increasing adoption of technology in financial services. Rand Merchant Bank (2019) found that individuals are more likely to start first-time savings and significantly increase their savings balances after FinTech activation. However, the study also observed that customers with low financial literacy are less likely to activate the tool in the first place. Overall, the results suggest that emerging FinTechs indeed have the potential to affect household saving behaviour (Becker, 2017:29). Efforts are made to improve savings within the country, but without major success. It is therefore urgent and vital to fulfil what this study aimed to achieve, which is to craft innovative ways to encourage household savings for financial wellbeing, especially for households in townships, and to spread it to the whole nation and other emerging economies. When examining the FinTechs in Kenya, the study found that Kenya has a low rate of financial inclusion; however, this improved in 2006, when M-PESA, the leading local mobile money transfer service, was introduced. Van Hove and Dubus (2019:18) indicated that only 18.5% of Kenyans used formal services (i.e., mostly bank accounts), 8.1% used semi-formal services (such as those provided by microfinance institutions), 35% used the informal sector (rotating savings and credit associations, etc.), and 38.3% were completely excluded. Van Hove and Dubus (2019:18) examined the uptake and use of M-PESA's mobile financial services in Kenya with the aim of exploring for which parts of the population the success of M-PESA as a money transfer mechanism resulted in higher financial inclusion, which is equated with being able to save. Their findings indicated that those who do not benefit

from the positive effects of FinTechs, such as the ability to receive more frequent and faster remittances and, ultimately, the ability to save in a formal account, were disproportionately non-educated, poor, and female.

## **2.12 THE THEORETICAL FRAMEWORK OF THE STUDY**

As indicated in the background of the study, this study explored various prominent theoretical frameworks such as the life cycle hypothesis, the precautionary savings theory, Ricardian equivalence, and the permanent income hypothesis. These theories are significant and were utilised in this study on innovative ways to encourage household savings for financial wellbeing with the purpose of indicating where the study intended to close gaps in knowledge and practice. The theoretical framework also assisted the study to elucidate the need for innovative ways to encourage household savings for financial wellbeing by enabling the study to provide a well-defined and proven argument.

### **2.12.1 The life cycle hypothesis**

Modigliani and Brumberg (1954) established the life cycle hypothesis during the 1950s. It is intensely utilised to interpret people's lifespan behaviour regarding their savings and spending. The main purpose of this hypothesis in this study is to ensure that households' standard of living is effectively improved by enabling and/or assisting them to institute a balance between saving and spending in different stages of life, which is commonly known as consumption smoothing. The life cycle hypothesis supposes that a person designs their lifetime spending based on the future income that they speculate they will earn. Many empirical studies have yielded fruitful results using the life cycle hypothesis. However, it is important to note that due to the

insufficiency of continuous empirical evidence for the life cycle hypothesis, it is sometimes attributed to “unrealistic assumptions”. In order to rectify these “unrealistic assumptions”, significant efforts have been made to address these assumptions, such as the introduction of the household as the decision-making unit, as well as by introducing uncertainty (Xiao, 1996). The understanding of the life cycle hypothesis is vital and relevant in this study because Shin and Kim (2018:396) explained that households that comprehend their current income as less compared to normal years are unlikely to save relative to those who view their income as the same as the reference point. The life cycle hypothesis is also significant in this study because the study sought to establish if the life cycle hypothesis’ assumption that households are rational and do forward planning is correct. In contrast to the life cycle hypothesis, behavioural economics indicate that households might avoid planning, which could be because they may lack the self-control to decrease their current spending and save more for the future. Behavioural economics therefore reiterate that life cycle planning could be easier if households have a large income.

### **2.12.2 The precautionary savings theory**

Precautionary savings occur when one postpones current consumption due to the unpredictability of the future. Uncertainties may include life uncertainties such as when people suspect that they may be retrenched from their employment, or when they experience ill health challenges that are anticipated to result in loss of income. In order to render insurance for these contingencies, people save a fraction of their income, which results from limiting their current consumption. Levenko (2020:130) conducted research on determinants of household savings in Europe with particular focus on the impact of labour income uncertainty. The study concurred with the precautionary

theory by establishing that the household savings rate was mainly persistent and driven largely by income growth and by changes in labour income uncertainty, which can be divided into two components: realised uncertainty and expectations of future uncertainty. According to Chen *et al.* (2021:333), these savings form a buffer against adverse financial shocks that households may come across and may be interpreted as people imposing liquidity constraints on themselves. Therefore, when innovative ways to encourage household savings are crafted, households will not only be expected to save only when expecting future financial difficulties, but they will also be expected to save as their household policy and habit. Pericoli and Ventura (2012:580) indicated that precautionary savings only resulted in 11% of total household savings. According to Dossche and Zlatanov (2020:5), household savings increased in European countries during the COVID-19 pandemic as a result of precautionary savings. Their study emphasised that higher levels of unemployment were forecasted, which led to many households depositing and saving their money to guard against the future speculated unemployment rate. The precautionary savings theory applies and is relevant to this study because better comprehension of this theory will enable better crafting of innovative ways to encourage household savings for financial wellbeing, which will allow households to save habitually and not as a precaution against speculated dire financial constraints.

### **2.12.3 Ricardian equivalence**

This theory was developed by David Ricardo during the 19<sup>th</sup> century and was detailed by Robert Barro, who was a professor at Harvard University (Buchanan & Barro, 1976:42). The theory explains how government spending is as a result of taxes that are collected, which means that the collected taxes form the government's revenue,

which is spent as government expenditure. The predicted future government spending is therefore based on future collected taxes, and has an effect on the whole economy. This theory does not support the ever-growing debt taken on by the government to enhance government spending, as it indicates that households will also bear the brunt of increasing government debt. The taxpayers will be negatively affected by increasing tax deductions. The higher the government debt, the more people will be taxed so that the government can repay its debt. The theory emphasises that household savings are mainly based on their speculated tax increases. According to Hameed, Ahmed and Salman (2020:1496), this theory renders the Keynes fiscal policy ineffective, based on the notion that a shortage in spending may result in favourable economic growth and output. The theory indicates that a government spending shortage is equal to spending out of the current taxes. The Ricardian equivalence theory is relevant and important in this study and in the crafting of innovative ways to encourage household savings because this theory indicates that government shortages or deficits equal spending out of current taxes. This implies that households that are tax payers would have savings to pay forecasted future taxes, which would lead to offsetting the macro-economic effects of increased government spending.

#### **2.12.4 The permanent income hypothesis**

Milton Friedman formulated this hypothesis in 1957, which indicates that people's spending is based on their forecasted income expectations inverse to consumption, which results in their current income after tax. This theory indicates that households spend at a level that is based on how continuously they speculate they will receive income. Households are therefore only able to save if they obtain income that is higher than their expectations of the permanent income they usually receive. Based on this

theory, when economic policies result in a rising income, it will not necessarily result in increasing consumption spending (Ireland, 1995:1). Innovative ways to encourage household savings are relevant to the permanent income hypothesis because the study aimed to inspire households to save irrespective of a change in their income. Innovative ways to save seek to enable households to effectively comprehend the purpose of savings and to outline ways that will make it possible for these households to have savings, whether the households' permanent income is constant or is speculated to diminish or increase.

## **2.13 CHAPTER SUMMARY**

This concluding part of the literature review summarises some of the findings and methodologies of various empirical studies on savings, particularly household savings and financial wellbeing, examined in the chapter. The summary reports the findings of both household-level (micro) empirical studies and national-level (macro) empirical studies. The literature review commenced by providing an overview of the South African economy, and examining the township economy by exploring its prospects and challenges. Several studies have focused on investigating the macro effects in relation to household savings and financial wellbeing (Prinsloo, 2000:2; Lerner, 1938:297; SARB, 2020:17; Chitema & Chitongo, 2020; Lydon & McIndoe-Calder, 2021; Aidoo-Mensah, 2018), which emphasised the importance of household savings at the macro level. Their findings were that households improve the country's overall investment and contribute to the growth of the economy. These studies were also able to identify determinants of household savings and financial wellbeing in South Africa as an emerging economy, as well as in other emerging economies and developed economies. The studies utilised various research methodologies and commonly

presented their findings using descriptive statistics and the vector error correction model, but rarely used ARDL.

Firth and Madinginye (2015:29), Van Hove and Dubus (2019:18), Cole *et al.* (2014), and many others, as indicated in the literature review, identified determinants of household savings and financial wellbeing at the household (micro) level. Their findings included lack of household savings, which was attributed to a lack of a household saving culture because South Africa identifies as a credit economy. Van Hove and Dubus (2019:18) identified an innovative way to encourage household savings through the introduction of M-PESA in Kenya. The intensive literature review did, however, identify a gap in terms of innovative ways to encourage household savings; hence the need to conduct this study and to craft innovative ways to encourage household savings.

## **CHAPTER 3:**

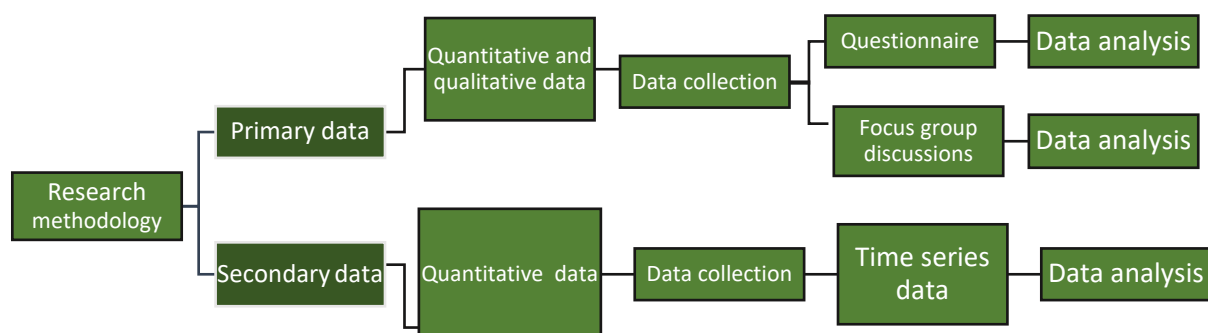
### **DATA AND METHODOLOGY**

#### **3.1 INTRODUCTION**

The literature review examined and discussed various previous research works. The literature review revealed a gap in terms of ways to encourage household savings. The purpose of this chapter is to demonstrate and expound the data collection and analysis of this study. As indicated earlier, the study sought to craft innovative ways to encourage household savings for financial wellbeing by using the Atteridgeville township as a case study and conducting a national analysis, using a mixed-methods approach. The mixed-methods approach combines both the qualitative and quantitative approaches, which enables a richer and a fuller scale of research and assists in balancing the limitations of each method. The mixed-methods approach allows the integration of both qualitative and quantitative approaches and benefits from the strengths of both methods. The advantage of the mixed-methods approach is that it allows a research study to have credibility and contextualisation. The former enables the study to be reliable as it utilises various methods of data collection on the same subject, while the latter enables the study to be put in a better context and allows a substantially detailed conclusion.

The convergent parallel method was utilised in this study, where both quantitative and qualitative data were collected and the analyses were conducted separately. After data collection and analysis, the results were compared in order to draw a tangible and satisfactory conclusion. A popular classification of the purposes of mixed-methods research was first introduced by Greene, Caracelli and Graham (1989) based on an analysis of published mixed-methods studies. This classification is still in use, as

indicated by Greene (2007:116) and Greene *et al.* (1989:259). This study collected primary data using a questionnaire and focus group discussions, while secondary time series data were collected from reports of the OECD, the World Bank, and the SARB (see Figure 3.1).



**Figure 3.1: Visual perspective of the study's research methodology**

Source: Researcher's own compilation

## 3.2 RESEARCH OBJECTIVES

The primary objective of the study was to craft innovative ways that can be used to encourage savings for financial wellbeing at the household level in townships, using Atteridgeville township as the case study, and at the national level, using secondary data.

The secondary research objectives were as follows:

- To determine the perceived effect demographic characteristics on household and demographic characteristics<sup>1</sup> in the study area and at the national level.
- To analyse the extent to which savings can be promoted among households.

<sup>1</sup> In this study, demographic characteristics refer to gender, age, marital status, household income, etc. These aspects are fully discussed in this chapter.

- To understand the determinants of household savings and using the results to construct innovative policy recommendations to encourage household savings for financial wellbeing in townships and at the national level.
- To identify innovative factors that promote household savings at the household (micro) level and the national (macro) level.

### **3.3 THE STUDY AREA AND SAMPLE SIZE FOR PRIMARY DATA COLLECTION**

To effectively analyse innovative ways to encourage household savings in townships, Atteridgeville township was chosen as the case study. It is situated in Pretoria, in the Gauteng province, which is recorded as the province with the largest population in South Africa (Stats SA, 2020:2). The reason for the selection of this study area is because Pretoria is the capital city of South Africa, and Atteridgeville is a prominent township with households that consist of people with different socio-economic statuses, who are assumed to have a low savings rate. These differences assist in allowing the results to enable a proper conclusion to be drawn on household savings; not only in Atteridgeville, but also as a reflection of the occurrences in other townships. In other words, innovative ways to encourage savings in households introduced in Atteridgeville will easily spread to other townships.

According to Stats SA (2012), Atteridgeville is a township with a population of 64 425 people in 16 456 households and 1 673 people per square kilometre. This led to the study calculating the sample size using Slovin's formula.

Slovin's formula, as indicated in Equation 3.1, was utilised to determine the sample size, as follows:

$$n = \frac{N}{(1+N\ell^2)} \quad (3.1)$$

$$n = \frac{16456}{(1 + 16456 \times 0.05^2)}$$

$$n = 390$$

Where:

n denotes the sample size;

N denotes the population size; and

(e) denotes the marginal error.

The results recommended a sample size of 390; the researcher therefore deemed 400 as a suitable sample size that had the advantage of providing accurate and reliable results.

### **3.4 RESEARCH DESIGN: PRIMARY DATA**

To examine the implications of the root causes of low household savings with the aim of finding solutions and crafting innovative ways to encourage household savings for financial wellbeing in the study area and beyond at the macro level, the study used the mixed-methods research approach.

#### **3.4.1 Questionnaire**

The questionnaire was designed for and administered to households with various socio-economic statuses in Atteridgeville township. The questionnaire contained both open- and closed-ended questions. The questionnaire also contained Likert-scale

questions which was developed by the researcher where the participants were asked to rate their satisfaction with financial institutions, which are commercial banks. The purpose of the questionnaire was to obtain relevant information to assist the researcher in crafting innovative ways to encourage household savings for financial wellbeing in townships. The questionnaire had four sections and took the participants between 25 and 30 minutes to complete. The participation was strictly voluntary; no one was forced to participate and no penalty was imposed upon withdrawal. The questionnaire was administered both online and onsite. The online questionnaire was presented on Google Forms. The online questionnaire was administered by asking the participant's cell phone number or email address in order to send the Google Forms link to them. The onsite questionnaire was printed out and distributed to various households, whereafter the data were manually entered onto Google Forms at a later stage to allow the data to be coded and analysed. The researcher observed that many households preferred the hard copies of the questionnaires that were distributed as it was more convenient for them to complete the questionnaire on paper. Some households were also not comfortable with receiving links on their devices. Based on the calculated sample size of 400, the questionnaires were distributed to 400 households, of which 387 completed questionnaires were returned. The response rate was 97%, which was sufficient for data analysis. The data analysis therefore consisted of 387 questionnaires.

#### **3.4.2 Focus group discussion**

The significance of crafting innovative ways to encourage household savings for financial wellbeing cannot be overemphasised. It was therefore fundamental that the research methodology was not limited to quantitative data only, but to consist of both

quantitative and qualitative data. To achieve the research objectives, the study also conducted focus group discussions, which were significant and useful in providing important information that the questionnaire might have omitted or that might have been problematic to gather using the questionnaire (see Annexure 3). The focus group discussions were useful in gaining an understanding of the root causes of low household savings in townships and to suggest innovative ways to encourage households to save for financial wellbeing. Focus group discussions that consisted of 23 participants from different backgrounds, such as academics, government officials, professionals, and local community members, were held. The first focus group discussion was conducted via Google Meet and was attended by 11 participants. The second focus group discussion was conducted with 12 participants. The participants made meaningful contributions to both the focus group discussions. The focus group discussions investigated the main reasons for low household savings in the country, which enabled the researcher to understand that the reasons for a lack of savings emanated from the effects of the apartheid era and other elements that resulted in the country's and households' lack of savings.

These discussions assisted in crafting innovative ways to encourage savings as such focus group discussions have never been held before. The crafted innovative ways will not only benefit households by encouraging them to save but it will also benefit the country because when citizens are not saving, the country's net savings are also affected negatively, which negatively affects the country's economic growth. The crafted innovative ways to encourage household savings will also assist the government in implementing solutions that will empower people to save for the betterment of the nation.

### **3.4.3 Descriptive analysis**

Descriptive analytical statistics were used to draw inferences from the collected data. This is mainly because the descriptive statistics enabled data visualisation, which allowed the data to be analysed and interpreted effortlessly and in a relevant and transparent manner.

The descriptive analytical statistics allowed the frequency distribution of numerical data. The data were analysed and the results were presented in different formats, such as socio-demographic tables, chi-square test tables, and bar graphs.

### **3.4.4 Multivariate regression model**

Following Rehman, Faridi and Bashir (2010), the research adopted the multivariate regression model, where multiple logistic regression and linear regression were used to analyse the relationship between the dependent and independent variables.

For multiple logistic regression, the dependent variables were household savings, which was a dummy variable. This variable was measured in the questionnaire as a yes or no answer where households were asked to indicate if they had savings.

For multiple linear regression, the dependent variable was monthly household savings in rands, and it was deemed a continuous variable. Logarithms were used for the continuous dependent variable to avoid extreme values and to enable reliable data analysis.

The broad multivariate regression model formula is as follows:

$$Y_{i=f(X_i)}, \text{ where } i=1...n \dots \quad (3.2)$$

Where:

$Y_i$  = explained variable, which is household savings in this study.

$X_1, X_2, X_3, \dots, X_n$  = explanatory variables in this study.

There were many independent variables, which enabled the use of multiple regression models, as demonstrated below:

$$Y_i = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_K X_K + \varepsilon_i \quad (3.3)$$

Where:

$\alpha$  = an intercept;

$\beta_K X_K$  = vectors of coefficients;

$X_1, X_2, X_3, \dots, X_n$  = explanatory variables; and

$\varepsilon_i$  = an error term.

#### **3.4.4.1 Model specification and variable description**

The chi-square was calculated. The variables that were found in the chi-square test to be significantly associated with household savings, except for socio-demographic variables, were included in the multivariate analysis models. These explanatory factors were drawn from the literature review.

Below is the model specification with explanatory factors. The same independent variables were used for both the multiple logistic and multiple linear regression.

$$\begin{aligned}
HS = & \alpha + \beta_1 Agegroup + \beta_2 FamSize + \beta_3 MonthlyIncRange + \beta_4 FinKw + \beta_5 FinResil \\
& + \beta_6 YearsEmpty + \beta_7 HSsatisfied + \beta_8 HSPrioritysavings \\
& + \beta_9 HSPrefersSOC + \beta_{10} HSConsumCovid19 + \beta_{11} HSEduLevel \\
& + \beta_{12} HSSectorWork + \varepsilon
\end{aligned}$$

A description of the independent variables for both the multiple linear and multiple logistics models is as follows:

- **Age group** was an independent variable that established the age effect on household savings. The variable also assisted with comprehending dependency, as to whether households with fewer dependants had more savings, and vice versa.
- **Family size (number)** was an independent variable with the purpose of determining the number of people in the household and its effect on household savings.
- **Monthly salary range** was an independent variable that assisted in determining if different incomes levels have an effect on different households' savings.
- **Financial knowledge** was an index independent variable where households' financial knowledge and their level of financial literacy were combined to establish the financial knowledge variable, which sought to establish whether being financially knowledgeable has a link with household savings for financial wellbeing.
- **Financial resilience** was an index variable where households' financial behaviour and their level of financial risk tolerance were combined to establish financial resilience.

- **Years of employment** was an independent variable used to establish the relationship between the number of years of employment in households and the households' savings.
- **Households participating in stokvels** was an independent variable to determine households' participation in stokvels and its relationship with household savings.
- **Households' satisfaction with household savings** was an independent variable to determine whether households that had savings were satisfied with their savings.
- **Type of household priority savings** was an independent variable used to establish the type of savings that households were likely to prioritise.
- **Households that prefer a state-owned bank** was an independent variable to determine if households would prefer a state-owned bank with the purpose of encouraging household savings by offering incentives for household savings.
- **Households' consumption during the COVID-19 pandemic** was an independent variable used to determine how household consumption was affected by the COVID-19 pandemic.
- **Highest level of education** was an independent variable that assisted to establish if there was a relationship between households' savings and education, as well as whether levels of education mattered.
- **Sector working in** was an independent variable used to determine whether the sectors that households work in had an impact on household savings.

### 3.5 RESEARCH DESIGN

In addition to the primary data methodology as outlined earlier, the study also utilised secondary data to allow in-depth analysis by including national-level variables that would assist in crafting innovative ways to encourage household savings for financial wellbeing. The secondary data comprised time series data that were collected from various authoritative sources such as the OECD, World Bank, and SARB. Data spanning the period 1980 to 2021 were utilised in this study. The year 1980 was chosen as the starting year to enable better comprehension of household savings trends during the apartheid era and in the post-apartheid era environment. The dummy variable was created to manage and allow the transition of household savings between the apartheid and the post-apartheid governments. The dependent variable was the household savings ratio, where the net household savings were calculated as the disposable income that had not been spent on final consumption plus the change in net equity of households in pension funds, which was measured as the percentage of disposable income. The dependent variable was taken from the SARB. The data was on annual basis.

The independent variables are outlined below:

- **Real house price (RHP)** was an independent variable utilised as an indicator to determine how property prices change over time. This indicator is an index with the base year 2020 and was taken from the OECD (1980 to 2021).
- **Old age dependency (ADR)** was a demographic factor that indicated the population aged above 65 as a percentage of the total population. It was measured as a percentage of the population over 65 and was taken from the World Bank (1980 to 2021).

- **GDP growth rate (GGR)** was an independent variable that is described as the annual percentage growth in the GDP per capita based on constant local currency, which was taken from the World Bank (1980 to 2021).
- **Inflation (INFL)** was an independent variable used to capture income uncertainty. Inflation is defined as the Consumer Price Index reflected as the annual percentage change in the cost of the average consumer of acquiring a basket of goods and services. This variable was taken from the World Bank (1980 to 2021).
- **Real interest rate (RIR)** was an independent variable used as the real long-term interest rate based on lagged GDP deflator inflation. It was measured as the annual real interest rate and was taken from the World Bank (1980 to 2021).
- **The dummy transvariable** was denoted as 0 from 1980 to 1993 and as 1 from 1993 to 2021, with the purpose of explaining structural breaks from the apartheid regime to the post-apartheid era and the impact on household savings.

### 3.5.1 Econometric model for national-level data

The econometric model utilised in the study was the ARDL model, as adapted from the work of Pesaran, Shin and Smith (1996) and the adjusted version by Pesaran, Shin and Smith (1999). The study used the ARDL methodology because (1) no work to the best of the researcher's knowledge has been conducted and the literature reviewed used it to investigate this kind of relationship in the case of South Africa and (2) it allowed the study to model variables that were of different orders of integration, which means that it could mix those variables that were stationary in their levels  $I(0)$  and those stationary after first differencing  $I(1)$ .

The basic ARDL model for co-integration testing that this study used is shown below:

$$\Delta X_t = \delta_0 + \sum \alpha_i k_i = 1\Delta X_{t-1} + \sum \alpha_2 k_i = 1\Delta Y_{t-1} + \delta_2 Y_{t-1} + v_{it} \quad (3.4)$$

$k$  is the ARDL model's maximum lag order and was chosen by the user. The F-statistic was carried out on the joint null hypothesis that the coefficients of the lagged variables  $(\delta_1 X_t - 1\delta_1 X_t - 1\text{or}\delta_1 Y_t - 1\delta_1 X_t - 1)$  are zero.  $(\delta_1 - \delta_2)$  and correspond to the long-term relationship, while  $(\alpha_1 - \alpha_2)$  represents the short-term dynamics of the model.

The hypothesis that the coefficients of the lag level variables are zero was tested. The null hypothesis of the non-existence of the long-run relationship is defined as follows:

- $H_0: \delta_1 = \delta_2 = 0$  (Null; i.e., long-run relationship does not exist)
- $H_1: \delta_1 \neq \delta_2 \neq 0$  (Alternative; i.e., long-run relationship exists)

The tests for co-integration provide a solid theoretical base for the presentation, testing, estimation, and modelling of co-integrated non-stationary time series variables. Co-integration analysis helps to avoid spurious results in modelling non-stationary data. It also provides an effective framework for testing and estimating long-run relationships from actual time series data (Utkulu, 1997:1). The ARDL approach provides numerous advantages (Nkoro & Uko, 2016:1), such as that the test is based on a single equation; consequently there are fewer parameters to estimate. The ARDL test also allows for the number of lags to be applied to each variable separately. Lastly, the great advantage of the ARDL approach in this regard is its ability to identify co-integrating vectors where there are multiple co-integrating vectors. The error correction model (ECM) can therefore be derived from the ARDL model through a

simple linear transformation. The derived ECM takes a sufficient number of lags to capture the data-generating process in general to specific modelling frameworks.

### **3.6 CHAPTER SUMMARY**

This chapter focused on the methodological framework of the study and provided insight into the empirical tools utilised to analyse the lack of household savings in the country with the focus on the micro and macro levels. The study used various data-collection tools such as a questionnaire, focus group discussion, and secondary national-level data in order to obtain detailed insight to enable crafting innovative ways to encourage household savings for financial wellbeing at both the micro and macro levels. The next chapter reports on the data analysis and subsequent findings.

## **CHAPTER 4:**

### **DATA ANALYSIS, FINDINGS, AND INTERPRETATIONS**

#### **4.1 INTRODUCTION**

As stated in the previous chapters, the aim of this study was to craft innovative ways to encourage household savings for financial wellbeing at the household level using Atteridgeville township as a case study and at the national level using secondary time series data. Following the analytical framework in Chapter 3, this chapter presents the data analysis, findings, and interpretations of both the primary and secondary data.

#### **4.2 PRIMARY DATA ANALYSIS AND INTERPRETATIONS**

In this chapter, the results of the primary data analysis are presented in tables and graphs. The aim of this chapter is to address the following research objectives: to evaluate the extent to which savings can be improved among households with different perceptions and demographic characteristics, and to identify innovative factors that promote household savings in the study area. This chapter reports the descriptive and inferential statistics results. Moreover, it presents the empirical analysis and findings drawn from the quantitative data, with the focus on addressing the underlying research objectives of the study.

The data were analysed using Stata version 17 (Statacorp, 2021:1). Both descriptive and inferential statistical analyses were conducted to understand the distribution of participants and their behaviour in terms of household savings. Chi-square tests were used to assess the percentages and significant differences in the participants' socio-demographic variables and behaviour in terms of household savings.

Multiple logistic regression and linear regression were used to assess the relationship between the dependent variables (household savings as the dummy variable and monthly household savings in rands as the continuous variable) and the independent variables (socio-demographics, family size, financial knowledge, financial resilience, households participating in stokvels, households' satisfaction with their savings, type of household priority savings, households that prefer a state-owned bank, and household consumption during the COVID-19 pandemic). The variables that were found by the chi-square test to be significantly associated with household savings (except for socio-demographic variables) were included in the multivariate analysis models. Logarithms were used for a continuous dependent variable to avoid extreme values.

#### **4.2.1 The participants' socio-demographic characteristics and their household behaviour in terms of savings**

The participants' socio-demographics and their behaviour in terms of household savings are discussed in this section (see Table 4.1).

**Table 4.1: Participants' socio-demographic characteristics and their household saving behaviour**

| <b>Variables</b>         | <b>Frequency (n=387)</b> | <b>Percentage (%)</b> |
|--------------------------|--------------------------|-----------------------|
| <b>Age group (years)</b> |                          |                       |
| 15-30                    | 138                      | 36.13                 |
| 31-45                    | 70                       | 18.32                 |
| 45-60                    | 163                      | 42.67                 |
| 60+                      | 11                       | 2.88                  |
| <b>Gender</b>            |                          |                       |
| Female                   | 225                      | 58.29                 |
| Male                     | 161                      | 41.71                 |
| <b>Marital status</b>    |                          |                       |
| Married                  | 183                      | 48.8                  |
| Widowed/divorced         | 16                       | 4.27                  |

| Variables                                                                              | Frequency (n=387) | Percentage (%) |
|----------------------------------------------------------------------------------------|-------------------|----------------|
| Never married                                                                          | 176               | 46.93          |
| <b>Monthly salary range (R)</b>                                                        |                   |                |
| Below 3 500                                                                            | 43                | 11.29          |
| 3 500 – 20 000                                                                         | 265               | 69.55          |
| >20 000                                                                                | 61                | 16.01          |
| >51 000                                                                                | 12                | 3.15           |
| <b>Years employed in the sector</b>                                                    |                   |                |
| 1-5 years                                                                              | 141               | 38.32          |
| 6-10 years                                                                             | 69                | 18.75          |
| 11-15 years                                                                            | 31                | 8.42           |
| >16 years                                                                              | 127               | 34.51          |
| <b>Participating in stokvels</b>                                                       |                   |                |
| Yes                                                                                    | 138               | 36.51          |
| No                                                                                     | 240               | 63.49          |
| <b>Household satisfaction with current household savings</b>                           |                   |                |
| Yes                                                                                    | 90                | 24             |
| No                                                                                     | 285               | 76             |
| <b>Household willing to attend meetings/seminars that encourage households to save</b> |                   |                |
| Yes                                                                                    | 370               | 98.14          |
| No                                                                                     | 7                 | 1.86           |
| <b>Type of priority household savings</b>                                              |                   |                |
| Emergency household savings                                                            | 323               | 85             |
| Pension/retirement savings                                                             | 25                | 6.58           |
| Education savings                                                                      | 24                | 6.32           |
| Stokvels                                                                               | 4                 | 2.11           |
| <b>Households that prefer the introduction of a state-owned bank</b>                   |                   |                |
| Yes                                                                                    | 280               | 74.07          |
| No                                                                                     | 98                | 25.93          |
| <b>Household savings during the COVID-19 pandemic</b>                                  |                   |                |
| Increased                                                                              | 59                | 15.78          |
| Decreased                                                                              | 169               | 45.19          |
| Remained unchanged                                                                     | 146               | 39.04          |
| <b>Household consumption during the COVID-19 pandemic</b>                              |                   |                |
| Increased                                                                              | 215               | 57.18          |
| Decreased                                                                              | 59                | 15.69          |
| Remained unchanged                                                                     | 102               | 27.13          |
| <b>Highest level of education</b>                                                      |                   |                |
| Below Grade 12                                                                         | 71                | 18.54          |
| Grade 12                                                                               | 192               | 50.13          |
| Certificate                                                                            | 34                | 8.88           |
| Diploma                                                                                | 28                | 7.31           |

| Variables                 | Frequency (n=387) | Percentage (%) |
|---------------------------|-------------------|----------------|
| Degree                    | 33                | 8.61           |
| Postgraduate              | 25                | 6.53           |
| <b>Sector employed in</b> |                   |                |
| Public sector             | 91                | 24.27          |
| Private sector            | 107               | 28.53          |
| Informal sector           | 164               | 43.73          |
| Other                     | 13                | 3.47           |
| <b>Breadwinner</b>        |                   |                |
| Yes                       | 324               | 85.04          |
| No                        | 57                | 14.96          |

A total of 387 participants completed the self-administered questionnaire. Table 4.1 indicates that the largest group of participants were aged between 45 and 60 years (42.67%), followed by participants aged between 15 and 30 years (36.13%) and between 31 and 45 years (18.32%). The greater proportion of the participants were female (58.29%). The majority of the participants were married (48.8%), followed by those who were never married (46.93%). Of the participants who participated in this study, those with a monthly salary of between R3 500 and R20 000 (69.55%) were in the higher proportion. The majority of the participants indicated that they had been employed in the same sector for between one and five years (38.32%), followed by more than 16 years (34.51%). The majority of the participants indicated that they did not participate in stokvels (63.49%) and were not satisfied with their current household savings (76%). However, most participants (98.14%) indicated that they were willing to attend meetings and/or seminars that encourage households to save. Furthermore, a high percentage of the participants reported that they prioritised saving for emergency household requirements (85%) and that they would prefer a state-owned bank to be introduced in the country (74.07%). The participants also reflected on how their household savings and household consumption were impacted by the COVID-19 pandemic. The majority indicated that their household savings decreased (45.19%)

and that their household consumption increased (57.18%). Regarding the highest level of education, just over half of the participants reported having passed Grade 12 (50.13%), followed by those who reported education below Grade 12 (18.54%). Many participants reported working in the informal sector (43.73%), followed by working in the private sector (28.53%) and the public sector (24.27%). Lastly, a large proportion of the participants (85.04%) indicated that they were the breadwinners in their households.

#### 4.3 COMPARISON OF HOUSEHOLD SAVINGS WITH SOCIO-DEMOGRAPHICS AND BEHAVIOUR IN TERMS OF HOUSEHOLD SAVINGS VARIABLES

This section summarises the household savings comparisons of all the participants. Table 4.2 presents the household savings by socio-demographics, while Table 4.3 indicates the behaviour in terms of household savings variables.

**Table 4.2: Household savings by the socio-demographics of the participants (chi-squared row %)**

| Household savings        | Not saving   | Saving       |              |
|--------------------------|--------------|--------------|--------------|
|                          | n (%)        | n (%)        | P-value      |
| <b>Age group (years)</b> |              |              | <b>0.014</b> |
| 15-30                    | 85 (62.04%)  | 52 (37.96%)  |              |
| 31-45                    | 27 (38.57%)  | 43 (61.43%)  |              |
| 46-60                    | 85 (52.80%)  | 76 (47.20%)  |              |
| 61+                      | 5 (45.45%)   | 6 (54.55%)   |              |
| <b>Gender</b>            |              |              | <b>0.399</b> |
| Female                   | 122 (54.71%) | 101 (45.29%) |              |
| Male                     | 79 (50.32%)  | 78 (49.68%)  |              |
| <b>Marital status</b>    |              |              | <b>0.156</b> |
| Married                  | 95 (52.20%)  | 87 (47.80%)  |              |
| Widowed/divorced         | 5 (31.25%)   | 11 (68.75%)  |              |
| Never married            | 98 (56.00%)  | 77 (44.00%)  |              |

|                                     |              |              |              |
|-------------------------------------|--------------|--------------|--------------|
| <b>Breadwinner</b>                  |              |              | <b>0.328</b> |
| Yes                                 | 167 (51.86%) | 155 (48.14%) |              |
| No                                  | 33 (58.93%)  | 23 (41.07)   |              |
| <b>Monthly salary range (R)</b>     |              |              | <b>0.000</b> |
| Below 3 500                         | 29 (67.44%)  | 14 (32.56%)  |              |
| 3 500 – 20 000                      | 159 (60.69)  | 103 (39.31%) |              |
| >20 000                             | 11 (18.03%)  | 50 (81.97%)  |              |
| > 51 000                            | 2 (16.67%)   | 10 (83.33%)  |              |
| <b>Years employed in the sector</b> |              |              | <b>0.000</b> |
| 1-5 years                           | 89 (63.12%)  | 52 (36.88%)  |              |
| 6-10 years                          | 25 (36.23%)  | 44 (63.77%)  |              |
| 11-15 years                         | 10 (32.26%)  | 21 (67.74%)  |              |
| >16 years                           | 72 (57.14%)  | 54 (42.86%)  |              |
| <b>Highest level of education</b>   |              |              | <b>0.000</b> |
| Below Grade 12                      | 42 (60.00%)  | 28 (40.00%)  |              |
| Grade 12                            | 130 (67.31%) | 62 (32.29%)  |              |
| Certificate                         | 9 (27.27%)   | 24 (72.73%)  |              |
| Diploma                             | 8 (28.57%)   | 20 (71.43%)  |              |
| Degree                              | 9 (27.27%)   | 24 (72.73%)  |              |
| Postgraduate                        | 4 (16.00%)   | 21 (84.00%)  |              |
| <b>Sector working in</b>            |              |              | <b>0.001</b> |
| Public sector                       | 35 (39.33%)  | 54 (60.67%)  |              |
| Private sector                      | 52 (48.60%)  | 55 (51.40%)  |              |
| Informal sector                     | 104 (68.02%) | 59 (35.98)   |              |
| Other                               | 6 (50%)      | 6 (50%)      |              |

As shown in Table 4.2, household savings were statistically significant with age group, monthly salary range, years employed in the sector, highest level of education, and the sector employed in. Participants aged between 31 and 45 years (61.43%) and 61 and older (54.55%) were more likely to have household savings compared to other groups. Moreover, those who earned a monthly salary of more than R20 000 (81.97%) and more than R51 000 (83.33%) had higher proportions of household savings than those who earned less than R20 000. Working for 11 to 15 years (67.74%) and six to 10 years (63.77%) in a sector was highly associated with household savings.

Additionally, having a postgraduate degree (84%) and working in the public sector (60.67%) also predicted household savings. There were no significant differences between household savings and other socio-demographic variables, such as gender, marital status, and being the breadwinner.

**Table 4.3: Behaviour in terms of household savings**

| Household savings                                                    | Not saving   | Saving       |              |
|----------------------------------------------------------------------|--------------|--------------|--------------|
|                                                                      | n (%)        | n (%)        | P-value      |
| <b>Participating in stokvels</b>                                     |              |              | <b>0.000</b> |
| Yes                                                                  | 39 (28.26%)  | 99 (71.74%)  |              |
| No                                                                   | 160 (67.23%) | 78 (32.77%)  |              |
| <b>Households' satisfaction with current household savings</b>       |              |              | <b>0.000</b> |
| Yes                                                                  | 21 (23.33%)  | 69 (76.67%)  |              |
| No                                                                   | 177 (62.32%) | 107 (37.68%) |              |
| <b>Attend community meetings on savings packages</b>                 |              |              | <b>0.185</b> |
| Yes                                                                  | 194 (53.74%) | 167 (46.26%) |              |
| No                                                                   | 5 (35.71%)   | 9 (64.29%)   |              |
| <b>Participate in community savings seminars</b>                     |              |              | <b>0.099</b> |
| Yes                                                                  | 196 (53.99%) | 167 (46.01%) |              |
| No                                                                   | 4 (30.77%)   | 9 (69.23%)   |              |
| <b>Attend meetings/seminars that encourage households to save</b>    |              |              | <b>0.038</b> |
| Yes                                                                  | 198 (53.80%) | 170 (46.20%) |              |
| No                                                                   | 1 (14.29%)   | 6 (85.71%)   |              |
| <b>Priority household savings</b>                                    |              |              | <b>0.001</b> |
| Emergency savings                                                    | 185 (57.28%) | 138 (42.72%) |              |
| Pension/retirement                                                   | 4 (17.39%)   | 19 (82.61%)  |              |
| Education savings                                                    | 9 (37.50%)   | 15 (62.50%)  |              |
| Stokvels (festive savings)                                           | 4 (50%)      | 4 (50%)      |              |
| <b>Households that prefer the introduction of a state-owned bank</b> |              |              | <b>0.037</b> |
| Yes                                                                  | 156 (56.12%) | 122 (43.88%) |              |
| No                                                                   | 43 (43.88%)  | 55 (56.12%)  |              |
| <b>Household savings during the COVID-19 pandemic</b>                |              |              | <b>0.537</b> |
| Increased                                                            | 31 (52.54%)  | 28 (47.46%)  |              |
| Decreased                                                            | 94 (55.95%)  | 74 (44.05%)  |              |
| Unchanged                                                            | 72 (49.66%)  | 73 (50.34%)  |              |
| <b>Household consumption during the COVID-19 pandemic</b>            |              |              | <b>0.000</b> |
| Increased                                                            | 133 (62.15%) | 81 (37.85%)  |              |
| Decreased                                                            | 25 (42.37%)  | 34 (57.63%)  |              |
| Unchanged                                                            | 40 (39.60%)  | 61 (60.40%)  |              |

Table 4.3 shows that a significantly higher proportion of the participants who were satisfied with the state of their household savings (76.67%) and who participated in stokvels (71.74%) were more likely to have household savings. Attending meetings and seminars that encourage households to save (46.2%) and having a preference for a state-owned bank to promote household savings in townships (43.88%) lowered the likelihood of having household savings. Household savings were highly associated with pension or retirement (82.61%) as the saving priority. The participants who indicated that their household consumption during the COVID-19 pandemic remained unchanged (60.40%) were more likely to have household savings than the other categories. No significant differences were observed between household savings and attending community meetings on savings packages, participating in community savings seminars, and household savings during the COVID-19 pandemic.

#### **4.4 MULTIPLE LOGISTIC REGRESSION AND LINEAR REGRESSION PREDICTIONS OF HOUSEHOLD SAVINGS**

Multiple logistic regression and linear regression models were performed to assess the associations between the dependent variables and the independent variables. The dependent variables were household savings (for logistic regression) and monthly household savings in rands (for linear regression). Variables found in the chi-square analysis to be significantly associated with household savings (except for socio-demographic variables) were entered into both models. These variables were socio-demographics, family size, financial knowledge, financial resilience, households participating in stokvels, households' satisfaction with their savings, type of household priority savings, households that prefer a state-owned bank, and household consumption during the COVID-19 pandemic.

#### 4.4.1 Multiple logistic regression

Table 4.4 summarises the results obtained from the multiple logistic regression of household savings and selected predictor values.

**Table 4.4: Multiple logistic regression of household saving and selected predictor variables**

| Variable                                                      | Odds ratio | Std. err. | z     | P-value | 95% CI    |       |
|---------------------------------------------------------------|------------|-----------|-------|---------|-----------|-------|
| Age group (years)                                             |            |           |       |         |           |       |
| 15-30                                                         | Ref.       |           |       |         |           |       |
| 31-45                                                         | 1.39       | 0.77      | 0.59  | 0.558   | 0.47      | 4.13  |
| 45-60                                                         | 2.50       | 1.45      | 0.57  | 0.116   | 0.80      | 7.81  |
| 60+                                                           | 3.64       | 3.76      | 1.25  | 0.212   | 0.48      | 27.62 |
| Family size (number)                                          | 1.03       | 0.13      | 0.23  | 0.015   | 0.01      | 1.31  |
| Monthly salary range                                          |            |           |       |         |           |       |
| Below R3 500                                                  | Ref.       |           |       |         |           |       |
| R3 500 – R20 000                                              | 1.31       | 0.70      | 0.51  | 0.613   | 0.46      | 3.77  |
| >R20 000                                                      | 4.92       | 3.46      | 2.26  | 0.024   | 1.25      | 19.54 |
| >R51 000                                                      | 2.36       | 2.90      | 0.70  | 0.485   | 0.21      | 26.18 |
| Financial knowledge                                           | 1.46       | 0.26      | 2.08  | 0.037   | 1.02      | 2.08  |
| Financial resilience                                          | 0.76       | 0.15      | -1.38 | 0.168   | 0.52      | 1.12  |
| Years of employment                                           |            |           |       |         |           |       |
| 1-5 years                                                     | Ref.       |           |       |         |           |       |
| 6-10 years                                                    | 2.28       | 1.25      | 1.48  | 0.140   | 0.76      | 6.78  |
| 11-15 years                                                   | 1.89       | 1.35      | 0.90  | 0.369   | 0.47      | 7.62  |
| >16 years                                                     | 1.42       | 0.90      | 0.56  | 0.047   | 0.41      | 4.90  |
| Participating in stokvels                                     |            |           |       |         |           |       |
| Yes                                                           | Ref.       |           |       |         |           |       |
| No                                                            | 0.31       | 0.10      | -3.50 | 0.000   | 0.16      | 0.60  |
| Households' satisfaction with current household savings       |            |           |       |         |           |       |
| Yes                                                           | Ref.       |           |       |         |           |       |
| No                                                            | 0.19       | 0.08      | -4.18 | 0.000   | 0.09      | 0.42  |
| Type of household priority savings                            |            |           |       |         |           |       |
| Emergency savings                                             | Ref.       |           |       |         |           |       |
| Pension/retirement savings                                    | 3.25       | 2.85      | 1.35  | 0.178   | 0.59      | 18.08 |
| Education savings                                             | 2.02       | 1.33      | 1.07  | 0.285   | 0.56      | 7.37  |
| Stokvels (festive savings)                                    | 0.16       | 0.20      | -1.48 | 0.138   | 0.01      | 1.80  |
| Households that prefer the introduction of a state-owned bank |            |           |       |         |           |       |
| Yes                                                           | Ref.       |           |       |         |           |       |
| No                                                            | 1.310862   | 0.5317487 | 0.67  | 0.505   | 0.5919273 | 2.90  |

| Variable                                           | Odds ratio | Std. err. | z     | P-value | 95% CI |       |
|----------------------------------------------------|------------|-----------|-------|---------|--------|-------|
| Household consumption during the COVID-19 pandemic |            |           |       |         |        |       |
| Increase                                           | Ref.       |           |       |         |        |       |
| Decrease                                           | 3.29       | 1.66      | 2.37  | 0.018   | 1.23   | 8.83  |
| Unchanged                                          | 1.47       | 0.68      | 0.82  | 0.410   | 0.59   | 3.65  |
| Highest level of education                         |            |           |       |         |        |       |
| Below Grade 12                                     | Ref.       |           |       |         |        |       |
| Grade 12                                           | 0.97       | 0.45      | -0.06 | 0.950   | 0.39   | 2.42  |
| Certificate                                        | 3.24       | 2.12      | 1.79  | 0.073   | 0.89   | 11.72 |
| Diploma                                            | 3.28       | 2.56      | 1.52  | 0.128   | 0.71   | 15.13 |
| Degree                                             | 4.80       | 3.94      | 1.91  | 0.056   | 0.96   | 24.00 |
| Postgraduate                                       | 7.23       | 7.24      | 1.98  | 0.048   | 1.02   | 51.48 |
| Sector working in                                  |            |           |       |         |        |       |
| Public sector                                      | Ref.       |           |       |         |        |       |
| Private sector                                     | 0.97       | 0.43      | -0.07 | 0.941   | 0.41   | 0.31  |
| Informal sector                                    | 1.13       | 0.53      | 0.27  | 0.036   | 0.46   | 2.82  |
| Other                                              | 0.40       | 0.52      | -0.71 | 0.876   | 0.03   | 5.11  |

CI = confidence interval

Age group, financial resilience, type of household priority savings, households that would prefer a state-owned bank, and type of sector the participants worked in did not predict household savings. However, family size, monthly salary range, financial knowledge, years of employment, households participating in a stokvel, households' satisfaction with their savings, households' consumption during the COVID-19 pandemic, and highest level of education were significantly associated with household savings. Participants who indicated earning more than R20 000 per month (AOR 4.92; 95% CI 1.25, 19.54;  $p = 0.024$ ) were almost five times more likely to have household savings than those who earned below R3 500. Participants who had high financial knowledge (AOR 1.46; 95% CI 1.02, 2.08;  $p = 0.037$ ) were more likely to have household savings. Participants who were not satisfied with the state of their household savings (AOR 0.19; 95% CI 0.09, 0.42;  $p < 0.001$ ) and who did not participate in stokvels (AOR 0.31; 95% CI 0.16, 0.60;  $p < 0.001$ ) had lower odds of having household savings than participants who were satisfied with the state of their

household savings and who participated in stokvels. Participants who reported that their household consumption decreased (AOR 3.29; 95% CI 1.23, 8.83;  $p = 0.018$ ) during the COVID-19 pandemic were more than three times more likely to have household savings than those who reported an increase in consumption. Lastly, having a postgraduate education (AOR 7.23; 95% CI 1.02, 51.48;  $p = 0.048$ ) predicted household savings more than having less than Grade 12 education.

#### 4.4.2 Multiple linear regression model

Table 4.5 presents a summary of the analysis of variance (ANOVA) results, while Table 4.6 indicates the results of the multiple linear regression model of monthly household saving and selected predictor variables.

**Table 4.5: Summary of analysis of variance (ANOVA)**

| Source   | SS       | Df  | MS     | Number of obs.         | 194    |
|----------|----------|-----|--------|------------------------|--------|
| Model    | 100.0743 | 13  | 7.6980 | F (13.180)             | 6.55   |
| Residual | 211.4278 | 180 | 1.1746 | Prob>F                 | 0.0000 |
| Total    | 311.5021 | 193 | 1.614  | R-squared              | 0.3213 |
|          |          |     |        | Adj. R-squared         | 0.2722 |
|          |          |     |        | Root mean square error | 1.0838 |

**Table 4.6: Multiple linear regression model of monthly household saving and selected predictor variables**

| Variable                                          | Coefficient | Std. err. | t     | P-value>t | 95% CI |      |
|---------------------------------------------------|-------------|-----------|-------|-----------|--------|------|
| Age group (years)                                 | 0.10        | 0.13      | 0.77  | 0.441     | -0.15  | 0.35 |
| Family size (number)                              | 0.13        | 0.06      | 2.16  | 0.032     | 0.01   | 0.25 |
| Monthly salary range                              | 0.13        | 0.15      | 0.88  | 0.378     | -0.16  | 0.42 |
| Financial knowledge                               | 0.15        | 0.08      | 1.89  | 0.040     | -0.01  | 0.30 |
| Financial resilience                              | -0.07       | 0.08      | -0.86 | 0.393     | -0.23  | 0.09 |
| Years of employment                               | 0.12        | 0.11      | 1.12  | 0.265     | -0.09  | 0.33 |
| Participating in stokvels                         | 0.50        | 0.20      | 2.55  | 0.011     | 0.11   | 0.89 |
| Household satisfaction with their current savings | -0.25       | 0.18      | -1.35 | 0.177     | -0.61  | 0.11 |

| Variable                                                      | Coefficient | Std. err. | t     | P-value>t | 95% CI |      |
|---------------------------------------------------------------|-------------|-----------|-------|-----------|--------|------|
| Type of household priority savings                            | 0.26        | 0.12      | 2.16  | 0.032     | 0.02   | 0.50 |
| Households that prefer the introduction of a state-owned bank | 0.92        | 0.21      | 4.37  | 0.000     | 0.50   | 1.33 |
| Household consumption during the COVID-19 pandemic            | 0.39        | 0.11      | 3.52  | 0.001     | 0.17   | 0.60 |
| Highest level of education                                    | 0.13        | 0.07      | 1.86  | 0.046     | -0.01  | 0.27 |
| Sector working in                                             | -0.05       | 0.12      | -0.44 | 0.660     | -0.28  | 0.18 |

Table 4.6 shows the socio-demographics and selected behaviour in terms of household savings attributes (independent variables) and monthly household savings (dependent variable). The independent variables in the model were significantly predictive of monthly household savings, as determined by the ANOVA statistics [ $F(13, 180) = 6.55$ ;  $p < 0.001$ ]. Moreover, the model's degree of predicting the dependent variable was  $R^2 = 0.321$  and the model's degree of explaining the variance in the dependent variable was  $\text{Adj } R^2 = 0.272$ . This shows that the model predicted the dependent variable very well. The coefficients of family size (number) ( $\beta = 0.13$ ;  $p = 0.032$ ), households participating in stokvels ( $\beta = 0.50$ ;  $p = 0.011$ ), type of household savings priority ( $\beta = 0.26$ ;  $p = 0.032$ ), households that prefer a state-owned bank ( $\beta = 0.92$ ;  $p < 0.001$ ), and household consumption during the COVID-19 pandemic ( $\beta = 0.39$ ;  $p = 0.001$ ) were significantly positively associated with monthly household savings. There was no significant relationship between age group (years), monthly salary range, financial knowledge, financial resilience, years of employment, household satisfaction with savings, highest level of education, sector employed in, and monthly household savings.

#### **4.5 FOCUS GROUP DISCUSSION OUTCOMES**

The researcher conducted focus group discussions as a tool to identify the root causes of the lack of household savings in South Africa with the purpose of allowing the participants to provide innovative ways to encourage household savings for financial wellbeing in townships and nationally. The focus group discussions were facilitated on an online platform (Zoom) with a total of 23 participants on two occasions. Eleven people attended the first meeting and the second meeting was attended by a total of 12 people. The participants came from various backgrounds such as the church, political organisations, academic backgrounds, ward councillors in Atteridgeville township, community members, the employed and unemployed, and the educated and the uneducated. The common factor among these individuals was that they all had experienced residing in various townships and have therefore all been confronted by the various socio-economic challenges prevalent in townships. The focus group discussions were facilitated by the researcher. The names and affiliations of a few of the participants who participated in the focus group discussions are as follows: Pastor Neo Musa Zonal, Partnership Manager at Christ Embassy Southern African Zone 2; Ms Lebogang Mulaisi, Head of Policy at the Congress of South African Trade Unions (COSATU); Ms Yolanda Khoza, statistician at Stats SA and deputy secretary at community ward; Mr Barry White, parliamentary officer at the National Education, Health and Allied Workers' Union (NEHAWU); Mr Makola, former Ward 71 counsellor for Atteridgeville; Mr Phasa, current Ward 71 counsellor for Atteridgeville; and Ms Tebogo Ngele, former master's student at TUT. Other participants from the community consisted of both the youth and the elderly. The focus group discussions effectively deliberated on the lack of household savings, identified the root causes of the lack of savings, and assisted in crafting innovative ways to encourage household savings for

financial wellbeing. The outcomes of the focus group discussions in narrative form follow.

The study identified the root causes of lack of household savings among households in townships through the focus group discussions. The study found that many townships are still characterised by the highest levels of poverty, unemployment, and inequality. Very few households in townships are considered to be of a better socio-economic status than other households. The study also established that households with a better socio-economic status are traced back to their previous generation, where the previous generation was also better off relative to others of their generation.

The discussions also indicated that black South African households have no substantial inheritance; all they inherited from the apartheid legacy was poverty and lack, and the results are still visible today. The discussions also identified households' lack of financial planning as a root cause of lack of household savings. Many households in townships were identified as spenders, as they spend their money as soon as it reaches their hands. Households are unable to prioritise how they spend their money, and without budgeting, it is nearly impossible to account for how money is spent.

The focus group discussions also considered the dynamics of households in townships such as that there are more female-headed households than male-headed households. The female household heads are the breadwinners and mainly work in the informal sector. Innovative ways to encourage household savings should therefore seek to ease the burden for female-headed households, as the study found that the average township household size is four people, which means it is the average number of people the female household head must care for.

The focus group discussions concluded that prior to 1994, apartheid was still prevalent in South Africa and many households were excluded from the South African formal financial system, few black South Africans were formally employed, and few black South Africans were educated. Black households were oppressed and did not have substance. Those who worked sent their meagre wages that they earned to their families at home. However, post-1994, the households started actively participating in the formal economy. They were exposed to working in the formal sector and gained financial liberation as they had the ability to take out loans from financial institutions. Households became indebted and could not consider savings. This is why the credit economy has been described as hindering households to save, and worsens households' ability to save.

Many households in townships obtain their income from the informal economy. They are hairdressers, street vendors, construction workers, retail workers, owners of spaza shops, and more. Few households work in the formal economy and the data collected by this study confirmed this statement. Their income is unstable, depending on the performance of the sector that they are employed in, and if there is an unexpected economic downturn, those sectors suffer.

South Africa has introduced a minimum wage of R3 500; however, the focus group discussions indicated that this was a minimum wage and not a living wage. Households can barely survive on the minimum wage; the introduction of a living wage is therefore urgently needed. The focus group discussions established that many households spend more of their income on transport than on their basic needs. Depending on the size and the dynamics of the households, the recommended living wage for households in townships should therefore be between R10 000 and R15 000.

The focus group discussions identified that the most common form of savings that many households were aware of and exposed to was stokvels. However, many township households are reluctant to participate in stokvels due to the level of risks that these schemes pose. Many people have been victims of stokvels such as “Kipi” and WhatsApp stokvel scams. They are also aware of and exposed to burial societies but not many households are aware of savings accounts for children and other savings initiatives that are provided by the commercial banks. The focus group discussions concluded that households in townships lack knowledge of household savings initiatives that are available to them, some households are not able to secure consistent income, and others lack discipline to save. The focus group discussions also noted that some households are characterised by not having much support beyond their immediate household and therefore cannot even take care of their basic needs.

In conclusion, the focus group discussions indicated that households are exposed to packages, initiatives, and policies on household savings; however, due to plentiful household savings scams that they have been exposed to or are aware of, households need to be properly informed of the advantages and disadvantages of these packages, initiatives, and policies to set them at ease.

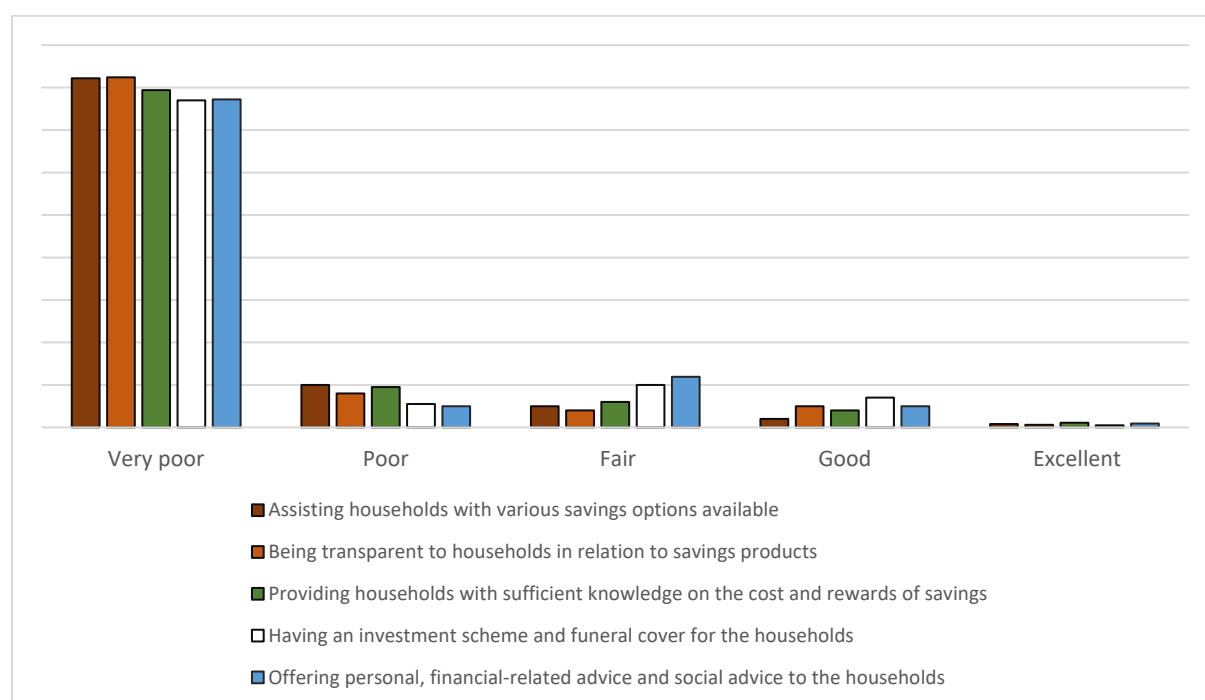
#### **4.6 RELATIONSHIP BETWEEN COMMERCIAL BANKS AND HOUSEHOLDS**

As presented in Annexure 1, Section 3 of the questionnaire focused on the relationship between financial institutions (commercial banks) and households. The purpose and significance of this section of the questionnaire was for township households to rate their level of satisfaction with the services they received from the various commercial banks located in their township, as well as to suggest areas of improvement for the

commercial banks, without mentioning the name of the bank. The relevance of this to this study was to discover the level of commercial banks' involvement in encouraging household savings for financial wellbeing in townships. Commercial banks in this study are referred to as financial institutions but the terms are used interchangeably.

#### 4.6.1 How financial institutions service households' various socio-economic issues

Figure 4.1 indicates that banks deliver very poor service to households in terms of various socio-economic issues.

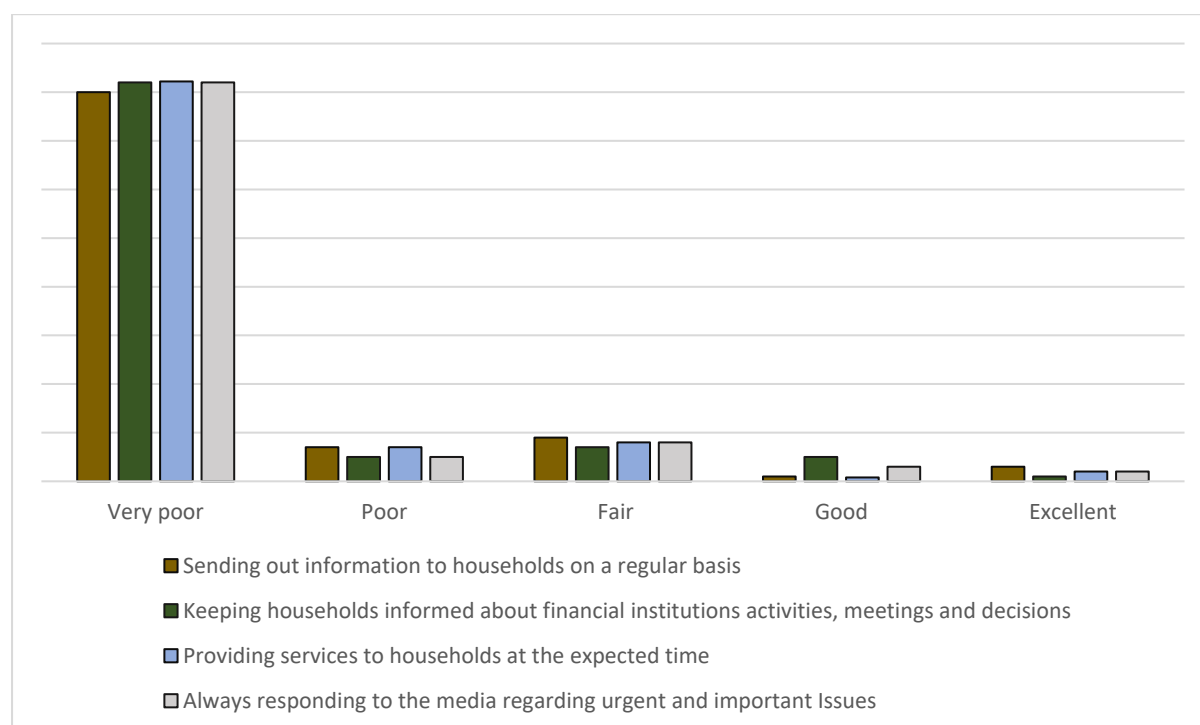


**Figure 4.1: Financial institutions' services related to households' socio-economic issues**

#### 4.6.2 How financial institutions communicate with township households

Figure 4.2 indicates that the financial institutions communicate very poorly with township households. The financial institutions do not send out information to households on a regular basis and they do not keep households informed of the

financial institutions' activities, meetings, and decisions. The data also show that financial institutions perform very poorly in terms of providing services to households at the expected time and correspondence in terms of responding to important issues that are of concern to households. Figure 4.2 illustrates how financial institutions communicate with households in townships.

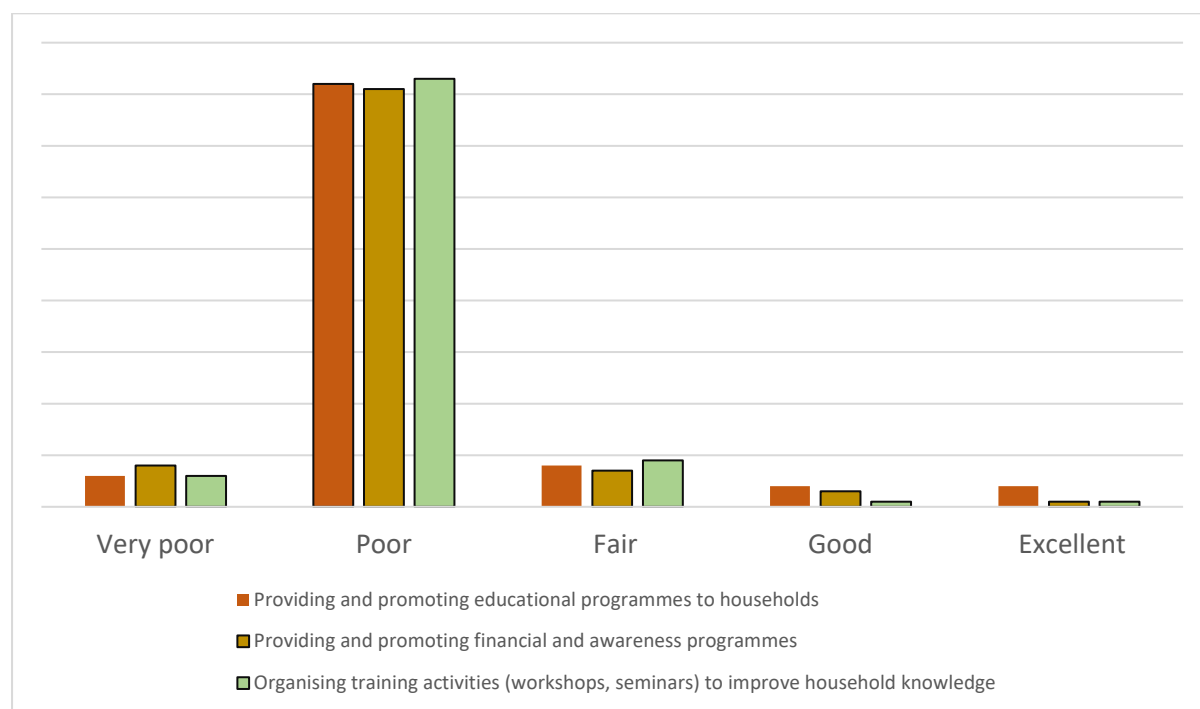


**Figure 4.2: Financial institutions' communications**

#### **4.6.3 How financial institutions provide educational services to township households**

Figure 4.3 shows that the financial institutions perform poorly with regard to offering financial education to households in townships. The financial institutions perform poorly in terms of providing and promoting educational programmes and financial awareness programmes to township households. The data also indicate that financial institutions perform poorly in terms of organising training activities such as workshops

and/or seminars aimed at households in townships. Figure 4.3 illustrates the data on how financial institutions provide educational services to township households.

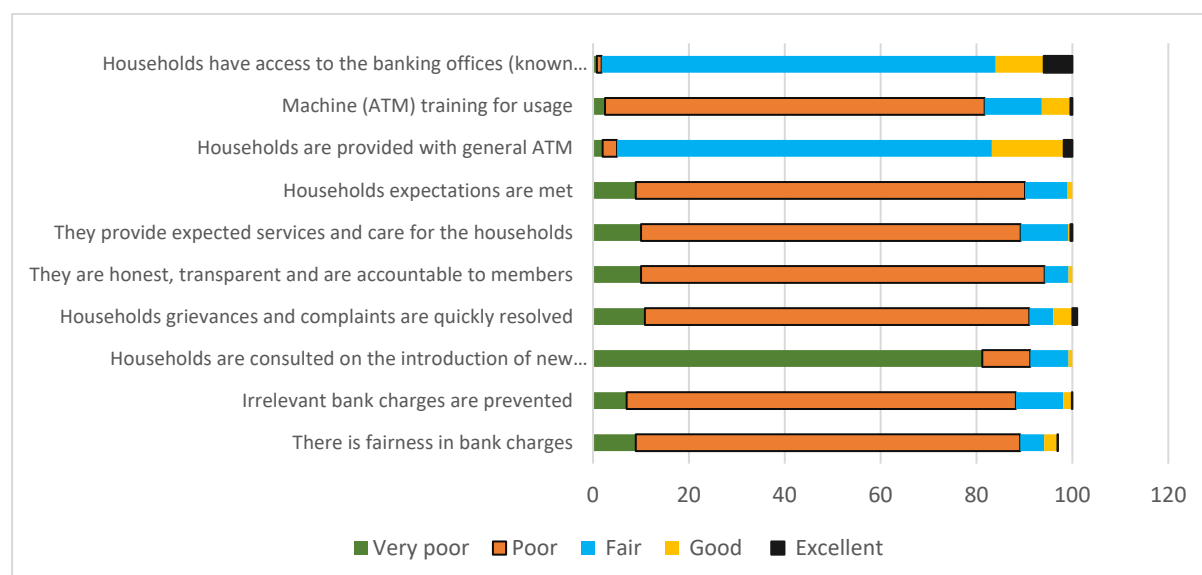


**Figure 4.3: Financial institutions' provision of educational services**

#### **4.6.4 Financial institutions' provision of personal and collective services to township households**

Figure 4.4 illustrates financial institutions' services offered to households in townships in terms of personal and collective services. The data show that financial institutions perform fairly in relation to township households having access to banking offices and knowing their bank's locations and contact details. However, financial institutions perform poorly in terms of training households how to use automated teller machines (ATMs) and providing a sufficient number of ATMs in townships. The households also indicated that financial institutions are poorly meeting their expectations, and they perform poorly in terms of providing the expected services and care to households. Financial institutions are poorly accountable and transparent to households, and they

respond poorly to grievances and complaints issued by households. The data also indicate that households are very poorly consulted when financial institutions introduce new technologies. There is also little fairness in terms of bank charges.



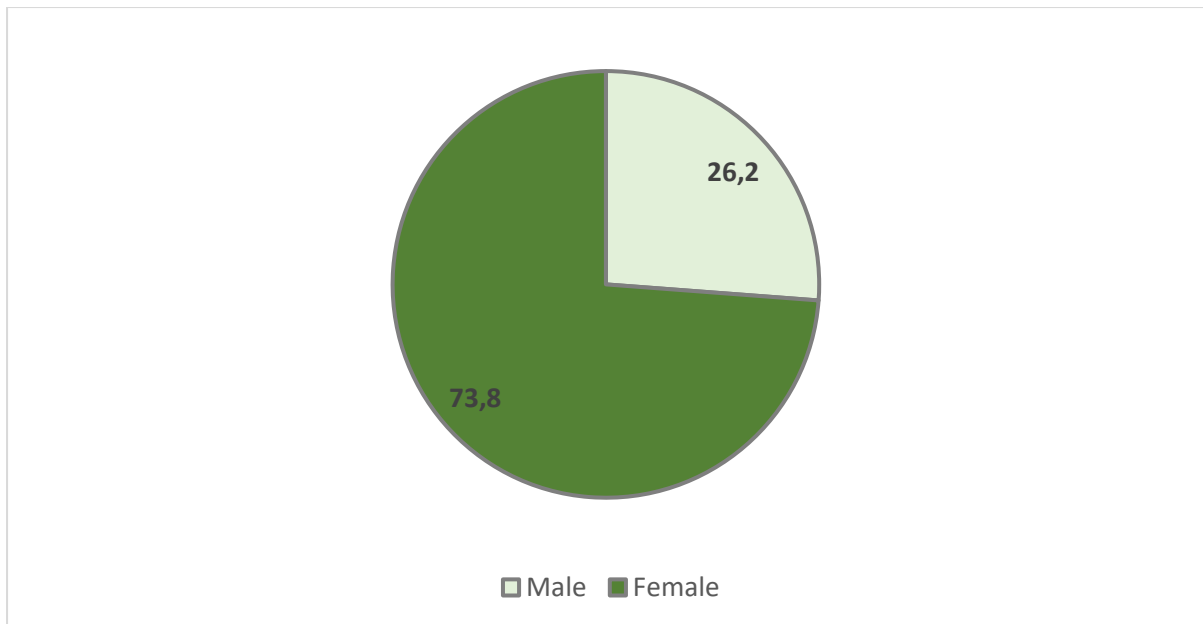
**Figure 4.4: Financial institutions' offers of personal and collective services to township households**

## 4.7 DESCRIPTIVE ANALYSIS OF VARIABLES NOT INCLUDED IN THE MULTIPLE LOGISTICS AND LINEAR MODELS

This section provides a descriptive analysis of the variables that were not included in the multiple logistics and multiple linear models.

### 4.7.1 Township household head's gender

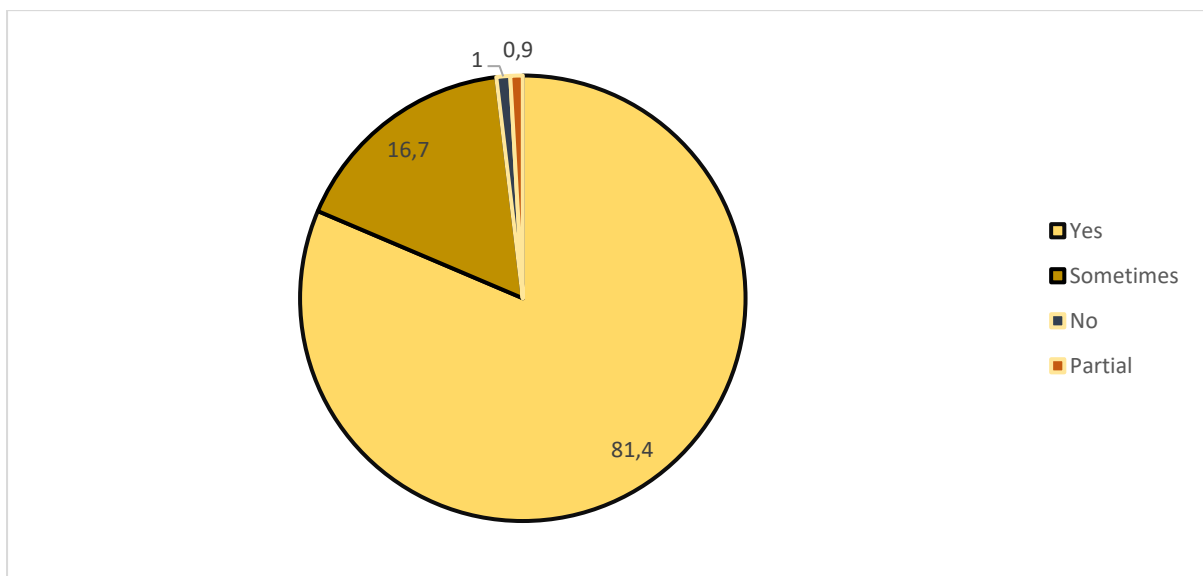
Figure 4.5 illustrates the gender of household heads in townships, namely that 73.8% of the households indicated that they were headed by females, while only 26.2% were headed by males.



**Figure 4.5: Household gender distribution**

#### **4.7.2 Township household breadwinners**

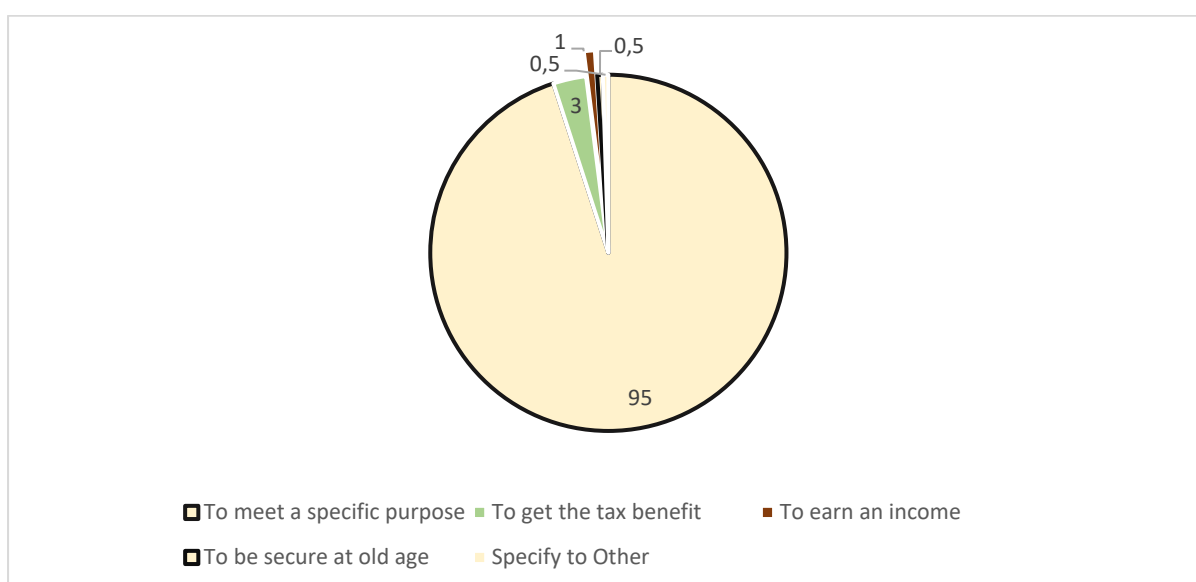
Figure 4.6 illustrates that 81.4% of the participants were breadwinners, followed by 16.7% who indicated that they were sometimes breadwinners (not all the time), 0.9% indicated that they were partial breadwinners, and only 1% indicated that they were not breadwinners.



**Figure 4.6: Household breadwinner distribution**

### 4.7.3 Motivation for household savings

Figure 4.7 indicates households' motivation to save, where 95% of the households indicated that their motivation to acquire household savings was to meet a specific purpose, followed by 3% of the households that indicated that their motivation was to obtain a tax benefit. Only 1% of the households indicated that their motivation to acquire household savings was to earn an income, while 0.5% of the households indicated that their motivation was for their households to be secure in old age and the other 0.5% indicated that their motivation was for other specific reasons.

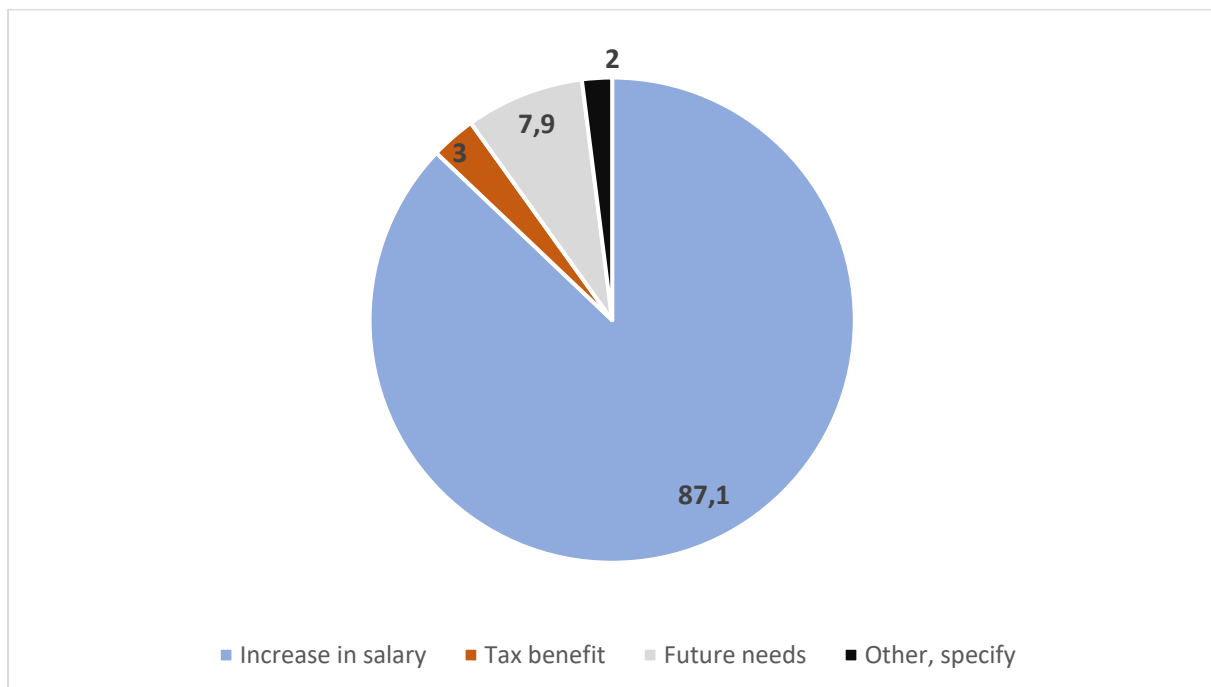


**Figure 4.7: Household saving motivations**

### 4.7.4 Factors that households consider to increase their savings

Figure 4.8 illustrates the factors that households considered to increase their savings. Of the households, 87.1% indicated that an increase in salary was a factor that they considered in order to increase their household savings, while 7.9% of the households indicated that they considered future needs as the reason to increase their savings. Three percent (3%) of the households indicated tax benefits as a factor to increase

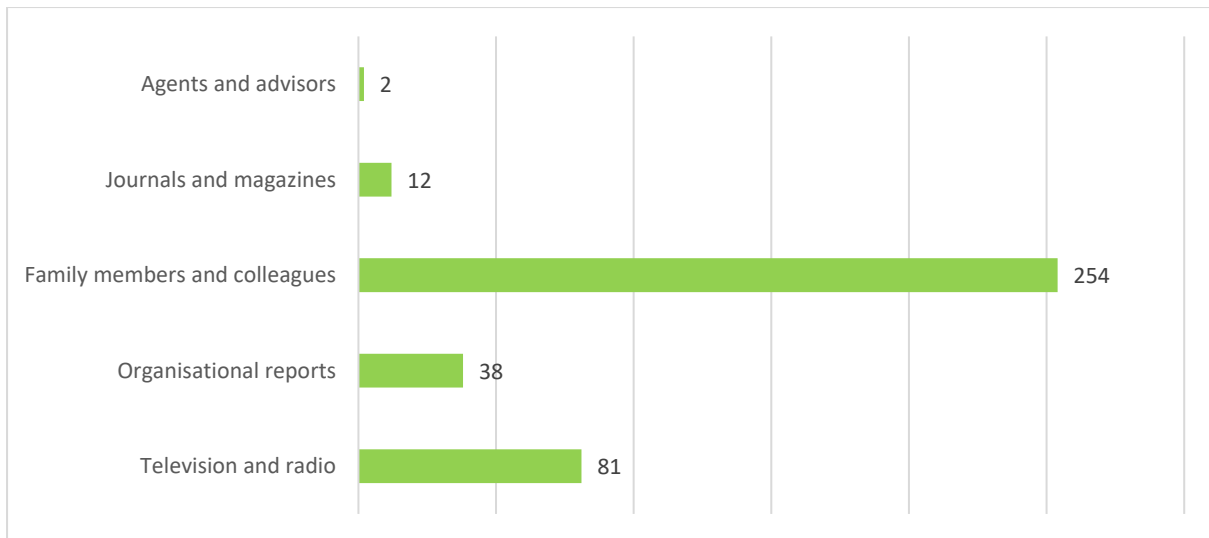
household savings, and 2% indicated that they considered other factors to increase their households' savings.



**Figure 4.8: Factors to consider an increase in household savings**

#### **4.7.5 Source of information on household savings**

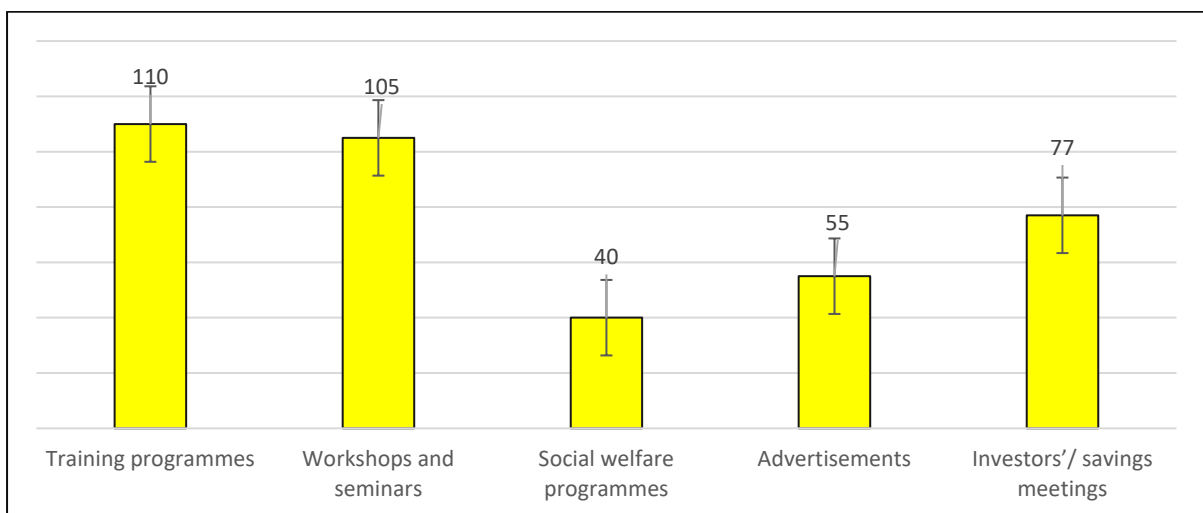
Figure 4.9 indicates the sources of where households obtain information on savings. Of the households, 254 indicated that they obtained information on savings from family members and colleagues, 81 households obtained information on savings from television and radio, 38 households obtained information on savings from organisational reports, while 12 and two households obtained information from journals and magazines and agents and advisors respectively.



**Figure 4.9: Where households obtain information on savings**

#### **4.7.6 Initiatives that households recommend to create awareness of savings**

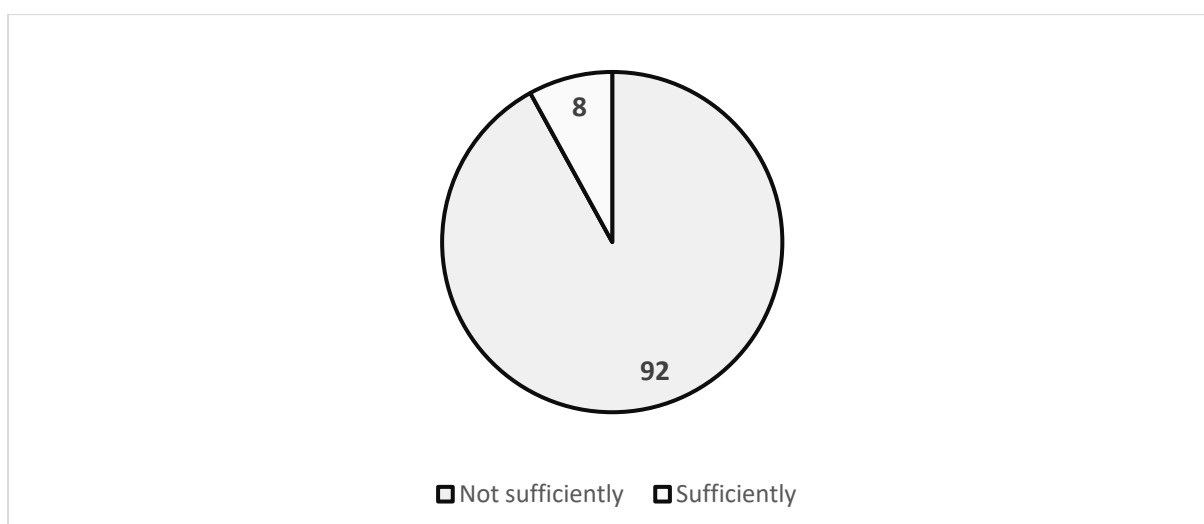
Figure 4.10 presents the initiatives that households recommended to create awareness of savings. One hundred and ten (110) households recommended training programmes as an initiative to create awareness of savings, 105 households recommended workshops and seminars to create household awareness of savings, 40 households recommended social welfare programmes, while 55 and 77 households recommended advertisements and investors'/savings meetings as an initiative to create awareness of savings respectively.



**Figure 4.10: Recommended initiatives to create awareness of savings**

#### **4.7.7 Whether existing household savings ensure households' financial wellbeing**

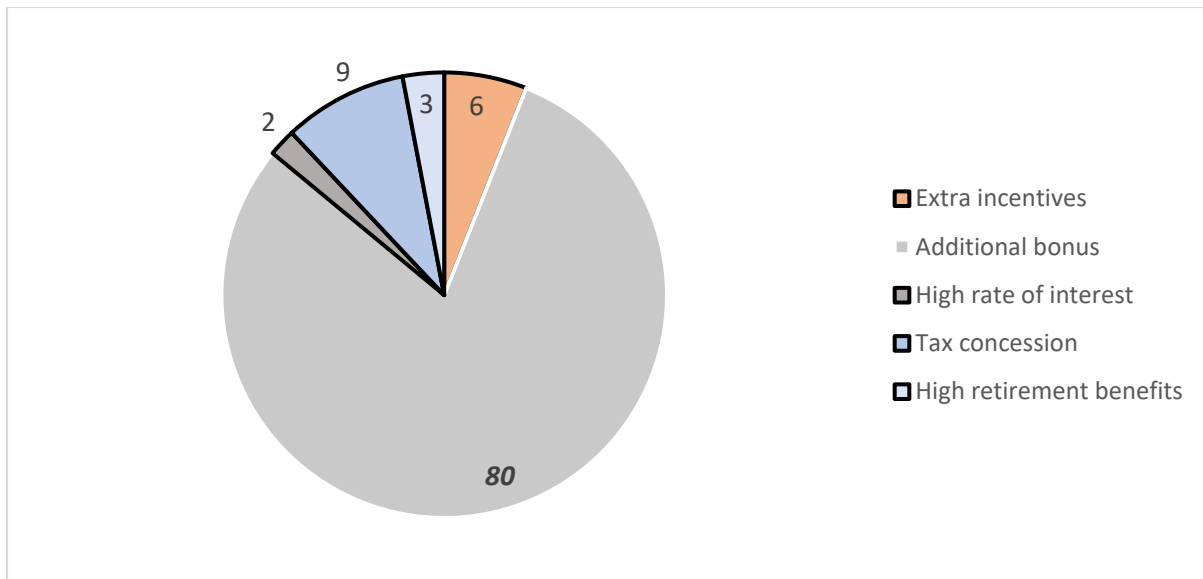
Figure 4.11 indicates whether the existing savings that households might have acquired ensured the households' financial wellbeing. Ninety-two percent (92%) of the households indicated that their existing savings did not sufficiently ensure their households' financial wellbeing, and only 8% of the households indicated that their existing savings sufficiently ensured their financial wellbeing.



**Figure 4.11: Existing household savings ensuring household financial wellbeing**

#### **4.7.8 Special provisions needed by households to boost their savings to ensure financial wellbeing**

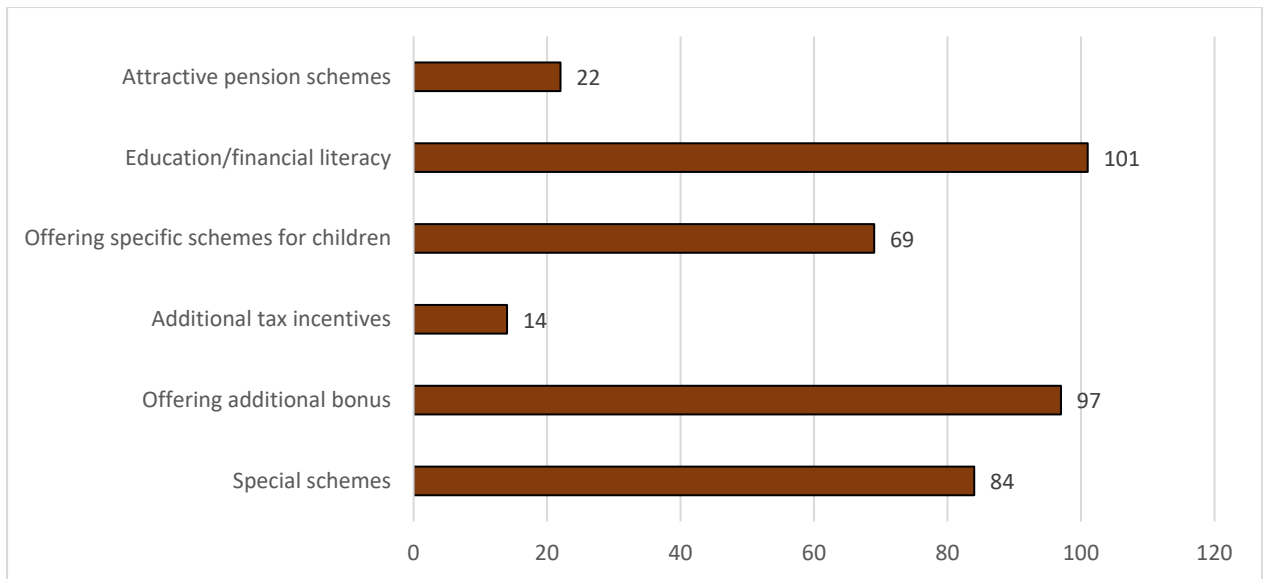
Figure 4.12 illustrates special provisions that households need to boost their savings to ensure their financial wellbeing. Eighty percent (80%) of the households indicated that they required additional bonuses as a special provision to boost their household savings, 9% indicated that they needed tax concessions as a special provision, 6% and 3% indicated that they required extra incentives and high retirement benefits to boost their household savings, and 1% indicated that they required a high interest rate as a special provision to boost their household savings.



**Figure 4.12: Special provisions to boost household savings**

#### **4.7.9 Suggestions offered by households to popularise savings schemes among township households**

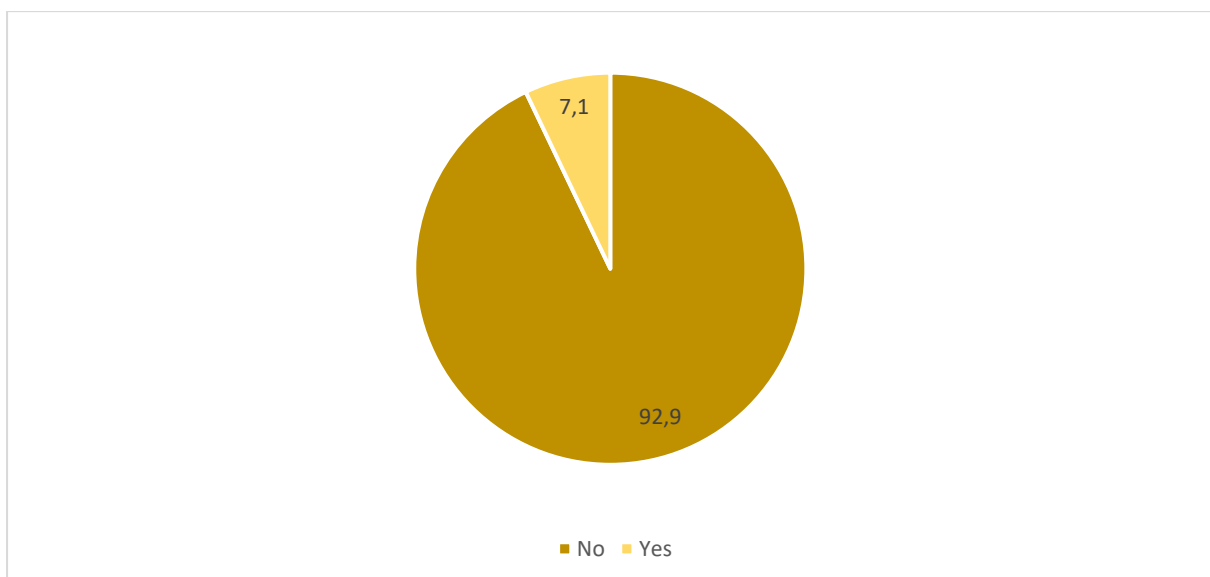
Figure 4.13 presents the suggestions made by the participants to popularise savings schemes among households. One hundred and one (101) households suggested that education/financial literacy can be utilised as a tool to popularise savings schemes among households, 97 suggested offering additional bonuses to popularise savings schemes, 84 indicated that initiating special schemes for households' savings requirements can be utilised as a tool for popularising savings schemes, 69 and 22 indicated that offering special schemes for children and attractive pensions schemes can popularise savings schemes among households respectively, and 14 households indicated that additional tax incentives can popularise savings schemes among households.



**Figure 4.13: Suggestions on popularising savings schemes among households**

#### **4.7.10 Whether savings provide a sense of future independence**

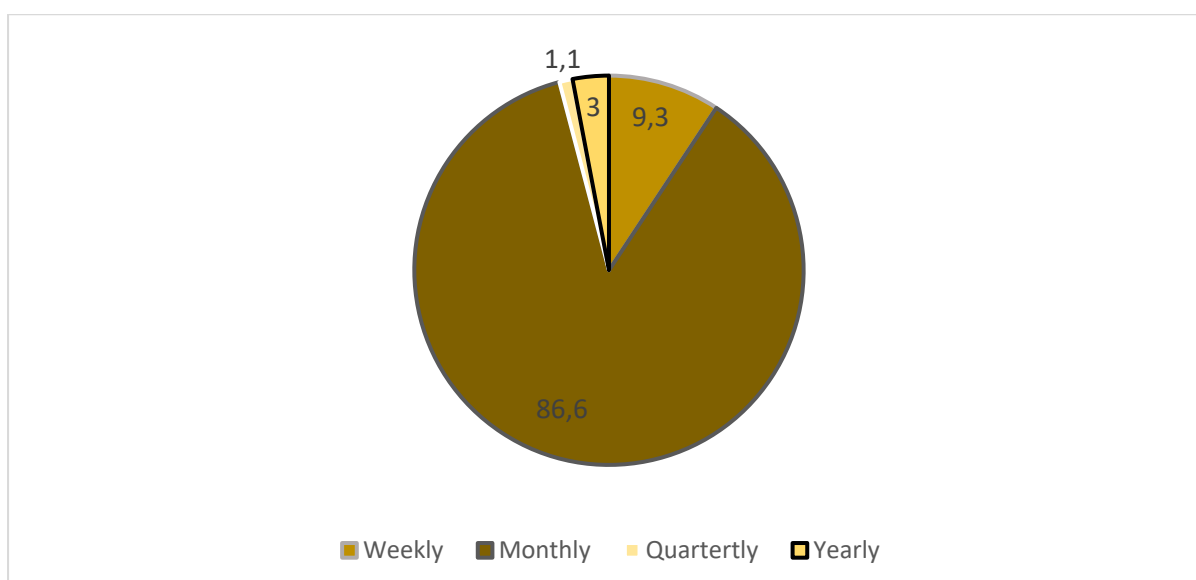
Figure 4.14 illustrates whether their savings gave the households a sense of future independence. It shows that 92.9% of the households indicated that their savings did not give them a sense of future independence, and only 7.1% of the households indicated that their savings gave them a sense of future independence.



**Figure 4.14: Participants' savings and sense of future independence**

#### 4.7.11 Frequency at which households save

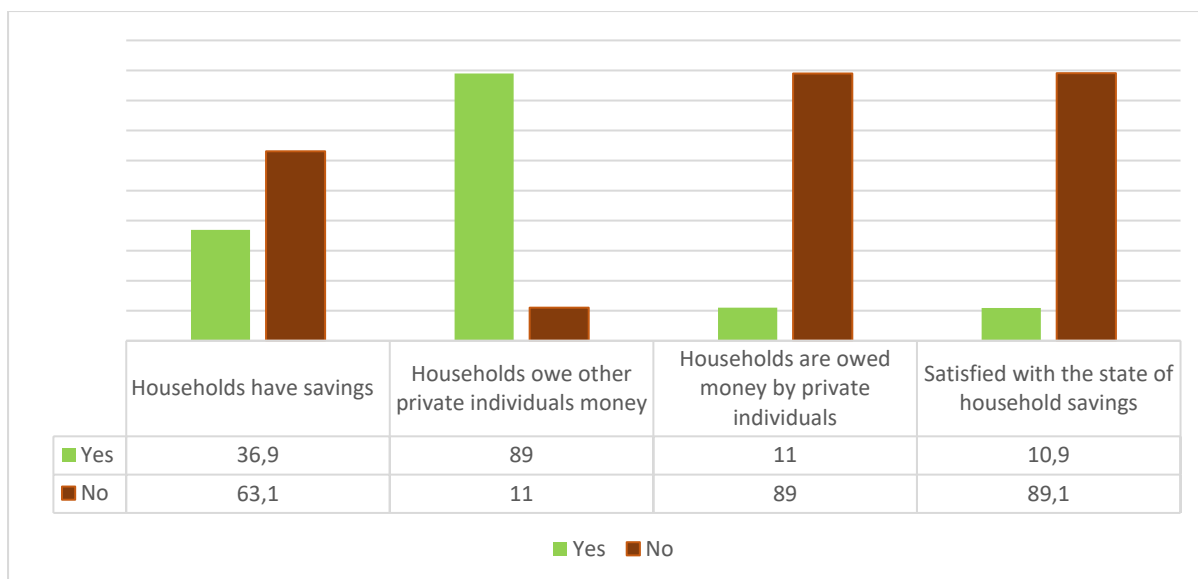
Figure 4.15 shows that 86.6% of the households indicated that they saved monthly, 9.3% saved weekly, and 3% and 1% of the households saved yearly and quarterly respectively.



**Figure 4.15: Frequency of household saving**

#### 4.7.12 Households' savings, debt, and satisfaction with their savings

Figure 4.16 depicts information on households and their savings. Of the households, 63.1% did not have any savings, while 36.9% of the households did; 89% of the households indicated that they owed other private individuals money, while 11% did not; 11% of the households indicated that they were owed money by other private individuals, while the other 89% indicated that they were not owed money by private individuals; and 89.1% of the households indicated that they were not satisfied with the state of their household savings, while 10.9% of the households were satisfied with the state of their household savings.



**Figure 4.16: Households’ savings, debt, and satisfaction with their savings**

## 4.8 NATIONAL-LEVEL (MACRO-ECONOMIC) DATA ANALYSIS

The study also collected secondary data. National-level (macro-economic) data were analysed to further explore household savings for financial wellbeing at the macro level. The purpose of this analysis was to ensure that the suggested innovative ways to encourage household savings for financial wellbeing do not only cater for a specific township with specific needs but also for the nation as well. This data allowed the study to be well informed when making its contribution to NDP Vision 2030, as well as to meet the UN’s SDGs 8 and 10 of promoting sustainable and inclusive economic growth and productivity, employment and decent work for all, and reducing inequalities.

### 4.8.1 The augmented Dickey-Fuller (ADF) and Phillips-Perron unit root tests

Table 4.7 depicts the results of the ADF and Phillips-Perron tests that were conducted to determine the order of integration of the variables because the first step when dealing with time series data is to check stationarity. When testing for stationarity utilising the ADF test, the study found that all the variables were non-stationary in their

levels and they only became stationary after the first difference  $I(1)$ . The study then conducted the second unit root test, namely the Phillips-Perron test, to confirm the results of the ADF test. The Phillips-Perron test confirmed that if variables were not stationary in their levels form, they all became stationary after they were differenced once  $I(1)$ . The two tests confirmed each other, and the results indicated no ambiguity regarding the order of integration. Given the mixed order of co-integration, the study proceeded to test for co-integration using ARDL bound tests in order to observe if the dependent variable was co-integrated with its independent variables; however, this test cannot be run if the dependent variable is stationary in its level form (Pesaran *et al.*, 1996). It is not a problem if the explanatory variable is  $I(0)$  as long as the dependent variable is  $I(1)$ . In addition, the bounds test cannot be run if any of the variables become stationary after second difference  $I(2)$  since the bounds testing assumes that the variables are either  $I(0)$  or  $I(1)$ . The long-run relationship between variables was observed after computing the W-statistic and F-statistic for testing the significance of the lagged variables in their level form in the error correction form of the underlying ARDL model. Yearly data from 1980 to 2021 were used and the system was allowed to choose the number of the lags, and it chose one lag.

**Table 4.7: Results of the unit root tests**

| ADF            |                     |               |              |       | Phillips-Perron     |              |              |       |
|----------------|---------------------|---------------|--------------|-------|---------------------|--------------|--------------|-------|
| Variables      | Trend and intercept | Intercept     | None         | Order | Trend and intercept | Intercept    | None         | Order |
| ADR            | -0.421864           | 1.49E-14      | 1.079206     |       | -0.822534           | 0.958068     | 1.716262     |       |
| P-value        | (0.9827)            | (0.9523)      | (0.9236)     | I(1)  | (0.9550)            | (0.9953)     | (0.9774)     | I(1)  |
| $\Delta$ ADR   | -9.403329***        | -9.502642***  | -9.486833*** |       | -8.389839***        | -6.516307*** | -6.244998*** |       |
| P-value        | (0.0000)            | (0.0000)      | (0.0000)     |       | (0.0000)            | (0.0000)     | (0.0000)     |       |
| GGR            | -5.010552 ***       | -5.135147***  | -3.690691*** | I(0)  | -4.996268***        | -5.124214*** | -3.650522*** | I(0)  |
| P-value        | (0.0011)            | (0.0001)      | (0.0005)     |       | (0.0012)            | (0.0001)     | (0.0005)     |       |
| HDEBT          | -3.307911           | -1.915651     | 0.305150     |       | -2.009470           | -1.653677    | 0.666568     |       |
| P-value        | (0.0796)            | (0.3220)      | (0.7690)     |       | (0.5789)            | (0.4467)     | (0.8560)     |       |
| $\Delta$ HDEBT | -3.786614**         | -3.744163 *** | -3.733645*** | I(1)  | -2.957383*          | -2.950807**  | -2.986679*** | I(1)  |
| P-value        | (0.0280)            | (0.0071)      | (0.0004)     |       | (0.0512)            | (0.0485)     | (0.0038)     |       |
| HDINCM         | -6.045472***        | -6.187016***  | -3.966301*** | I(0)  | -6.044970***        | -6.188271*** | -3.966301*** |       |
| P-value        | (0.0001)            | (0.0000)      | (0.0002)     |       | (0.0001)            | (0.0000)     | (0.0002)     | I(0)  |
| INFL           | -1.131865           | -0.768603     | -1.620419    |       | -2.921920           | -1.437991    | -1.552086    |       |
| P-value        | (0.9096)            | (0.8162)      | (0.0983)     |       | (0.1665)            | 0.5544       | 0.1121       |       |
| $\Delta$ INFL  | -5.580962***        | -3.157151**   | -2.790619*** | I(1)  | -10.76642***        | -9.360456*** | -6.271525*** | I(1)  |
| P-value        | (0.0003)            | (0.0311)      | (0.0066)     |       | (0.0000)            | (0.0000)     | (0.0000)     |       |
| RHP            | -3.335263           | -2.102131     | -0.211797    |       | -1.666683           | -1.305679    | -0.035403    |       |
| P-value        | (0.0752)            | (0.2449)      | (0.6035)     |       | (0.7479)            | 0.6179       | (0.6651)     |       |
| $\Delta$ RHP   | -3.716347**         | -3.655103***  | -3.700797*** | I(1)  | -3.488224*          | -3.445181**  | -3.501957*** | I(1)  |
| P-value        | (0.0330)            | (0.0089)      | (0.0005)     |       | (0.0544)            | (0.0150)     | (0.0009)     |       |
| RIR            | -5.015072***        | -5.340215***  | -2.666730*** | I(0)  | -4.999314***        | -5.280675*** | -2.826856*** | I(0)  |
| P-value        | (0.0011)            | (0.0001)      | (0.0089)     |       | (0.0012)            | (0.0001)     | (0.0058)     |       |
| HSR            | -2.026855           | -1.188047     | -2.757260    |       | -3.173930           | -2.730131    | -2.769031*** |       |
| P-value        | 0.5688              | 0.6700        | 0.0070       |       | (0.1038)            | (0.0777)     | (0.0068)     |       |
| $\Delta$ HSR   | -6.210968***        | -5.818906***  | -5.924528*** | I(1)  | -8.495179***        | -8.743470*** | -8.720148*** | I(1)  |
| P-value        | (0.0001)            | (0.0000)      | (0.0000)     |       | (0.0000)            | (0.0000)     | (0.0000)     |       |

HDEBT = Household debt; HDINCM = Household disposable income; HSR = Household savings ratio. Note: The null hypothesis for the ADF and the Phillips-Perron is that the variable contains a unit root, while the alternative hypothesis is that the variable is stationary. Rejection of the null hypothesis implies that the variable is non-stationary.  $p < 0.001^{***}$ ,  $p < 0.05^{**}$ ,  $p < 0.10^*$ .

#### 4.8.2 The co-integration tests

The co-integration test hypothesis is as follows:

- $H_0: \delta_1 = \delta_2 = 0$  (Long-run relationship does not exist)
- $H_1: \delta_1 \neq \delta_2 \neq 0$  (Long-run relationship exists)

The bound test results are presented in Table 4.8. By using the critical value bounds as computed by Pesaran, Shin and Smith (2001), the F-statistic is greater than the upper bound critical value at the 10%, 5%, 2.5%, and 1% levels. The null hypothesis of no co-integration is thus rejected and the long-run relationship therefore exists.

**Table 4.8: Results of the bound test**

| Dependent variable: HSR |          |              |      |      |
|-------------------------|----------|--------------|------|------|
| Test statistic          | Value    | Significance | I(0) | I(1) |
| F-statistic             | 5.697443 | 10%          | 1.95 | 3.06 |
| K                       | 8        | 5%           | 2.22 | 3.39 |
|                         |          | 2.5%         | 2.48 | 3.7  |
|                         |          | 1%           | 2.79 | 4.1  |

\*Critical values were obtained from Pesaran *et al.* (2001)

#### 4.8.3 The autoregressive distributed lag (ARDL) model

##### 4.8.3.1 The ARDL model estimates

As indicated in Table 4.9, the results showed that the household savings ratio (-1), GDP growth rate, household debt, household disposable income, household disposable income (-1), inflation, real house price, and real house price (-1) were statistically significant in explaining household savings for financial wellbeing, while old age dependency was not statistically significant in explaining household savings for financial wellbeing. The diagnostic test revealed that the model was a good model. In Table 4.9, the dummy transvariable was statistically significant. This indicates that

the structural break from the apartheid regime and post-apartheid government had an impact on household savings. These structural breaks had an impact on the financial wellbeing of households.

**Table 4.9: The ARDL model estimates**

| ARDL model estimates (1, 0, 0, 0, 0, 0, 1, 0, 0) |             |           |           |
|--------------------------------------------------|-------------|-----------|-----------|
| Variable                                         | Coefficient | T- value  | P-value   |
| HSR (-1)                                         | 0.514528    | 3.840727  | 0.0006*** |
| ADR                                              | 0.046220    | 0.185218  | 0.8543    |
| GGR                                              | -0.239360   | -2.711969 | 0.0110*** |
| GGR (-1)                                         | -0.062443   | -2.015424 | 0.0529*** |
| HDEBT                                            | 0.273355    | 3.562012  | 0.0013*** |
| HDINCM                                           | -0.085624   | -1.186847 | 0.2446    |
| HDINCM (-1)                                      | -0.114095   | -4.526303 | 0.0001*** |
| INFL                                             | 0.107476    | 3.836727  | 0.0006*** |
| RHP                                              | 0.041679    | 0.936892  | 0.3563    |
| RHP (-1)                                         | -0.913283   | -1.192235 | 0.2425    |
| RIR                                              | 5.130902    | 1.812512  | 0.0799*   |
| DUMMY<br>TRANSV.                                 | 0.514528    | 3.840727  | 0.0006*** |
| C                                                | 0.046220    | 0.185218  | 0.8543    |
| F-statistic 32.41156 (0.000000)***               |             |           |           |
| Durbin-Watson stat 2.172181                      |             |           |           |
| Jarque- Bera 0.216185 (0.897544)                 |             |           |           |

NB: p-values in parenthesis; \*, \*\*, and \*\*\* indicate the level of significance at 10%, 5%, and 1% respectively.

#### 4.8.4 The long-run coefficient and the error correction model (ECM)

Table 4.10 shows the findings of the ECM of the (1, 0, 0, 0, 0, 0, 1, 0, 0), including all the model statistics that are relevant to this research. The study found that the speed of adjustment or the error correction coefficient (ECT) was highly significant and held the expected sign. The ECT was also within the range of (0 to -1), which removes doubts of the model being misspecified. The dummy transvariable was used to explain the transition from the apartheid government to the post-apartheid government, where

the study found that it was not significant in explaining household savings. The old age dependency and the dummy transvariable therefore do not explain their impact on household savings. In the long run, the GDP growth rate, household debt, inflation and real house price were statistically significant, with a negative relationship with household savings.

**Table 4.10: Long-run coefficients and the ECM for the ARDL model**

| <b>Long-run coefficients and ECM for the ARDL model (1, 0, 0, 0, 0, 0, 1, 0, 0)</b> |                    |                   |                    |                  |
|-------------------------------------------------------------------------------------|--------------------|-------------------|--------------------|------------------|
| <b>Variable</b>                                                                     | <b>Coefficient</b> | <b>Std. error</b> | <b>T-statistic</b> | <b>Prob.</b>     |
| D (ADR)                                                                             | 0.046220           | 0.249546          | 0.185218           | 0.8543           |
| D (GGR)                                                                             | -0.239360          | 0.088261          | -2.711969          | 0.0110***        |
| D (HDEBT)                                                                           | -0.062443          | 0.030982          | -2.015424          | 0.0529***        |
| D (HDINCM)                                                                          | 0.273355           | 0.076742          | 3.562012           | 0.0013***        |
| D (INFL)                                                                            | -0.085624          | 0.072144          | -1.186847          | 0.2446           |
| D (RHP)                                                                             | -0.114095          | 0.025207          | -4.526303          | 0.0001***        |
| D (RIR)                                                                             | 0.041679           | 0.044486          | 0.936892           | 0.3563           |
| DUMMY TRANSV.                                                                       | -0.913283          | 0.766027          | -1.192235          | 0.2425           |
| <b>ECM</b>                                                                          | <b>-0.485472</b>   | <b>0.133966</b>   | <b>-3.623843</b>   | <b>0.0011***</b> |

#### **4.8.5 Diagnostic tests**

As shown in Table 4.11, the diagnostics and the Lagrange multiplier (LM) test indicated the absence of autocorrelation at two lags, while there was no evidence of heteroskedasticity as indicated by the White test. The Jarque-Bera normality test suggests that the residuals were normally distributed.

**Table 4.11: Results of the diagnostic tests**

| Test                    | $H_0$                                  | Test statistics           | p-value  | Conclusion                                                                   |
|-------------------------|----------------------------------------|---------------------------|----------|------------------------------------------------------------------------------|
| Jarque-Bera             | $H_0$ : Normally distributed residuals | 0.216185                  | 0.897544 | Failure to reject $H_0$ . Residuals are normally distributed.                |
| Ljung-Box Q             | $H_0$ : No serial correlation          | (6) = 6.4920              | 0.370    | Failure to reject $H_0$ . No first-order serial correlation up to sixth lag. |
| Breusch-Godfrey LM Test | $H_0$ : No serial correlation          | $nR^2(2) = 2.003151$      | 0.9597   | Failure to reject $H_0$ . No autocorrelation up to second order.             |
| ARCH-LM                 | $H_0$ : No heteroscedasticity          | $nR^2(2) = 0.229106$      | 0.8918   | Failure to reject $H_0$ . No heteroscedasticity up to second order.          |
| White                   | $H_0$ : No heteroscedasticity          | $nR^2$ (no CT) = 8.882128 | 0.5433   | Failure to reject $H_0$ . No heteroscedasticity.                             |
| Ramsey RESET            | $H_0$ : No misspecification            | $LR(1) = 1.541235$        | 0.2144   | Failure to reject $H_0$ . No misspecification.                               |

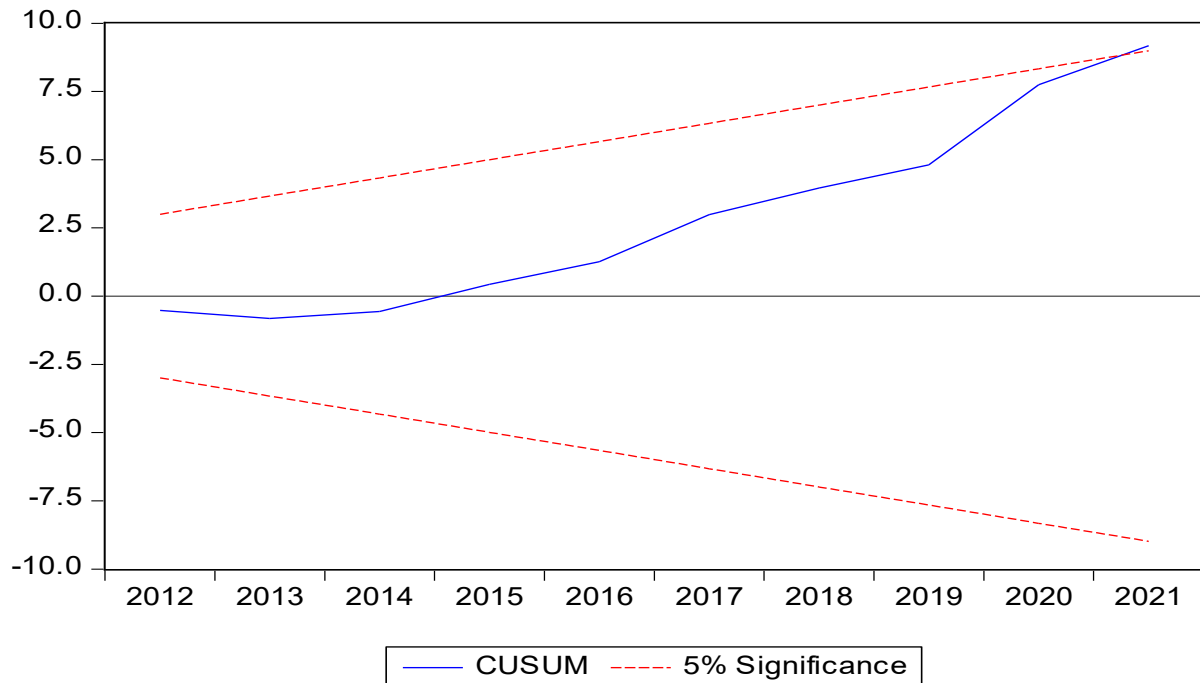
RESET = Regression equation specification error test

#### 4.8.6 Stability tests

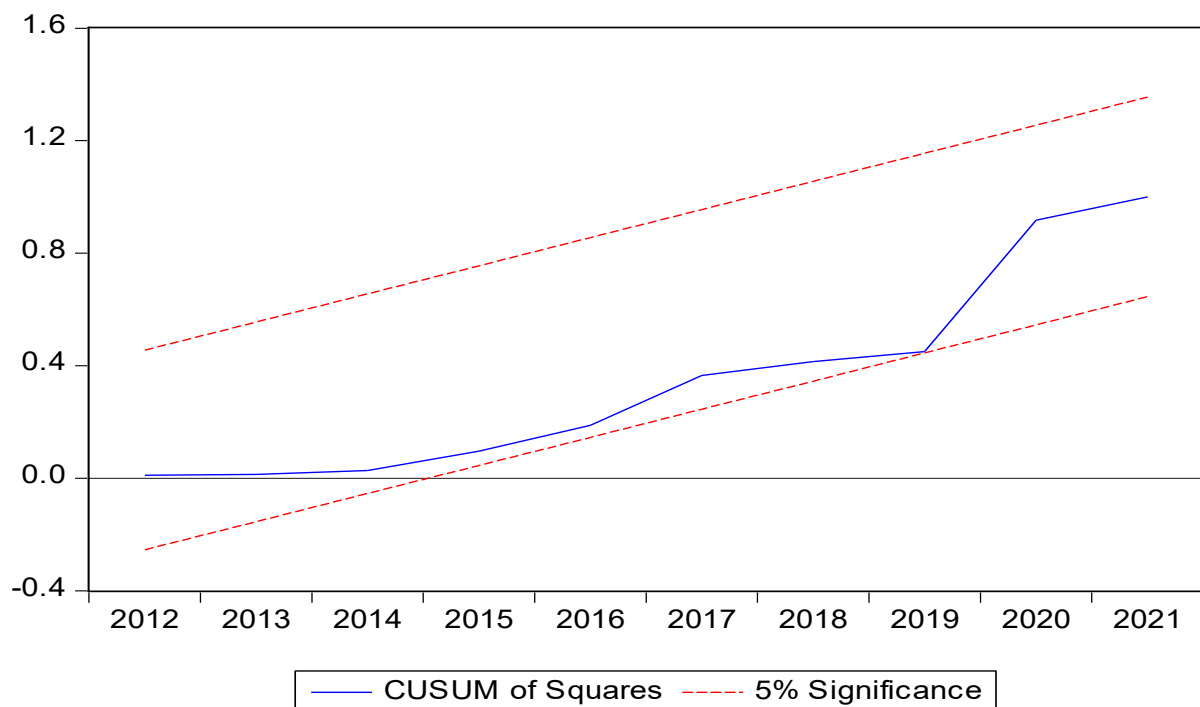
Various tests are available to test for stability. The most popular test is the cumulative sum of recursive residuals (CUSUM) and the cumulative sum of squared recursive residuals (CUSUMSQ) tests. The CUSUM test is based on the plot of the sum of the recursive residuals. If the sum of the recursive residuals is located on the outside of the critical bounds, it can be concluded that there is a structural break at the point to which the sum of the recursive moves towards the bound. The CUSUMSQ test functions similarly to the CUSUM test but the test plots the squared sum of recursive residuals. Both the CUSUMSQ and CUSUM tests were used to determine the parameter stability once the ECM model had been estimated.

As illustrated in Figures 4.17 and 4.18, the stability of the model was tested using the CUSUM and CUSUMSQ tests. Both the CUSUM and CUSUMSQ tests indicated that

the model was stable because the calculated blue line fell within the critical bounds of the 5% level of significance. The study therefore concluded that the model was statistically stable and there were no structural breaks.



**Figure 4.17: Cumulative sum of recursive residuals (CUSUM) test**



**Figure 4.18: Cumulative sum of squared recursive residuals (CUSUMSQ) test**

## **4.9 OVERALL DETAILED DISCUSSION OF ALL THE RESULTS AS PER THE LITERATURE REVIEW AND OBJECTIVES**

### **4.9.1 Discussion of the multiple logistic and linear regression model findings**

The results of the multiple logistic and linear regression modelling indicated that socio-demographics have an influence on household savings. Based on the literature review, these results contradict the findings that many households in the townships do not have savings, irrespective of their socio-economic status (OECD, 2020:1) and the findings concur with Lusardi *et al.*'s (2011:1) findings by confirming that although financial fragility and lack of household savings are more severe among low-income households with low educational attainment, families with many dependants, and the unemployed, both the multiple logistic and linear models found that the educational level and the monthly income range had an impact on household savings. The study's findings based on the multiple logistic and linear models agree with the literature (Frisancho, 2016; Aydermir, 2021; Reyers, 2019) on households and financial literacy by supporting the importance of financial knowledge as a factor that influences household savings. Based on the findings, the study emphasised the significance of encouraging household savings, especially in the area of financial knowledge. Financial knowledge is important for all South Africans to assist in alleviating financial distress, as it may also assist in increasing household income in times of need. When people start working at an early age and are better informed financially, their life cycles depict better financial results. This was also confirmed by literature on the theoretical framework of the life cycle hypothesis (Modigliani & Brumberg, 1954), which was established in the 1950s and is still utilised to interpret people's lifespan conduct in terms of their savings and spending. Matuku and Kaseke (2014:509) and Coetzee and Cross (2002:12) conducted research on informal community savings and credit groups

on a cyclical or rotational basis and found that stokvels are important self-help strategies that assist with tackling challenges such as income inequality and poverty in various communities. This study's findings also indicated that households that participate in stokvels have more savings than those who do not participate in stokvels for various reasons. The study also found that family size, household satisfaction with household savings, type of household priority savings, households that prefer a state-owned bank, and household consumption during the COVID-19 pandemic were significant variables that influence household savings. Kibet *et al.*'s (2017:137) findings concur with this study's findings on the importance of the dependency ratio as a contributor to household savings. The literature (Saqid *et al.*, 2016:54; Nghiem Le *et al.*, 2022:42) also indicated the importance of employment and age as the most positive factors that contribute to household savings.

#### **4.9.2 Discussion of the findings on financial institutions (commercial banks)**

The overall findings on commercial banks were that they performed poorly in terms of offering personal and collective services to households in townships, how they service households in terms of various socio-economic issues, and how they communicate with and provide educational services to households in townships. There are no initiatives to encourage individuals and households to save; instead the banks promote indebtedness to people. Roberts *et al.*'s (2018) findings are congruent with this study's findings that overindebtedness remains an issue, with more than 50% of South Africa's credit-active consumers being regarded as overindebted. Since 2009, the value of consumer credit increased steadily as more individuals acquired credit. However, various sources such as the National Treasury (2021a:7) and Nawabzada (2017:4) valued the importance of commercial banks and indicated that they are an effective

and large financial sector in an emerging economy relative to other developing countries' financial sectors.

#### **4.9.3 Discussion of the results of the national-level data analysis**

The overall findings from the analysis of national-level data (secondary data) indicated a positive relationship between household disposable income and household savings. Ralarala and Masipa (2021:28), Chitema and Chitongo (2020), Zwane *et al.* (2016:215), and Simleit *et al.* (2011) conducted research on determinants of household savings, and their findings corroborate this study's findings. The dummy transitional (dummy trans) variable was not able to explain its impact on household savings in the long run. In the long run, the GDP growth rate, household debt, inflation rate, and real house prices were statistically significant and indicated a negative relationship with household savings. Based on the literature review, Simleit *et al.*'s (2011) findings corroborate this study's finding that household savings in South Africa are influenced by factors such as interest rates as they cause household savings to decline. Drawing observations from other developing countries with similar traits as South Africa, using the case of Kosovo in the study by Ribaj and Mexhuani (2021) on savings and economic growth in developing countries, their study also indicated a long-run negative relationship between household savings and the GDP growth and inflation rates. This study's findings also indicated a negative relationship between household savings and household debt. The OECD's 2022 report concluded that households with assets save more than households with liabilities. Joubert and Van der Merwe (2021:8-10) also indicated that households with wealth were in a better position to acquire household savings than households with debt. Prinsloo (2002:63) defined household liabilities as obligations or responsibilities arising from borrowing

money or taking goods or services on credit against an obligation to pay later and concluded by concurring with the OECD (2022) regarding the negative relationship between household debts and household savings.

## **CHAPTER 5:**

### **SUMMARY, CONCLUSION, AND POLICY IMPLICATIONS**

#### **5.1 OVERALL SUMMARY**

This study sought to craft innovative ways that can be used to encourage household savings for financial wellbeing at the micro level using Atteridgeville Township as the case study, as well as at the national level. To achieve this main objective, the study pursued secondary objectives. One of these objectives was that the study analysed the extent to which savings can be improved among households with different perceptions and demographic characteristics in the study area in order to identify innovative factors that promote household savings, to construct innovative policy inputs to encourage household savings for financial wellbeing in the study area and beyond and to perceive the effect of commercial banks/financial institutions on household saving . The study was able to meet all its objectives, which is mainly because the study collected both micro- and macro-level data. The researcher also conducted focus group discussions as an additional data-collection method to allow the study to achieve its objectives and to craft innovative ways to encourage household savings for household and national financial wellbeing.

Based on the macro- and micro-level data and the focus group discussion outcomes, the study discovered that there is indeed a lack of household savings in the country. The households that demonstrated some level of savings noted that their savings have been declining at all levels. The study also examined the active role of commercial banks in assisting township households to save. The findings indicated very poor involvement by commercial banks, poor communication and explanations of their

products and services, and a lack of educational services offered to households, such as financial literacy programmes.

The secondary data were analysed using the ARDL model. The outcomes were congruent with the outcomes of the questionnaires and the focus group discussions, which all indicated the huge challenge of a lack of household savings in South Africa and confirmed the need for this study to provide solutions by crafting innovative ways to encourage household savings for financial wellbeing.

## **5.2 INNOVATIVE WAYS TO ENCOURAGE HOUSEHOLD SAVINGS**

This section presents innovative ways to encourage household savings for financial wellbeing at both the macro and micro levels and as a roadmap to achieve the NDP's mandate on savings by 2030. These are the main contribution that this study makes to the existing body of knowledge.

Many households indicated the need for a state-owned commercial bank. The study therefore suggests the introduction of a state-owned commercial bank as an innovative way to encourage household savings. Given the need for the government, through the relevant national department, to construct innovative policy inputs to encourage household savings for their financial wellbeing, the state-owned bank should offer packages that cater for households in townships that work in different sectors and earn different salaries, including those earning less than South Africa's national minimum wage rate. The FDIC (2020:9) illustrated the strength of initiatives that are channelled to cater for community needs as a key method to support community/local small businesses and to enable a saving culture in those communities. Comes *et al.* (2018:238), Mizrak and Dastan (2018:235), and Steinert *et al.* (2018:240) found that the availability of financial products through credit

encourages household savings. The introduction of a state-owned bank will reduce household dependency on credit and enable and encourage household savings as the state-owned bank will offer products that cater to specific community needs.

As indicated in Chapter 4, the study discovered that financially literate households are more likely to have household savings than those that are not financially literate. The NPC (2012:297) also emphasised that building national capabilities requires quality ECD. This study identified the need to introduce a financial literacy curriculum into ECD as an innovative way to encourage household savings for financial wellbeing in the long term. This will allow children to develop an understanding and a habit of savings at an early age. Such educational programmes should also be expanded to cater for the secondary and tertiary educational sectors. The study findings indicated that township households are eagerly awaiting these programmes. Households and community members can be encouraged to participate in financial literacy programmes by awarding them certificates of attendance and completion. Frisancho (2016:12) conducted research on the relationship between household savings and financial literacy, which emphasised the importance of encouraging household savings, especially among low-income households as it contributes to urgent financial relief. Reyers (2019:336) emphasised that financial literacy is a remedy that can result in household savings that will enable households to understand and be in charge of their finances. Sibanda and Sibanda (2016:10) found that financial literacy skills are limited in South Africa.

Another innovative way to encourage household savings for financial wellbeing is that the government, through the National Treasury, should allocate incentives for household savings. These incentives can be implemented by providing tax incentives

to saving households. According to Cole *et al.* (2014), PLS products should be introduced to South Africa to initiate and encourage savings by low-income households. The National Treasury currently offers Treasury bonds that start from R1 000. Many township households are not aware of this particular savings product. The data also indicated that, on average, households save R1 000. There is a need to create awareness of currently existing products that can be utilised to encourage households to save for financial wellbeing. The study identified the need for educational seminars/workshops on the importance of savings and categories of household savings that households can gain access to, based on their needs, as an innovative way to encourage household savings. Aydermir (2021) found that the majority of women, pertaining to their savings and financial behaviour, were significantly impacted by awareness of and education on savings.

Financial institutions, which were defined as commercial banks in this study, should cater for township households by offering them services specific to townships and their traits. Commercial banks should incentivise households to encourage them to save as a way of giving back to previously disadvantaged communities and correcting the damages caused by the apartheid era, as indicated in the focus group discussion findings. The problems of low household savings are partly attributed to the colonial disenfranchisement policies that excluded a high percentage of the general public from the mainstream economy and barring them from formal savings instruments (Zwane *et al.*, 2016:209). Kloppers and Pienaar (2014:692) also indicated that apartheid policies were detrimental to many people's livelihoods, primarily black people, and plundered their higher-yielding assets, especially livestock and land.

As noted in the study findings, the sector in which households work has an impact on their savings. Therefore, at the macro level, the government and its stakeholders should develop a policy that enforces workplaces to enable compulsory household savings packages for the households they employ, irrespective of the nature of the employment contract. For example, workplaces should enable all employees to save a portion of their income. The workplace can work together with the employees to ensure that every employee saves at least 1% of their salary, which should exclude employees' provident or pension funds. This should be compulsory and employees should be able to access their savings based on the nature of the savings package they have; for example, festive/Christmas savings, school savings, emergency savings, and other vital types of household savings. In 2021, South African savings for households was 2% of the GDP (National Treasury, 2021a:3). This 2% predominately comprised compulsory savings in the form of retirement funds. The SARB (2022:1) also indicated that the only savings that many households are able to have are compulsory savings.

Another innovative way to encourage household savings for financial wellbeing at both the macro and micro level is the need for advertisements on various platforms on various household savings packages. The advertisements can be on billboards in townships, and on community and national radio and television stations.

The study emphasised, based on its findings, that households are willing to acquire household savings and are keen to be taught about savings so that they can cultivate a saving culture and the discipline that will allow them to save. Township households are willing to allow themselves to be trained and to participate in financial literacy training programmes if offered to them.

### **5.3 CONCLUSION**

The South African economy is confronted with financial hardships and is characterised by unemployment, poverty, and inequality. NDP Vision 2030 was developed with the objective for the country to annihilate and lessen inequality by 2030 through uniting South Africans, unleashing the energies of its citizens, growing an inclusive economy, building capabilities, and enhancing the ability of the state and leaders to work together to solve complex problems (NPC, 2012:1). The NDP has the goal of resolving South Africa's lack of savings at both the macro and micro levels and indicates that by 2030, South Africa should have a comprehensive system of social protection that includes social security grants, mandatory retirement savings, risk benefits (such as unemployment, death, and disability benefits), and voluntary retirement savings (NPC, 2012:53). As indicated in the literature review, and confirmed by the study's findings, the only savings that the citizens are able to point out are compulsory savings enabled by their workplaces. The study concluded that the NDP is currently failing to fulfil its goal of increasing national savings from 16% of the GDP to 25% (NPC, 2012:69), and it is unable to meet its goals of creating incentives that will encourage a culture of individual saving for various reasons (NPC, 2012:73). This study's findings can be adopted by the government and its policymakers to fulfil the mandate of improving savings at both the macro and micro levels. An increase in savings will reduce the country's reliance on foreign capital inflows, which reinforce the oligopolistic nature of the economy since foreign investors invest in existing high-profit firms. Skills constraints push up the premium for skilled labour, which induces a large gap between the salaries of skilled and unskilled people and further increases levels of inequality. According to the NDP, the rate of investment to GDP is expected to increase from 17% to 30% by 2030 (NPC, 2012:127). This will happen incrementally, with substantial

contributions from a gradually expanding pool of domestic savings, of which household savings are also part. This study therefore has positive implications, as it is a tool that can be used to achieve South Africa's NDP 2030 goals, and it can also contribute to financial inclusion and to the UN's SDGs 8 and 10 of promoting sustainable and inclusive economic growth, productivity, and decent work for all, and reducing inequalities. It is not surprising that the promotion of household savings in developing countries, especially in South Africa, is both vital and important. Simleit *et al.* (2011:10) also acknowledged the importance of the role of the government in boosting savings by indicating that the presence of a partial offset between household savings and government savings also has important implications for the effectiveness of using the fiscal position of the South African government to boost savings.

#### **5.4 POLICY IMPLICATIONS**

The outcomes of this study, in the form of innovative ways to encourage household savings for financial wellbeing at both the household (micro) and national (macro) levels, are very significant and may be adopted by policymakers as a tool that they can utilise to comprehend the root causes of low, and in many instances lack of, household savings. Policymakers can also adopt these innovative ways to encourage household savings for household and national financial wellbeing as a vehicle to achieve the goals of the NDP in terms of increasing household and other types of savings. This study can therefore be adopted and debated by the National Economic Development and Labour Council as it is a vehicle where the government, labour, businesses, and community organisations meet with the aim to cooperate, through problem solving and negotiation, on economic, labour and development issues and related challenges facing the country.

## **5.5 SUGGESTIONS FOR FURTHER RESEARCH**

The study was able to achieve all its objectives. However, the study's main limitation was finances, which limited the study to collecting primary data in only one township. It is therefore suggested that further research be conducted that focuses on other townships. A follow-up cross-sectional study can also be conducted in the Atteridgeville township to identify areas of improvements that the households might have recorded after this study was conducted. The study used a dummy variable as the transitional variable to define the structural break in household savings and financial wellbeing before and after the apartheid era in South Africa. The study suggests that more variables be adopted to explain the structural break.

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# ANNEXURES

## Annexure 1: Questionnaire



### QUESTIONNAIRE ON HOUSEHOLDS AND SAVINGS

The purpose of this questionnaire is to obtain relevant information that will assist in crafting innovative ways to encourage household savings for financial wellbeing in townships. The questionnaire has **FOUR** sections and it should take between 25 and 30 minutes to complete. Please note that this is an anonymous and confidential questionnaire; therefore you cannot be identified. The participation is strictly voluntary; no one should be forced to participate and no penalty can be imposed upon withdrawal. It is very important for us to learn your opinions. Your survey responses will be strictly confidential and will be used for statistical analysis only. The data from this research will be reported only in the aggregate. Your information will be coded and will remain confidential.

**Instructions:** All sections are vital to ensure innovative ways to encourage savings are identified. If you do not know exact answers, please give your estimations.

## SECTION 1: DEMOGRAPHIC DETAILS

(Please mark the correct answer with an X)

|                                                                          |                |               |                   |                 |           |                        |       |
|--------------------------------------------------------------------------|----------------|---------------|-------------------|-----------------|-----------|------------------------|-------|
| 1. What is your gender?                                                  |                |               |                   | Male            | Female    | Prefer not to disclose |       |
| 2. Are you a breadwinner?                                                |                |               |                   | Yes             |           | No                     | Other |
| 3. What is your age group?                                               |                | 15-30         | 31-45             | 45-60           | 61+       |                        |       |
| 4. Which sector are you working in?<br>*Sub-question (Area of expertise) |                | Public sector | Private sector    | Informal sector |           |                        |       |
| 5. What is your monthly salary range?                                    |                | Below R3 500  | R3 500 to R20 000 | >R20 000        | >R51 000  |                        |       |
| 6. What is your highest level of education?                              | Below Grade 12 | Grade 12      | Certificate       | Diploma         | Degree    | Postgraduate           |       |
| 7. How many years have you been employed in this sector?                 |                | 1 to 5 years  | 6 to 10 years     | 11 to 15 years  | 16+ years |                        |       |
| 8. Nature of employment?                                                 | Permanent      | Temporary     |                   | Self-employed   |           |                        |       |

## SECTION 2: HOUSEHOLDS AND SAVINGS

(Please mark the correct answer with an X)

|                                                                                                                                                          |  |        |         |           |          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------|---------|-----------|----------|
| 1. Does your household have any type of saving(s)?<br>If yes, kindly indicate the saving amount in rand.....                                             |  |        |         | Yes       | No       |
| 2. How often do you save?                                                                                                                                |  | Weekly | Monthly | Quarterly | Annually |
| 3. Where are the household savings kept?<br>In the bank? <input type="checkbox"/> In the house? <input type="checkbox"/> Other? <input type="checkbox"/> |  |        |         |           |          |
| 4. Does your household owe other private individuals money?<br>If yes, how much? .....                                                                   |  |        |         | Yes       | No       |
| 5. Is your household owed money by private individuals?<br>If yes, how much? R.....                                                                      |  |        |         | Yes       | No       |
| 6. Are you satisfied with the state of your households' savings? If not, provide reason(s).                                                              |  |        |         | Yes       | No       |
| 7. Are you participating in community savings or stokvels?                                                                                               |  |        |         | Yes       | No       |
| 8. Would you like to be made aware of new ways to encourage your household savings?                                                                      |  |        |         | Yes       | No       |
| 9. Would you attend community meetings on savings packages?                                                                                              |  |        |         | Yes       | No       |

|                                                                                       |     |    |
|---------------------------------------------------------------------------------------|-----|----|
| 10. Would you like to participate in community savings seminars?                      | Yes | No |
| 11. Would you attend meetings/seminars that help to encourage your household to save? | Yes | No |
| 12. What type of household savings will you prioritise?                               |     |    |
| A. Emergency household savings                                                        | A   |    |
| B. Pension / retirement savings                                                       | B   |    |
| C. Education savings                                                                  | C   |    |
| D. Stokvels (festive savings)                                                         | D   |    |
| E. Other (Please specify): .....                                                      | E   |    |

### SECTION 3: HOUSEHOLDS' RELATIONSHIPS WITH FINANCIAL INSTITUTIONS

In this section, households should rate their level of satisfaction with the services they receive from financial institutions, and also suggest areas where they feel the financial institutions can improve. The relevance of this section is to discover whether financial institutions are proactive in encouraging household savings.

Please use the scale below to reflect on how you feel about financial institutions. In the last row, mention areas of improvement that you want financial institutions to consider. ***Please mark the correct answer with an X.***

|                  |             |             |             |                  |
|------------------|-------------|-------------|-------------|------------------|
| 1. Very Poor [1] | 2. Poor [2] | 3. Fair [3] | 4. Good [4] | 5. Excellent [5] |
|------------------|-------------|-------------|-------------|------------------|

|                                                                                                       |     |     |     |     |     |
|-------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|
| <b><i>Financial institutions should service households on socio-economic issues by:</i></b>           |     |     |     |     |     |
| 1. Assisting households with various savings options available.                                       | [1] | [2] | [3] | [4] | [5] |
| 2. Being transparent to households in relation to savings products.                                   | [1] | [2] | [3] | [4] | [5] |
| 3. Providing households with sufficient knowledge on the costs and rewards of savings.                | [1] | [2] | [3] | [4] | [5] |
| 4. Providing an investment scheme and funeral cover for households.                                   | [1] | [2] | [3] | [4] | [5] |
| 5. Offering personal, financial-related advices and social advice to households.                      | [1] | [2] | [3] | [4] | [5] |
| Please suggest how financial institutions can improve servicing its members on socio-economic issues: |     |     |     |     |     |
|                                                                                                       |     |     |     |     |     |

|                                                                                                                    |     |     |     |     |     |
|--------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|
| <b><i>Financial institutions' offers are communications by:</i></b>                                                |     |     |     |     |     |
| 1. Sending out information to households on a regular basis                                                        | [1] | [2] | [3] | [4] | [5] |
| 2. Keeping households informed of financial institutions' activities, meetings, and decisions                      | [1] | [2] | [3] | [4] | [5] |
| 3. Consulting households and receiving a mandate on issues regarding negotiations                                  | [1] | [2] | [3] | [4] | [5] |
| 4. Providing services to households on time as expected                                                            | [1] | [2] | [3] | [4] | [5] |
| 5. Always responding to the media on urgent and important issues                                                   | [1] | [2] | [3] | [4] | [5] |
| Please suggest on how financial institutions can improve on communication issues:                                  |     |     |     |     |     |
| <b><i>Financial institutions provide educational services by:</i></b>                                              |     |     |     |     |     |
| 1. Providing and promoting educational programmes to households.                                                   | [1] | [2] | [3] | [4] | [5] |
| 2. Providing and promoting financial and awareness programmes.                                                     | [1] | [2] | [3] | [4] | [5] |
| 3. Organising training activities (workshops, seminars) to improve households' knowledge.                          | [1] | [2] | [3] | [4] | [5] |
| Please suggest how financial institutions can improve on providing educational services:                           |     |     |     |     |     |
| <b><i>Financial institutions offer personal and collective services by making sure that:</i></b>                   |     |     |     |     |     |
| 1. There is fairness in bank charges.                                                                              | [1] | [2] | [3] | [4] | [5] |
| 2. Irrelevant bank charges are prevented.                                                                          | [1] | [2] | [3] | [4] | [5] |
| 3. Households are consulted on the introduction of new technologies by financial institutions.                     | [1] | [2] | [3] | [4] | [5] |
| 4. Households grievances and complains are quickly resolved.                                                       | [1] | [2] | [3] | [4] | [5] |
| 5. They are honest, transparent, and accountable to members.                                                       | [1] | [2] | [3] | [4] | [5] |
| 7. They provide expected services and care for the households.                                                     | [1] | [2] | [3] | [4] | [5] |
| 8. Households' expectations are met.                                                                               | [1] | [2] | [3] | [4] | [5] |
| 9. Households are provided with general ATM training for usage.                                                    | [1] | [2] | [3] | [4] | [5] |
| 10. Households have access to bank offices (known location, contact details, and working hours).                   | [1] | [2] | [3] | [4] | [5] |
| Please suggest how financial institutions can improve on providing personal and collective services to households: |     |     |     |     |     |

## SECTION 4: ADDITIONAL TRIANGULATION QUESTIONS

1. Marital status

| Item                      | Tick the correct one |
|---------------------------|----------------------|
| Married                   |                      |
| Widowed                   |                      |
| Never married             |                      |
| Not interested to specify |                      |

2. Please indicate your **family size** in number\_\_\_\_\_.

3. According to 2021 OECD Better Life Index, the South African average yearly household income is R140 070. Given this estimation, what is your annual income? R\_\_\_\_\_.

4. Do savings give you a sense of future independence? Why?

|                       |    |
|-----------------------|----|
| Yes                   | No |
| Why? (Please specify) |    |

5. Do you think the government should introduce a state-owned commercial bank that promotes household saving in your township?

|                       |    |
|-----------------------|----|
| Yes                   | No |
| Why? (Please specify) |    |

6. The OECD Better Life Index indicates that the 2021 South African average monthly income is R11 672.50. Given this estimation, what is your monthly savings?

|        |        |                 |
|--------|--------|-----------------|
| Amount | R_____ | Other (specify) |
|--------|--------|-----------------|

7. The South African savings rate has been declining for over 10 years; the South African Reserve Bank's 2021 fourth-quarter statistics indicate that the South African national savings rate is 18% of the GDP. However, the household savings rate in South Africa decreased to 0.70% in the fourth quarter of 2021 from 1.20% in the third quarter of 2021. Given this estimation, what is your annual savings?

|        |        |                 |
|--------|--------|-----------------|
| Amount | R_____ | Other (specify) |
|--------|--------|-----------------|

8. If a person earns R10 000 a month and he/she saves R1 000, it means that the percentage of saving is 10%. Given this formula, roughly what percentage of your monthly income do you save?

|            |         |                 |
|------------|---------|-----------------|
| Percentage | _____ % | Other (specify) |
|------------|---------|-----------------|

9. What are your motivators for savings?

| Item                       | Tick the correct one |
|----------------------------|----------------------|
| To meet a specific purpose |                      |
| To get a tax benefit       |                      |
| To earn an income          |                      |
| To be secure in old age    |                      |
| Other (Please specify):    |                      |

10. Which factors do you consider for an increase in savings?

| Item                   | Tick the correct one |
|------------------------|----------------------|
| Increase in salary     |                      |
| Tax benefit            |                      |
| Additional             |                      |
| Future needs           |                      |
| Other (please specify) |                      |

[AWARENESS]

11. Where do you get information for your savings?

| Item                          | Tick the correct one |
|-------------------------------|----------------------|
| Television and radio          |                      |
| Organisational reports        |                      |
| Family members and colleagues |                      |
| Journals and magazines        |                      |
| Agents and advisors           |                      |

12. Please **assess** yourself on the following:

| Item                                   | Very poor | Poor | Fair | Good | Excellent |
|----------------------------------------|-----------|------|------|------|-----------|
| Your financial behaviour               |           |      |      |      |           |
| Your level of financial literacy       |           |      |      |      |           |
| Your financial knowledge               |           |      |      |      |           |
| Your level of financial risk tolerance |           |      |      |      |           |

13. Which of the following initiatives do you recommend to create awareness among working women/men about saving?

| Item                        | Tick the correct one |
|-----------------------------|----------------------|
| Training programmes         |                      |
| Workshops and seminars      |                      |
| Social welfare programmes   |                      |
| Advertisements              |                      |
| Investors'/savings meetings |                      |

14. Do you feel your existing savings amount will ensure your financial wellbeing in the future?

| Item             | Tick the correct one |
|------------------|----------------------|
| Not sufficiently |                      |
| Sufficiently     |                      |

15. Which of these special provisions do you need to boost your savings?

| Item                     | Tick the correct one |
|--------------------------|----------------------|
| Extra incentives         |                      |
| An additional bonus      |                      |
| High interest rate       |                      |
| Tax concession           |                      |
| High retirement benefits |                      |

16. What suggestion would you like to offer to popularise savings schemes among working women/men?

| Item                                   | Tick the correct one |
|----------------------------------------|----------------------|
| Special schemes                        |                      |
| Offering an additional bonus           |                      |
| Additional tax incentives              |                      |
| Offering specific schemes for children |                      |
| Education/financial literacy           |                      |
| Attractive pension schemes             |                      |

17. What is your household's saving culture? Please explain briefly.

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18. This last question is meant to measure the **income shock resilience** of households during the COVID-19 pandemic or any other income shock that happened in the last 12 to 24 months.

18.1 How did your savings assist in mitigating the income loss during the COVID-19 pandemic? Please explain briefly.

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18.2 What happened to your **savings** during the COVID-19 pandemic?

| Item      | Tick the correct one | Roughly by what percentage |
|-----------|----------------------|----------------------------|
| Increased |                      |                            |
| Decreased |                      |                            |
| Unchanged |                      |                            |

18.3 What happened to your **consumption** during the COVID-19 pandemic?

| Item      | Tick the correct one | Roughly by what percentage |
|-----------|----------------------|----------------------------|
| Increased |                      |                            |
| Decreased |                      |                            |
| Unchanged |                      |                            |

**THE END. Thank you for your cooperation and your valuable time.**

## **Annexure 2: Focus group discussion questions**



### **Innovative Ways to Encourage Household Savings for Financial Wellbeing**

The purpose of this focus group discussion is to investigate the main reasons for low household savings in the country. This will also enable us to understand where the main reasons for lack of savings emanate from. Has the apartheid era left negative socio-economic after-effects that led to the country's lack of savings and household savings? The overall objective of the focus group discussion is to assist in crafting innovative ways to encourage savings as this has never been done before.

The focus group will consist of 20 to 30 participants (divided into two groups) from different backgrounds, such as academics, government employees, professionals, and local community opinion leaders. The crafting of innovative ways will not only benefit the households by encouraging them to save, but will also benefit the country because when citizens are not saving, the country's economy is negatively affected. The crafted innovative ways to encourage household savings will also assist the government in implementing solutions that will enable people to save for the future.

#### **1. Focus group discussion questions**

- 1.1 What do you think are the main reasons for lack of household savings?
- 1.2 How would you describe household lack of savings / low savings before and after 1994?

- 1.3 What do you think are the main source(s) of income for households in townships?
- 1.4 What income level will be able to cover households' immediate monthly basic needs?
- 1.5 Are households aware of different savings products and initiatives available to them, both at the community level and bank level? (The purpose is to determine their financial literacy level.)
- 1.6 What is the main difficulty that hinders households to save and what can be done to ensure that they save at the desirable level? Please explain.
- 1.7 Are you aware of any policies designed for encouraging household savings, if any, and what can be done to ensure that those policies are effective in various communities?
- 1.8 What are the benefits of savings that households are exposed to?

## Annexure 3: Ethical clearance letter



Tshwane University  
of Technology

Faculty of Economics and Finance Research Ethics Committee [FREC-ECO]

The TUT Research Ethics Committee is a registered Institutional Review Board (IRB 00005968) with the US Office for Human Research Protections (IORG# 0004997) (Expires 19 Jan 2014). Also, it has Federal Wide Assurance for the Protection of Human Subjects for International Institutions (FWA 00011501) (Expires 31 Jan 2014). In South Africa it is registered with the National Health Research Ethics Council (REC-160509-21). The FREC-ECO is a subcommittee of the TUT Research Ethics Committee

6 June 2022

Baloyi NP

Ref #: FREC 2022/10-ECO  
Name: Baloyi NP  
Student 212074217

**Title:** Innovative Ways to Encourage Household Savings for Financial Wellbeing: The Case of Atteridgeville Township

**Programme:** Master of Business Studies in Finance

**Supervisor:** Dr MF Zerihun

Thank you for submitting your proposal for ethics clearance.

The Faculty of Economics and Finance Research Ethics Committee reviewed the documents.

The committee approved the ethical clearance on condition that any questionnaires should be submitted for ethical clearance.

The committee wishes you well with your research endeavours.

Kind regards

Chair: Prof HE Klingelhöfer  
Faculty Research Ethics Committee

[Ref#: 2022/012-ECO]

cc Supervisor/HoD/etc.



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## Annexure 4: Turnitin report

### INNOVATIVE WAYS TO ENCOURAGE HOUSEHOLD SAVINGS FOR FINANCIAL WELLBEING IN SOUTH AFRICA: A TOWNSHIP AND NATIONAL LEVEL EMPIRICAL ANALYSIS

#### ORIGINALITY REPORT

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#### PRIMARY SOURCES

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| <b>2</b> | <b>ojs.amhinternational.com</b><br>Internet Source | <b>1%</b> |
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| <b>4</b> | <b>csp.treasury.gov.za</b><br>Internet Source      | <b>1%</b> |
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## **Annexure 5: Curriculum vitae**

**Nhlonipho Prosperity Baloyi**

**Prosperitynhlonipho@gmail.com**

**078 659 0824**

### **PROFESSIONAL SUMMARY**

I am a dedicated, hardworking, vibrant young woman with more than seven years' experience in public institutions, government departments, and trade unions. I am currently a research officer at NEHAWU, and my role involves extensive policy alignment and monitoring in the labour market, public service, and the Fourth Industrial Revolution. I am a competent researcher with excellent data-collection skills, data-modelling skills, and data-analysis skills. I have good leadership qualities obtained from assuming different leadership roles.

### **EDUCATION**

#### **MASTER OF BUSINESS STUDIES IN FINANCE (NQF Level 9)**

**Tshwane University of Technology** Ga Rankuwa campus

**Year:** 2021 – currently

**Dissertation:** Households and Savings

#### **BCOM (HONOURS) ECONOMICS**

**University of South Africa** Pretoria campus

**Year:** 2021

**Dissertation:** The Issue of Foreign Aid in Developing Countries

## **BTECH: ECONOMIC MANAGEMENT ANALYSIS**

**Tshwane University of Technology** Pretoria campus

**Year:** 2015

**Research project:** The Impact of Oil Prices in the Automotive Industry

## **NATIONAL DIPLOMA: ECONOMIC MANAGEMENT ANALYSIS**

**Tshwane University of Technology** Pretoria campus

**Year:** 2012-2014

\*Top 5 in the Faculty of Economics and Finance

## **GRADE 12**

Saulridge Secondary School, Atteridgeville

## **WORK EXPERIENCE**

**Research Officer: Policy Development Unit**

**NEHAWU | October 2017 to present**

- Provide support to the union through analysing and disseminating information on a wide range of economic issues relevant to the union.
- Conduct and utilise research findings to impart expert knowledge to the union structures and members, and to actively assist the union in formulating economic campaigns.
- Monitor policy work on the economy, labour market, public service, Fourth Industrial Revolution, poverty, and inequality.
- Engage with other unions within COSATU, progressive research bodies, and allied non-governmental organisations, and to represent the union in relevant engagement forums, governmental, and statutory structures.

- Conduct workshops or seminars and help prepare submissions for the union on economic and public finance policy legislation, and assist the union in formulating press statements and writing articles on economic and public finance matters affecting the union and the society at large.
- Presented own research paper (4IR in the Public Service) at the Public Service Summit 2022; and was adopted as resolutions.

### **Junior Economist (Economic Impact and Policy Alignment Unit)**

#### **Department of Public Enterprises | March 2016 - September 2017**

- Provide support to the Economic Policy Alignment DDG: EIPA preparation and execution of economic policy analysis.
- Analysing the SOCs' financial statements and conducting impact assessments of the SOCs in the DPE Group, and to provide support to the Economic Policy DDG: EIPA in preparation and execution of economic policy modelling.
- Provide support on Industry-specific research conducted.
- Drafting submissions and secretarial-related administrative work.
- Support coordination and monitoring of the implementation of Operation Phakisa initiatives with the aim to facilitate regular progress reporting to the Minister and respective Operation Phakisa delivery units.

### **Economics Tutor**

#### **Tshwane University of Technology | April 2015 - December 2015**

- Facilitating weekly tutorial sessions; clarifying lecture content and assisting with tutorial assignments.
- Invigilation and marking assessments.
- Teaching the students the important concepts of the subject and discovering the problem areas around the subject.

- Alerting the lecturer about those problem areas and outsourcing additional subject-related information.
- Intermediating between the students and lecturer, and assisting to reach the determined pass rate goal.

### **Student Assistant: Librarian**

#### **Tshwane University of Technology | March 2014 - March 2015**

- Maintaining proper filing of shelves and Issuing books to students.
- Receiving (checking in) and checking out books and updating the system with new editions; receiving receipts of fines cancelling, and utilising the work on Millennium.

### **TRAINING**

- Research analytical skills, project management, business writing, and organising skills.
- Microsoft Office (MS Word, MS Excel, MS PowerPoint, MS Outlook, Internet Explorer, and E-Views).
- TIPS Industrial Policy Training and Nyukela Public Service Pre-entry Programme.

### **AWARDS AND LEADERSHIP**

- Member of the NEDLAC Public Finance Monitoring Chamber Committee (2018 to present)
- A member of NEDLAC Trade and Industry Committee (2018 to present)
- Top 5 TUT Economics graduates, 2014
- Most dedicated mentor – East Res, 2014
- Best-performing mentor – East Res, 2014

## **SKILLS**

- Analytical and modelling
- Risk assessment
- Stakeholder engagement
- Project management and administration
- Organisational skills economic research and reporting

## **REFERENCES**

Mr Dzingai Chapfuwa

Department of Public Enterprises

Chief Economist

[dchapfuwa@gmail.com](mailto:dchapfuwa@gmail.com)

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Tshwane University of Technology

Head of Economics Department

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NEHAWU

Head: Policy Development Unit

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082 455 2614 / 011 833 2902